

The Balancing Act in Causal Inference^{*}

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Abstract

The idea of covariate balance is at the core of causal inference. In randomized experiments, it is an expected outcome of randomization, whereas in observational studies balance is a goal we strive to approximate through adjustment methods such as matching and weighting. Inverse propensity weights play a central role because they are the unique set of weights that balance the covariate distributions of different treatment groups. We discuss two broad approaches to estimating these weights: the more traditional one, which fits a propensity score model and then uses the reciprocal of the estimated propensity score to construct weights, and the balancing approach, which estimates the inverse propensity weights essentially by the method of moments, finding weights that achieve balance in the sample. We connect ideas from the causal inference, sample surveys, and semiparametric estimation literatures, with particular attention to the role of balance as a sufficient condition for robust inference. We bridge practical approaches to matching in causal inference with the theoretical framework of semiparametric theory. We focus on the inverse propensity weighting and augmented inverse propensity weighting estimators for the average treatment effect and consider generalizations for a broader class of problems including policy evaluation and the estimation of individualized treatment effects.

Keywords: Causal Inference; Matching Methods; Propensity Score; Observational Studies; Weighting Methods

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1 Introduction

The idea of covariate balance is at the core of causal inference. Intuitively, when estimating the effect of a treatment, we want to compare “like with like”: if two groups are similar in every respect except in assignment to treatment, then we can attribute differences in outcomes to the treatment itself, as opposed to other confounding factors (Cochran, 1965).

This intuition is widespread. However, any two groups will differ in some respect, so when we talk about a like with like comparison, we rely on a consensus about which differences matter and which can be ignored. For example, we might focus on imbalances in the means of covariates, or in their marginal distributions, or in summaries of their joint distribution. In this paper, we explore the role covariate balance plays in statistical inference about treatment effects; its relationship to common notions of balance; and connections to the causal inference, sample surveys, and semiparametrics literatures.

Inverse propensity weights play a central role because they are the unique set of weights that balance the covariate distributions of different treatment groups. However, outside randomized experiments, these weights are unknown and must be estimated.

The most common approach to estimating inverse propensity weights is to estimate the conditional probability of treatment, or propensity score, and then take its reciprocal. With some estimation methods, this approach can be unstable, especially when this probability is close to zero, and can lead to poor balance in the sample at hand when the propensity score is not estimated accurately. Researchers often address these limitations with post-hoc fixes, e.g., truncating the extreme weights that arise from propensity score estimates that are too near zero or checking balance and re-estimating the propensity score if it is poor.

An alternative approach is to find weights that optimize balance in the data at hand. Because balance is a defining property of the inverse propensity weights, this is essentially a method of moments estimate of the inverse propensity weights. Researchers typically estimate such weights by solving a constrained optimization problem, e.g., by finding weights that restrict the maximum imbalance in covariate means between groups. Balance checks are therefore built in: we choose weights to pass the checks, rather than performing them post-hoc. Examples of proposals in line with this approach include Athey et al. (2018), Breidt, Opsomer, et al. (2017), Chan et al. (2016), Deville and Särndal (1992), Graham et al. (2012), Hainmueller (2012), Hirshberg and Wager (2021), Imai and Ratkovic (2014), Kallus (2020a), Li et al. (2018), Tan (2020), Wong and Chan (2018), Zhao (2019), and Zubizarreta (2015).

We will organize our discussion around a specific form of balance that is necessary and sufficient for straightforward inference based on the widely-used inverse probability weighting (IPW) and augmented inverse probability weighting (AIPW) estimators. For IPW, we need balance in the mean outcome conditional on treatment and covariates, and for AIPW, balance in the errors we make when estimating that conditional mean. This happens when we estimate the inverse propensity weights and this conditional mean function sufficiently well, an idea that gives rise to the concept of double-robustness, and also when we balance every function in a set that contains the conditional mean, an idea that gives rise to balance checks and weights optimized to achieve balance in this sense. In particular, many widely-used balance checks for weights are based on the *maximal imbalance* that would occur for a conditional mean in a set chosen to encode structural assumptions about it. This gives us a practical framework to link intuition and assumptions about the data to appropriate balance criteria to check or optimize.

2 Framework

2.1 Setup

We posit the existence of independent and identically distributed tuples $(X_i, W_i, Y_i(0), Y_i(1))$ randomly drawn from a population, where $X_i \in \mathbb{R}^p$ are observed covariates, $W_i \in \{0, 1\}$ is a binary treatment indicator, and $Y_i(0), Y_i(1) \in \mathbb{R}$ are the potential outcomes under control and treatment, respectively (Neyman, 1990 [1923]; Rubin, 1974). Under the Stable Unit Treatment Value Assumption (SUTVA; Rubin, 1980), we can write the observed outcome Y_i as $Y_i = (1 - W_i)Y_i(0) + W_iY_i(1)$. Since the tuples are randomly drawn from a population and we are interested in population-level estimands, we generally drop the subscript i .

We are primarily interested in the *Average Treatment Effect* (ATE), defined as

$$\tau := E[Y(1) - Y(0)]. \tag{1}$$

To ease exposition, we separate it into two components, $\psi_1 := E[Y(1)]$ and $\psi_0 := E[Y(0)]$, and focus on estimating the treatment-specific mean ψ_1 ; we can estimate ψ_0 analogously.

We work with the assumption of strong ignorability (Rosenbaum & Rubin, 1983).

Assumption 1 (Ignorability). $W \perp (Y(0), Y(1)) \mid X$

Assumption 2 (Overlap). $e(x) > 0$ where $e(x) := E[W \mid X = x]$ is the *propensity score*

With Assumptions 1-2, the treatment-specific mean is the average conditional mean outcome.

$$\psi_1 = E[\mu_1(X)] \quad \text{where} \quad \mu_w(x) := E[Y \mid W = w, X = x]. \quad (2)$$

The main challenge when estimating ψ_1 is that, outside of randomized experiments, the mix of units that receive one treatment tends to differ from the mix that receive another. Thus, the average outcome among those who receive the treatment reflects selection into treatment as well as the treatment's effect. Under Assumption 1, the observed covariates explain the salient differences between these groups, so eliminating selection effects reduces to a covariate adjustment problem.

2.2 Inverse propensity weighting

A celebrated result in causal inference is that the inverse propensity weighted average of the outcomes is an unbiased estimator of ψ_1 under Assumptions 1 and 2 (Rosenbaum, 1987):

$$\psi_1 = E[\mu_1(X_i)] = E\left[\frac{W_i}{e(X_i)}\mu_{W_i}(X_i)\right] = E\left[\frac{W_i}{e(X_i)}Y_i\right]. \quad (3)$$

In fact, this result holds more generally: the inverse propensity weights are the unique weighting function that, for *every* function $f(x)$, equates weighted averages of f over the treated units to unweighted averages over the study population. That is, $\gamma^{\text{ipw}}(x) = 1/e(x)$ is the only function satisfying

$$E[W_i\gamma^{\text{ipw}}(X_i)f(X_i)] = E[f(X_i)] \quad \text{for all bounded } f(x). \quad (4)$$

Because the solution is unique, Equation 4 also *defines* the inverse propensity weights γ^{ipw} . We call Equation 4 the *population balance property*: inverse propensity weights adjust for differences (in the averages of all functions f of the covariates) between the subpopulation receiving treatment and the overall population. Setting $f = \mu_1$ recovers the unbiasedness result above (3). The law of large numbers implies a corresponding *sample balance property*:

$$\frac{1}{n} \sum_{i=1}^n W_i \gamma^{\text{ipw}}(X_i) f(X_i) \approx \frac{1}{n} \sum_{i=1}^n f(X_i) \quad \text{for any bounded } f(x). \quad (5)$$

When using an estimate of the inverse propensity weights γ^{ipw} , it is common to check that sample balance indeed holds for some set of functions. One common approach, in which we check that each covariate has a similar average in the weighted treatment group and in the whole sample, looks at balance in the set of coordinate functions $f(x) = x_j$. Another, in which we compute the marginal histograms of each covariate for the two groups and compare them using the Kolmogorov-Smirnov statistic, looks at balance in a set of additive functions; see Appendix C.2 for further discussion. Balance checks like this have a long history in observational causal inference (Cochran, 1965).

2.3 Imbalance and estimation error

Sample balance is not just a signal we use to appraise the accuracy with which we have estimated the inverse propensity weights. It is a key determinant of the error of estimators that use them. As starting point, we consider the error of a general weighting estimator,

$$\hat{\psi}_1 = \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) Y_i. \quad (6)$$

Throughout, we will consider only design-based weights $\hat{\gamma}(X_i)$, i.e., weights that depend on the study’s *design* — the treatment W_i and covariates X_i — but not on the outcomes Y_i .¹ The error of this estimator is, in terms of $\varepsilon_i := Y_i - \mu_{W_i}(X_i)$,

$$\hat{\psi}_1 - \psi_1 = \underbrace{\frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \mu_1(X_i) - \frac{1}{n} \sum_{i=1}^n \mu_1(X_i)}_{\text{imbalance in } \mu_1} + \underbrace{\frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \varepsilon_i}_{\text{noise}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \mu_1(X_i) - \psi_1}_{\text{sampling variation}}, \quad (7)$$

The first term in Equation 7 is the imbalance in the conditional mean $\mu_1(X_i)$. The second and third terms are a weighted average of noise and irreducible sampling variation respectively. Since these two terms both have mean zero conditional on the design $\mathcal{D} = \{(W_i, X_i) : i \leq n\}$, the estimator’s design-conditional bias $E[\hat{\psi}_1 - \psi_1 \mid \mathcal{D}]$ is the imbalance in μ_1 .

What makes estimation challenging is that the conditional mean function μ_1 is unknown. One natural approach is to think of μ_1 as belonging to some set of functions $\mathcal{M}$, e.g., the set of linear functions, which serves as a *model*. This is not a model for the propensity score or a parametric likelihood—it is a set of functions that we believe contains the true conditional mean μ_1 , and whose geometry determines how well our weights can control bias. When μ_1 belongs to this set, the absolute design-conditional bias is bounded by the largest value it could take on for any $\mu_1 \in \mathcal{M}$, which we call the *maximal imbalance*, $\text{imbalance}_{\mathcal{M}}(\hat{\gamma})$:

$$|\text{design-conditional bias}| \leq \text{imbalance}_{\mathcal{M}}(\hat{\gamma}) := \max_{\mu_1 \in \mathcal{M}} \left| \frac{1}{n} \sum_{i=1}^n \mu_1(X_i) - \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \mu_1(X_i) \right|. \quad (8)$$

Controlling maximal imbalance is effective at controlling bias if we choose the model well.

2.4 Augmented Inverse Propensity Weighting

A prominent generalization of the IPW estimator augments it with an adjustment involving an estimate of the conditional mean function μ_1 fit via regression on the treated sample (Robins et al., 1994). This augmented IPW (AIPW) estimator uses the regression estimator $\hat{\mu}_1$ to impute and adjust for imbalance in μ_1 that is left over after weighting, or equivalently uses weighting to adjust for imbalance in the regression residuals left over after imputation.

$$\hat{\psi}_1^{\text{aug}} := \underbrace{\frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) Y_i}_{\text{weighting estimator}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - W_i \hat{\gamma}(X_i) \hat{\mu}_1(X_i)\}}_{\text{bias correction via imputation}} \quad (9)$$

$$= \underbrace{\frac{1}{n} \sum_{i=1}^n \hat{\mu}_1(X_i)}_{\text{imputation estimator}} + \underbrace{\frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \{Y_i - \hat{\mu}_1(X_i)\}}_{\text{bias correction via weighting}} \quad (10)$$

Though it is traditional to use a modeling approach to estimate the weights used in the AIPW estimator, many recent proposals combine this bias correction with control of maximal imbalance. Examples include *approximate residual balancing* (Athey et al., 2018), *augmented minimax linear estimation* (Hirshberg & Wager, 2021), and *automatic double machine learning* (Chernozhukov, Newey, & Singh, 2018).

The augmented estimator has an error decomposition analogous to that of a pure weighting estimator. In it, imbalance in the *regression error function* $\delta\mu_1 = \mu_1 - \hat{\mu}_1$ replaces imbalance in μ_1 itself.

$$\hat{\psi}_1^{\text{aug}} - \psi_1 = \underbrace{\frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \delta\mu_1(X_i) - \frac{1}{n} \sum_{i=1}^n \delta\mu_1(X_i)}_{\text{imbalance in } \delta\mu} + \underbrace{\frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \varepsilon_i}_{\text{noise}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \mu_1(X_i) - \psi_1}_{\text{sampling variation}}. \quad (11)$$

¹Furthermore, as implied by our notation, we will consider only weights that are functions of the covariate X_i . This is almost universal among design-based weights, there being no design-based reason to distinguish units with identical covariate and treatment status by assigning them different weights, but is not always indicated notationally the way we do here: some write $\hat{\gamma}_i$ instead of $\hat{\gamma}(X_i)$ for the weight of the i th unit.

Intuitively, imbalance with augmentation should be smaller than without because the regression error function δm should be smaller than the conditional mean function μ_1 itself. This generalizes our previous error decomposition for $\hat{\psi}_1$: the IPW estimator is AIPW with $\hat{\mu}_1 = 0$, in which case $\delta m = \mu_1$ and the error decomposition reduces to Equation 7. Thus, maximal imbalance can play a similar role in our thinking; however, in this case we will want our model $\mathcal{M}$ to contain the regression error $\delta\mu$, not μ_1 itself. For brevity, our discussion will be in terms of the more general augmented estimator and the regression error $\delta\mu$, noting when we require that the regression estimator $\hat{\mu}_1$ is or is not zero.

Remark 1. One popular alternative to AIPW is the *targeted maximum likelihood estimator* (TMLE), an imputation estimator that uses a specialized regression estimator $\hat{\mu}_1$ so that bias correction is unnecessary. To make this happen, $\hat{\mu}_1$ is chosen so that the bias correction term in Equation 10 is small or even exactly zero. We can do this, for example, using inverse-probability weighted least squares regression in a regression model that includes an unconstrained intercept. The resulting least squares predictor $\hat{\mu}_1$ minimizes weighted squared error among all functions in the model and therefore among the subset $\{\hat{\mu}_1(\cdot) + t : t \in \mathbb{R}\}$ of translations of $\hat{\mu}_1$, and therefore satisfies a zero-derivative condition

$$0 = \left. \frac{d}{dt} \right|_{t=0} \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \{Y_i - \hat{\mu}_1(X_i) - t\}^2 = -2 \times \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \{Y_i - \hat{\mu}_1(X_i)\}.$$

that implies our bias correction term in (10) is exactly zero. For the purpose of our discussion below, this *is* an AIPW estimator. The bias correction is there; it just happens to be zero.

3 Estimating inverse propensity score weights

In observational studies, the inverse propensity weights γ^{IPW} are unknown and must be estimated. The two equivalent definitions of γ^{IPW} , via the explicit form $\gamma^{\text{IPW}}(x) = 1/e(x)$ and via the population balance property (4), suggest two broad types of estimators for inverse propensity weights. Traditional modeling approaches estimate $e(x)$ by regressing treatment status on covariates. Balancing estimators instead directly estimate weights $\gamma(x)$ that satisfy the sample balance property (5). We turn to these now.

3.1 Modeling approach to weighting

By far, the most common approach to weighting is to take an estimate of the propensity score, $\hat{e}(x)$, and use its reciprocal to compute inverse propensity weights, $\hat{\gamma}(X_i) = 1/\hat{e}(X_i)$. Researchers typically estimate $e(x)$ via maximum likelihood, e.g., with a logistic regression of W on X , though other learning approaches are used as well. The sample balance property (5) implies that if the estimated propensity score $\hat{e}(x)$ converges to the true propensity score $e(x)$, $\hat{\psi}_1$ converges to ψ_1 . This approach suffers from two major drawbacks.

The first is instability. Observations with small estimated probabilities of treatment tend to dominate the analysis. To some extent, this is inevitable, as we need to give rarely-treated observations large weights to ensure representativeness of the weighted sample. However, small errors in estimating treatment probability can lead to dramatic differences in the weights. If two units are assigned to treatment with probability .05, but we estimate that probability to be .01 for one unit and .09 for the other, the first unit gets nine times the weight that the second does, though our probability estimate is off by only .04 in both cases.

The second is that the estimated weights $1/\hat{e}(X_i)$ may not satisfy the sample balance property in Equation 5 with much (or any) accuracy, which can lead to bias. This can occur when the estimated weights are inaccurate, e.g., due to misspecification or a small sample size. In practice, it is often difficult to estimate conditional treatment probabilities, and the instability brought on by inversion compounds this problem.

To deal with the first drawback, investigators sometimes *trim* the inverse propensity weights, taking $\hat{\gamma}(X_i) = \min\{T, 1/\hat{e}(X_i)\}$ for some threshold T (Crump et al., 2009). To deal with the second, investigators resort to balance checks like those discussed in Section 2.2. If imbalance is large, they often fit a new propensity score model, repeating this process until they find one that achieves good balance. This is a good accompaniment to trimming, which can introduce imbalances that are not present with the untrimmed weights $\hat{\gamma}(X_i) = 1/\hat{e}(X_i)$.

3.2 Balancing approach to weighting

An alternative approach is to find weights $\hat{\gamma}$ by solving for weights that satisfy the sample balance property (5), sometimes known as *balancing weights*. While different in its implementation, this is also an inverse propensity weighting estimator: we can interpret balancing weights as a *method of moments* estimate of the inverse propensity weights. Specifically, as we discuss in Section 2.2, the inverse propensity score weights γ^{ipw} are the unique solution to the population balance moment conditions (4). The balancing approach is to find weights $\hat{\gamma}$ that satisfy the corresponding sample balance moment conditions (5), i.e., weights for which $\text{imbalance}_{\mathcal{M}}(\hat{\gamma})$ is small for some model $\mathcal{M}$.

Typically, this is done by solving an optimization problem to choose a weighting function $\hat{\gamma}(\cdot)$ to trade off imbalance and some measure of the complexity of the weights,

$$\hat{\gamma} = \underset{\gamma}{\operatorname{argmin}} \left\{ \text{imbalance}_{\mathcal{M}}^2(\gamma) + \frac{\sigma^2}{n^2} \sum_{W_i=1} \gamma(X_i)^2 \right\}, \quad (12)$$

where σ is a tuning parameter, or more generally, for increasing convex functions ζ and χ ,

$$\hat{\gamma} = \underset{\gamma}{\operatorname{argmin}} \left\{ \zeta\{\text{imbalance}_{\mathcal{M}}(\gamma)\} + \frac{1}{n^2} \sum_{W_i=1} \chi\{\gamma(X_i)\} \right\}. \quad (13)$$

In the familiar special case that the weights control the *maximal imbalance* for the linear model with ℓ_1 -bounded coefficients, $\mathcal{M} = \{\beta \cdot x : \|\beta\|_1 \leq 1\}$, the corresponding imbalance measure is the largest imbalance in the individual elements of the covariate vector x :

$$\text{imbalance}_{\mathcal{M}}(\gamma) = \max_{j=1, \dots, p} \left| \frac{1}{n} \sum_{i=1}^n X_{ij} - \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) X_{ij} \right| \text{ for } \mathcal{M} = \{\beta \cdot x : \|\beta\|_1 \leq 1\}. \quad (14)$$

The special case (12) has a natural *minimax* interpretation: the weights minimize the worst-case design-conditional mean squared error for $\mu_1 \in \mathcal{M}$ and $\text{var}[Y_i | X_i, W_i] \leq \sigma^2$. Furthermore, when the model $\mathcal{M}$ is an absolutely convex set, they have an equivalent dual characterization as a least-squares estimate of the inverse propensity score. That is, the weights are the function of the covariates $\hat{\gamma}(X_i)$ that solves a penalized least squares problem,

$$\min_{\gamma} \underbrace{\frac{1}{n} \sum_{i=1}^n W_i \{\gamma^{\text{ipw}}(X_i) - \gamma(X_i)\}^2}_{\text{mean squared error}} - \underbrace{\left\{ \frac{1}{n} \sum_{i=1}^n \gamma(X_i) - \frac{1}{n} \sum_{i=1}^n W_i \gamma^{\text{ipw}}(X_i) \gamma(X_i) \right\}}_{\text{mean zero "noise"}} + \frac{\sigma^2}{n} \|\gamma\|_{\mathcal{M}}^2. \quad (15)$$

The penalty is proportional to the square of the *gauge* of the model, $\|\gamma\|_{\mathcal{M}} = \inf\{\alpha > 0 : \gamma \in \alpha\mathcal{M}\}$ for $\alpha\mathcal{M} = \{\alpha m : m \in \mathcal{M}\}$, which is the factor α by which we must *inflate* the model to get a set that contains γ .² The balancing property of the inverse propensity weights implies the middle term has mean zero, so up to this noise-like term, the minimax weights estimate γ^{ipw} by minimizing penalized mean squared error.

Many existing approaches can be written as examples of Equation 13 with various choices of ζ and χ , including *entropy balancing* (Hainmueller, 2012); the *stable balancing weights* approach (Zubizarreta, 2015); the *lasso minimum distance estimator* of the inverse propensity score (Chernozhukov, Newey, & Singh, 2018); and the *regularized calibrated propensity score estimator* (Tan, 2020) (see Wang and Zubizarreta, 2020). In fact, even imputing ψ_1 by averaging the predictions of least squares or ridge regressions of Y on X fit on the treated units can be written in this form: they are weighted averages of observations Y_i with weights solving the problem above (12) for $\sigma^2 = 0$ and $\sigma^2 > 0$ respectively (Chattopadhyay & Zubizarreta, 2021; Kallus, 2020b).

²The gauge of an absolutely convex set of functions of x is a norm or seminorm on functions of x . For example, for the linear model with ℓ_1 -bounded coefficients $\mathcal{M} = \{\beta \cdot x : \|\beta\|_1 \leq 1\}$, the gauge is the ℓ_1 -norm of the coefficients of $\gamma(x) = \beta \cdot x$: $\|\gamma\|_{\mathcal{M}} = \|\beta\|_1$. It is infinite if $\gamma(x)$ is not linear in x , i.e., if $\gamma(x)$ is not in the space spanned by the set of functions in the model. If our model is defined as the unit ball of some norm on functions, i.e. if $\mathcal{M} = \{m : \|m\| \leq 1\}$, then its gauge is just that norm $\|\cdot\|$.

The general case (13) has a qualitatively similar dual characterization, in which the inverse propensity weights $\gamma(X_i) = \eta_\chi\{g(X_i)\}$ are parameterized in terms of an *index function* $g(x)$, and the penalty is imposed on the index. For example, if the model were the set of linear functions with ℓ_1 -bounded coefficients $\{\beta \cdot x : \|\beta\|_1 \leq 1\}$, this would be a generalized linear model $\gamma(X_i) = \eta_\chi(\beta \cdot X_i)$ with a penalty scaling with $\|\beta\|_1$. In practice, χ is often chosen so that the *link function* η_χ is the reciprocal of a logit link, so when balancing we are fitting a familiar generalized linear model for the propensity score with a sparsity-inducing penalty. See Hirshberg (2026) for additional discussion of these dual characterizations.

This connection between the details in the specification of a balancing problem and an implicit model for the inverse propensity score has been discussed at length, often with a focus on the case that the model $\mathcal{M}$ is a linear one (e.g., Zhao, 2019). These details are less salient with a fully nonparametric model $\mathcal{M}$, as it is not really possible to get the link η_χ wrong when the space of functions spanned by our model contains good approximations to all functions g . For these models, irrespective of our choice of χ , the index function $\hat{g}_\chi$ solving the appropriate generalization of the dual (15) will simply converge to $g_\chi = \eta_\chi^{-1} \circ \gamma^{\text{ipw}}$, so $\eta_\chi \circ \hat{g}_\chi$ will converge to γ^{ipw} .

4 Asymptotic behavior

Typically, when imbalance in $\delta\mu$ is asymptotically negligible relative to the estimator’s standard error, $\hat{\psi}_1^{\text{aug}}$ is asymptotically normal. To be more precise, following Athey et al. (2018, see Corollary 6), when the weights $\hat{\gamma}(X_i)$ are not too concentrated on any one observation i and the noise $\varepsilon_i = Y_i - \mu_{W_i}(X_i)$ is not too heavy-tailed, negligible imbalance justifies the use of the standard Wald-type confidence interval $\hat{\psi}_1^{\text{aug}} \pm z_{\alpha/2} \sqrt{\hat{V}}$, where $\hat{V}$ is a natural variance estimator and $z_{\alpha/2}$ is a quantile of the standard normal distribution. We state a result for estimating $\tilde{\psi}_1 = \frac{1}{n} \sum_{i=1}^n \mu_1(X_i)$.³

Proposition 4.1. Let $\Gamma := \sqrt{n^{-1} \sum_{i=1}^n W_i \hat{\gamma}(X_i)^2}$ be the root-mean-squared weight. Suppose that each squared weight is negligible compared to their sum $n\Gamma^2$, and assume that the noise sequence $\varepsilon_1 \dots \varepsilon_n$ is uniformly square-integrable conditional on the design. If the imbalance in $\delta\mu$ (see Equation 11) is $o_p(n^{-1/2}\Gamma)$, then

$$\hat{\psi}_1^{\text{aug}} - \tilde{\psi}_1 = \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \{Y_i - \mu_1(X_i)\} + o_p(n^{-1/2}\Gamma). \quad (16)$$

Furthermore, if $\hat{\mu}_1$ is a consistent estimator of μ_1 , then

$$\frac{\hat{\psi}_1^{\text{aug}} - \tilde{\psi}_1}{\sqrt{\hat{V}}} \rightarrow N(0, 1) \quad \text{for} \quad \hat{V} = \frac{1}{n^2} \sum_{i=1}^n W_i \hat{\gamma}(X_i)^2 \{Y_i - \hat{\mu}_1(X_i)\}^2. \quad (17)$$

If imbalance in μ_1 is $o_p(n^{-1/2}\Gamma)$, the same results hold with $\hat{\psi}^{\text{ipw}}$ in place of $\hat{\psi}^{\text{aug}}$.

In Appendix B we include a more formal and general statement and a simple proof. In essence, because the noise term dominates the error decomposition (11) and is a sum of conditionally independent mean-zero random variables, the estimate satisfies a central limit theorem. We will from this point on focus on the case that the root mean squared weight Γ is constant order, so the condition that imbalance is $o_p(n^{-1/2}\Gamma)$ stated in Proposition 4.1 reduces to it being $o_p(n^{-1/2})$. This is the case when we have sufficient overlap (Assumption 2’) and our weights $\hat{\gamma}$ are a consistent estimator of γ^{ipw} ; we will have more extreme weights when overlap is poor, and can retain a noise-dominated error decomposition with a proportionately more imbalance.

When we assume sufficient overlap, a clear story of optimality emerges.⁴

³By focusing on accuracy in estimating $\tilde{\psi}_1$, a sample analog of ψ_1 , we drop the sampling variation term from the error decomposition in Equation 11. In a sense, because all we know about the distribution of X_i is summarized by the sample $X_1 \dots X_n$, it is inevitable that every estimator of ψ_1 is an estimate of $\tilde{\psi}_1$ as well, and a more precise one at that. See Imbens (2004).

⁴Sufficient overlap rules out the possibility that rare treatment assignments have an undue influence on the estimand. It is equivalent to a continuity property: if two conditional mean outcome functions μ and $\tilde{\mu}$ are close in the sense that $E\{\mu_W(X) - \tilde{\mu}_W(X)\}^2$ is small, then the corresponding treatment-specific means $\psi_1 = E[\mu_1(X)]$ and $\tilde{\psi}_1 = E[\tilde{\mu}_1(X)]$ must be close as

Assumption 2' (Sufficient Overlap). $E[1/e(X)] < \infty$.

When this holds, all *regular* estimators are asymptotically equivalent (i.e. differ by an $o_p(n^{-1/2})$ term) and therefore have the best possible asymptotic variance (see, e.g., van der Vaart, 2000, Chapter 25). In essence, regular estimators are ones that work as expected even if our assumptions about the data's distribution are slightly incorrect. In particular, they are equivalent to an instance of $\hat{\psi}_1^{aug}$ with the ideal weights $\hat{\gamma} = \gamma^{ipw}$ and outcome regression $\hat{\mu}_1 = \mu_1$. We get a regular estimator when we achieve negligible imbalance using weights that balance a sufficiently rich model $\mathcal{M}$.

Definition 1 (Nonparametric model). A model $\mathcal{M}$ is *nonparametric* if it can be inflated to approximate any square integrable function: for every m with $\|m\|_{L_2(P)} < \infty$ and $\epsilon > 0$, there exists $\tilde{m}$ with $\|\tilde{m}\|_{\mathcal{M}} \leq \alpha$ satisfying $\|m - \tilde{m}\|_{L_2(P)} < \epsilon$.

By using a model like this, we control the imbalance in every square integrable function m to some degree (i.e. $o_p(1)$ imbalance), as the maximal imbalance over the inflated model $\alpha\mathcal{M}$ is α times the maximal imbalance over the model itself. This ensures that small errors in our modeling assumptions do not completely change our estimator's behavior. See Appendix B for details.

On an intuitive level, optimality also boils down to balance, but in a different function: the inverse-probability-weighted conditional variance $m(x) = \gamma^{ipw}(x)v_1(x)$ for $v_1(x) = \text{var}[Y_i | W_i = 1, X_i]$. We can see this by comparing the conditional variance of an estimator with arbitrary weights $\hat{\gamma}$ to the one with the ideal weights γ^{ipw} . Using the identity $a^2 - b^2 = (a - b)^2 + 2(a - b)b$, we can characterize it as proportional to the sum of a variance-weighted mean-squared distance between the two sets of weights and the *difference in imbalance* we get in the function $f = \gamma^{ipw}v_1$ with the two sets of weights.

$$\begin{aligned} & n \times (V^{\hat{\gamma}} - V^{\gamma^{ipw}}) \\ &= \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i)^2 v_1(X_i) - \frac{1}{n} \sum_{i=1}^n W_i \gamma^{ipw}(X_i)^2 v_1(X_i) \\ &= \frac{1}{n} \sum_{i=1}^n W_i \{\hat{\gamma}(X_i) - \gamma^{ipw}(X_i)\}^2 v_1(X_i) + \frac{2}{n} \sum_{i=1}^n W_i \{\hat{\gamma}(X_i) - \gamma^{ipw}(X_i)\} \gamma^{ipw}(X_i) v_1(X_i) \end{aligned} \quad (18)$$

where, adding and subtracting the unweighted average of f ,

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n W_i \{\hat{\gamma}(X_i) - \gamma^{ipw}(X_i)\} \gamma^{ipw}(X_i) v_1(X_i) \\ &= \underbrace{\left\{ \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) f(X_i) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right\}}_{\text{imbalance in } f = \gamma^{ipw}v_1 \text{ with estimated weights } \hat{\gamma}} - \underbrace{\left\{ \frac{1}{n} \sum_{i=1}^n W_i \gamma^{ipw}(X_i) f(X_i) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right\}}_{\text{imbalance in } f = \gamma^{ipw}v_1 \text{ with ideal weights } \gamma^{ipw}}. \end{aligned} \quad (19)$$

When we use a model that is a sufficiently rich model in the sense described, both imbalance terms will be $o_p(1)$, so this variance difference effectively reduces to the mean squared distance term. This implies that we cannot improve on the variance we get with the weights γ^{ipw} without sacrificing balance in the function $f = \gamma^{ipw}v_1$. If we reverse the roles of $\hat{\gamma}$ and γ^{ipw} in the equation above, we see that if both sets of weights also balance the random function $f(x) = \hat{\gamma}(x)v_1(x)$, which follows from control of maximal imbalance together with a mild regularity condition on the weights (see Appendix B), then we cannot improve upon the variance achieved with $\hat{\gamma}$ either. Furthermore, because in this case the variance difference must converge to zero, it follows that the estimated weights $\hat{\gamma}$ must converge to the inverse probability weights γ^{ipw} .

We can reduce variance by using weights that balance a less rich model. In particular, this is possible when we use a model that we cannot inflate to approximate $\gamma^{ipw}v_1$, e.g. if our model contains only linear functions of x and $\gamma^{ipw}v_1$ is nonlinear. But this comes with a significant risk: if $\hat{\mu}_1$ deviates from μ_1 by as

well. This can be seen using the equivalent characterization $\psi_1 - \tilde{\psi}_1 = E[\{W/e(X)\}\{\mu_W(X) - \tilde{\mu}_W(X)\}]$; the Cauchy-Schwarz inequality implies that this is bounded by the square root of $E\{W/e(X)\}^2 E\{\mu_W(X) - \tilde{\mu}_W(X)\}^2 = E\{1/e(X)\} E\{\mu_W(X) - \tilde{\mu}_W(X)\}^2$. Without sufficient overlap or strong assumptions on the (semi)parametric form of $\mu_w(x)$, there does not exist a regular estimator of ψ_1 (A. van der Vaart, 1991).

little as $1/\sqrt{n}$ in a direction our model cannot approximate, we will get non-negligible bias. We are gaining efficiency by using balancing weights that converge to a *projection* of γ^{ipw} onto the span of the functions in the model $\mathcal{M}$ rather than γ^{ipw} itself, and these weights do not balance errors $\hat{\mu}_1 - \mu_1$ in directions outside this span. By using a model like this, we can get a lower variance than we'd get using the ideal weights γ^{ipw} , but we pay for it with a substantial loss of robustness (Hirshberg & Wager, 2021, Remark 3).

5 Balance measures and models

5.1 Balancing binary covariates

5.1.1 Models with no interactions

In the balancing approach of Section 3.2, we attempt to control imbalance in $\delta\mu$ by controlling the maximal imbalance for candidates in a model $\mathcal{M}$. We now discuss possible choices for this model and connections to balance criteria in the literature. We begin with the case where we have p binary covariates $X_{i1}, \dots, X_{ip} \in \{0, 1\}$, for example indicating employment status or high school education. One option is to balance covariates marginally, which is equivalent to balancing a linear model without interactions. This approach is implicit in typical balance checks in observational studies that compare the fraction of units in the treated group with each attribute to the corresponding fraction in the whole sample. In the linear model that puts no constraints on the coefficients β , controlling maximal imbalance requires exact marginal covariate balance.

$$\text{imbalance}_{\mathcal{M}}(\gamma) = \begin{cases} 0 & \text{if } \frac{1}{n} \sum_{i=1}^n X_{ij} - \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) X_{ij} = 0 \text{ for all } j \in 1 \dots p \\ +\infty & \text{otherwise} \end{cases} \quad (20)$$

for $\mathcal{M} = \{m(x) = \beta \cdot x : \beta \in \mathbb{R}^p\}$.

In a linear model that constrains the ℓ_1 norm of the coefficients, maximal imbalance is the largest maximal imbalance in any one covariate.

$$\text{imbalance}_{\mathcal{M}}(\gamma) = \max_{j \in 1 \dots p} \left| \frac{1}{n} \sum_{i=1}^n X_{ij} - \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) X_{ij} \right| \quad (21)$$

for $\mathcal{M} = \{m(x) = \beta \cdot x : \|\beta\|_1 \leq 1\}$

Finding weights to balance marginal proportions has a long history in survey statistics, where it is known as *raking* (e.g., Deming & Stephan, 1940; Deville & Särndal, 1992). In causal inference such balance measures are commonly used to assess covariate balance across treatment groups (e.g., Cochran, 1965; Rosenbaum & Rubin, 1985; Rosenbaum et al., 2007). Generally, if the conditional mean outcome μ_1 is a linear function $x \cdot \beta$ of the covariates, its bias conditional on the design $\mathcal{D} = \{(W_1, X_1) \dots (W_n, X_n)\}$ is

$$\beta \cdot \left(\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) X_i \right). \quad (22)$$

Thus, weights that exactly balance each covariate marginally will produce an unbiased estimator if $\mu_1(x) = x \cdot \beta$ for any coefficients β . If these coefficients satisfy an ℓ_1 norm constraint $\|\beta\|_{\ell_1} \leq B$, Hölder's inequality implies this conditional bias is bounded by B times $\text{imbalance}_{\mathcal{M}}(\gamma)$ from Equation 21. And more generally, if they satisfy an ℓ_q norm bound $\|\beta\|_{\ell_q} \leq B$ on the coefficients, Hölder's inequality implies that this bias is bounded by the $\ell_{q'}$ norm of the vector of all marginal imbalances where q' is the Hölder conjugate of q . Henceforth, we will focus on models that constrain the ℓ_2 norm of the coefficients, which corresponds to a maximal imbalance that is the square root of the sum of all squared marginal imbalances.

However, unless μ_1 is linear, balancing the margins alone is insufficient to remove bias. To see this, consider the decomposition of μ_1 into its best linear approximation $\tilde{\mu}_1(x) := \tilde{\beta} \cdot x$ and the residuals $r(x) := \mu_1(x) - \tilde{\mu}_1(x)$. Balancing margins leaves additional *approximation error* based on the difference between the approximation $\tilde{\mu}_1$ and μ_1 itself. Generally, the design-conditional bias is

$$\underbrace{\tilde{\beta} \cdot \left(\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) X_i \right)}_{\text{marginal imbalance}} + \underbrace{\left(\frac{1}{n} \sum_{i=1}^n r(X_i) - \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) r(X_i) \right)}_{\text{imbalance in the approximation error}}. \quad (23)$$

We can reduce approximation error by balancing a *basis expansion* $\phi(x)$ that includes interactions; however, this leaves us with two competing modeling goals. We would like to balance an expressive enough model to give a good approximation of $\mu_1(x)$ and also like to ensure that we are able to get reasonable balance for that model.

5.1.2 Models with strong interactions

Balancing models with interactions is the best course of action. For example, we might believe that the implications of being both employed *and* a high school graduate is more than the sum of its parts, so we would want to balance the interaction between the two. More formally, for our vector of p binary covariates $X_i = (X_{i1}, \dots, X_{ip})$ and multi-index vector $j = (j_1, \dots, j_p) \in \{0, 1\}^p$, let X_i^j be the covariate interaction $X_{i1}^{j_1} \cdot \dots \cdot X_{ip}^{j_p}$, so the interaction of being employed and a high school graduate corresponds to an index vector j with ones in those two coordinates, say with $j_1 = j_2 = 1$, and zeros elsewhere. In this notation, letting sums over j range over $\{0, 1\}^p$ implicitly, when $\mu_1(x) = \sum_j \beta_j x^j$ design-conditional bias is

$$\sum_j \beta_j \left(\frac{1}{n} \sum_{i=1}^n X_i^j - \frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) X_i^j \right). \quad (24)$$

What must be done to control this bias is analogous to what we saw in our discussion of imbalance measures for linear-in-covariates models (22). To control bias without assuming any constraint on the coefficients β , we must have exact balance in each interaction X_i^j , and Hölder's inequality implies bounds on this bias for coefficients satisfying ℓ_q norm bounds. For example,

$$\begin{aligned} \text{imbalance}_{\mathcal{M}}^2(\gamma) &= \sum_j \left| \frac{1}{n} \sum_{i=1}^n X_i^j - \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) X_i^j \right|^2 \\ \text{for } \mathcal{M} &= \{m(x) = \sum_j \beta_j x^j : \sum_j \beta_j^2 \leq 1\}. \end{aligned} \quad (25)$$

For low-dimensional covariates, this type of balance measure is commonly used in sample surveys, with exact balance of fully-interacted models being called *post-stratification*.

5.1.3 Models with weak interactions

The challenge with balancing higher-order interactions is that the set of all interactions to balance grows exponentially in the number of covariates, making it infeasible to balance everything well. Thus, the curse of dimensionality often prevents us from exactly balancing all higher-order interactions. Even if we allow for approximate balance, the imbalance measure in Equation 25 gives the same importance to the marginal imbalance in a covariate and the imbalance in the p^{th} order interaction $X_{i1} \cdot X_{i2} \cdot \dots \cdot X_{ip}$. This is appropriate if we think the effects of high-order interactions may be as large as the main effects, but models in which effects tend to be smaller for higher order interactions tend to be more plausible.

To encode the assumption that main effects are larger than the effects of first-order interactions, which are larger than those for second-order interactions, and so on, we can control the maximal imbalance over a model in which the coefficients are in an ellipse that narrows for higher-order interactions, so the coefficient on X_i^j must be less than a corresponding radius $\lambda_j^{-1/2}$ that decreases as a function of the interaction order $\sum_{k=1}^p j_k$. This measure of imbalance is used, for example, in Ben-Michael et al. (2021, Equation 9).

$$\begin{aligned} \text{imbalance}_{\mathcal{M}}^2(\gamma) &= \sum_j \lambda_j \left| \frac{1}{n} \sum_{i=1}^n X_i^j - \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) X_i^j \right|^2 \\ \text{for } \mathcal{M} &= \left\{ m(x) = \sum_j \beta_j x^j : \sum_j \beta_j^2 / \lambda_j \leq 1 \right\}. \end{aligned} \quad (26)$$

Taking $\lambda_j = 0$ corresponds to assuming away the presence of an effect of x^j , so we do not penalize imbalance in the corresponding interaction, whereas taking $\lambda_j = \infty$ corresponds to assuming no limits on

the effect, so we must balance the corresponding interaction perfectly. In Section 5.1.1 above, we discussed the case in which λ_j is nonzero only for main effects.

We can get a qualitatively similar model by restricting our coefficients to any shape that is more elongated in some directions than others, e.g. a rectangle, which corresponds to a bound on $\|\lambda^{-1/2}\beta\|_{\ell_q}$ for $q = \infty$. The ellipse is a common choice because, as with the use of ℓ_2 norms in other contexts such as least squares vs. least absolute deviations, it simplifies computation and analysis.

5.1.4 Nonbinary discrete and coarsened continuous covariates

The discussion above can be applied to discrete-valued but non-binary covariates using the common practice of *one-hot encoding*, i.e., expanding a k -valued covariate X_{ij} into a vector of $k - 1$ binary covariates $1(X_{ij} = 2) \dots 1(X_{ij} = k)$. However, if these discrete covariate are coarsened versions of continuous covariates, we may be missing an important source of bias. Our bias characterization (24) is based on the assumption that μ_1 is a function of the coarsened covariates rather than the underlying continuous-valued ones. For this reason, it can be better to work directly with models for functions of continuous-valued covariates.

5.2 Balancing continuous covariates

5.2.1 The basis expansion perspective

The key change when we work with continuous covariates is that interactions are not the only source of complexity: functions of even one continuous covariate can be complex and nonlinear. In essence, we face the curse of dimensionality even in one dimension, as non-parametric models require an *infinite* basis. For example, the natural generalization of the previous model (26) is a Taylor series model, in which we sum over all powers of each covariate ($j \in \mathbb{N}^p$), instead of the binary ones ($j \in \{0, 1\}^p$). In this model, we can think of higher-order terms as being the ones for which the vector j is large, e.g. both $x_1 \dots x_4$ and x_1^4 , whether they are interactions or powers of one covariate. More generally, we can use a similar model specified in terms of any set of basis functions $\phi_1(x), \phi_2(x), \dots$ and corresponding radii $\lambda_1^{-1/2}, \lambda_2^{-1/2}, \dots$

$$\text{imbalance}_{\mathcal{M}}^2(\gamma) = \sum_j \lambda_j \left| \frac{1}{n} \sum_{i=1}^n \phi_j(X_i) - \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) \phi_j(X_i) \right|^2 \quad \text{for} \quad (27)$$

$$\mathcal{M} = \left\{ m(x) = \sum_j \beta_j \phi_j(x) : \sum_j \beta_j^2 / \lambda_j \leq 1 \right\}.$$

To better isolate first order effects from higher order ones in a sequential manner, it is common to use a basis of *orthogonal polynomials* instead of the Taylor series-inspired basis above; for example, the Hermite polynomials $1, x_1, x_2, x_1^2 - 1, x_2^2 - 1, x_1 x_2$, written here for two covariates. This isolation can make specifying λ_j a bit more intuitive. Another option, motivated by Fourier series, is to use an orthogonal basis of cosines of different frequencies. We cannot balance cosines of all frequencies accurately just like we cannot balance all polynomials, so we use models that restrict effects associated with high frequency terms.

As before, taking $\lambda_j = 0$ corresponds to assuming away any effect for $\phi_j(x)$, so $\text{imbalance}_{\mathcal{M}}(\gamma)$ is unaffected by imbalance in ϕ_j , and taking $\lambda_j = \infty$ corresponds to assuming no limits on that effect and therefore requires exact balance in ϕ_j .

5.2.2 The function space perspective

While the basis expansion perspective can be useful when thinking about the balancing problem, specifying an infinite-dimensional model $\mathcal{M}$ directly in terms of basis functions and scale factors (ϕ_j, λ_j) is hard to do well. We can also view things from another perspective, focusing on the properties of the functions rather than their basis expansion. The theory of reproducing kernel Hilbert spaces (RKHSes) provides one approach for doing so. Subject to some constraints, the model $\mathcal{M}$ in Equation 27 is the unit ball $\{m(x) : \|m\|_{\mathcal{M}} \leq 1\}$

of an RKHS, and furthermore, the unit ball of every RKHS can be described like this (see, e.g., Cucker & Zhou, 2007, Chapters 2 and 4).⁵

One of the most common ways to characterize functions is their smoothness. For this, it is natural to work with Sobolev spaces: RKHSes defined in terms of averages of a function's derivatives. We can characterize the unit ball of a *Sobolev norm* in terms of the coefficients of Fourier series expansions. For example, these two Sobolev norms for functions on the unit square $[0, 1]^2$, which we can characterize equivalently in terms of the coefficients β_j in the function's cosine basis expansion $m(x) = \sum_{j \in \mathbb{N}^2} \beta_j \cos(\pi j_1 x_1) \cos(\pi j_2 x_2)$.

$$\|m\|_{\text{iso}}^2 = \int_{[0,1]^2} |m(x_1, x_2)|^2 + \left| \frac{\partial}{\partial x_1} m(x_1, x_2) \right|^2 + \left| \frac{\partial}{\partial x_2} m(x_1, x_2) \right|^2 \quad (28)$$

$$= \sum_{j \in \mathbb{N}^2} \beta_j^2 / \lambda_j \text{ with } 1/\lambda_j = 1 + \pi^2 j_1^2 + \pi^2 j_2^2. \quad (29)$$

$$\|m\|_{\text{dom}}^2 = \|m\|_{\text{iso}}^2 + \int_{[0,1]^2} \left| \frac{\partial^2}{\partial x_1 \partial x_2} m(x_1, x_2) \right|^2 \quad (30)$$

$$= \sum_{j \in \mathbb{N}^2} \beta_j^2 / \lambda_j \text{ with } 1/\lambda_j = 1 + \pi^2 j_1^2 + \pi^2 j_2^2 + \pi^4 j_1^2 j_2^2 \quad (31)$$

In these models, our assumptions are encoded in the derivatives included in the norm. If we expect μ_1 to be relatively smooth, including higher order derivatives would encode this assumption. If we believe that most of the complexity lies along the coordinate axes, i.e., that a function tends to vary more with one covariate than with a mixture of a few, including higher order mixed partial derivatives, as we do in $\|m\|_{\text{dom}}$, encodes this assumption. Additive models take this idea to an extreme, entirely forbidding variation that is not simply a sum of what we see along the axes.⁶

So far, we have focused on functions defined on $\mathbb{R}^2$ for notational simplicity. However, these concepts naturally extend to higher-dimensional settings. The subject of nonparametric models is vast; see Appendix C for further discussion.

Remark 2 (Invariance and equivariance). The average treatment effect $\tau = \psi_1 - \psi_0$ is translation invariant; it is not affected by the way the outcomes are centered. It aids our intuition to use estimators $\hat{\psi}_1$ and $\hat{\psi}_0$ with a corresponding *equivariance property*: if our estimate $\hat{\psi}_w$ is a when $\mu_w(x) = m(x)$, then it should be $a + b$ when $\mu_w(x) = m(x) + b$. The IPW and AIPW estimators have this property when the weights $\hat{\gamma}$ average to one, and it is common practice to ensure translation invariance by normalizing them to average to one, using $\hat{\gamma}(X_i) = \hat{e}(X_i)^{-1} / \{n^{-1} \sum_{i=1}^n W_i \hat{e}(X_i)^{-1}\}$ instead of $\hat{\gamma}(X_i) = \hat{e}(X_i)^{-1}$. This equivariance property arises naturally from our balance criterion when our model includes all translations of every function in it, i.e., if $m(\cdot) + t \in \mathcal{M}$ whenever $m \in \mathcal{M}$. In models like this, the imbalance is infinite if we do not have equivariance.

$$\text{imbalance}_{\mathcal{M}}(\gamma) = \begin{cases} \text{imbalance}_{\mathcal{M}_0}(\gamma) & \text{if } \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) = 1 \\ \infty & \text{if } \frac{1}{n} \sum_{i=1}^n W_i \gamma(X_i) \neq 1 \end{cases} \quad \text{for}$$

$$\mathcal{M} = \{m(x) + t : m \in \mathcal{M}_0, t \in \mathbb{R}\}.$$

We can add these translations to the Sobolev models described above by simply dropping the term without derivatives from the corresponding norm, e.g. $\mathcal{M} = \{m : \rho_{\text{iso}}(m) \leq 1\}$ for

$$\rho_{\text{iso}}(m)^2 = \int_{[0,1]^2} \left| \frac{\partial}{\partial x_1} m(x_1, x_2) \right|^2 + \left| \frac{\partial}{\partial x_2} m(x_1, x_2) \right|^2 \quad (32)$$

$$= \sum_{j \in \mathbb{N}^2} \beta_j^2 / \lambda_j \text{ with } 1/\lambda_j = \pi^2 j_1^2 + \pi^2 j_2^2. \quad (33)$$

⁵Specifically, these constraints are that the basis ϕ_j is orthonormal with respect to some probability measure and the decay factors λ_j have a finite sum.

⁶We get a series expansion of the set of functions of the form $\sum_k m_k(x_k)$ with $\|m\|_{\text{iso}} \leq 1$ by making $1/\lambda_j = \infty$ for terms involving mixed partials rather than simply increasing it as we do for $\|m\|_{\text{dom}}$.

5.3 Other considerations

Constraining the weights to be non-negative and average to one disallows *extrapolation* outside of the convex hull of the data (Hirshberg, 2026). Intuitively, using weights that interpolate the data limits the extent of biases due to misspecification of the model $\mathcal{M}$ relative to what is possible with extrapolation. In particular, this ensures that the estimate is sample bounded, i.e. that it lies between the minimum and maximum observed values of the outcome. This rules out nonsensical estimates when outcomes are known to be bounded, such as for binary and survival outcomes.

Such constraints can be enforced by choosing a weight complexity penalty χ in Equation 13 that blows up outside the allowed range. A quadratic penalty $\chi(\gamma) = \gamma^2$ on the domain $[0, \infty)$ enforces non-negativity. The negative entropy penalty $\chi(\gamma) = \gamma \log \gamma - \gamma$ enforces $\gamma \geq 0$, and $\chi(\gamma) = \gamma \log \gamma - \gamma + 1$ enforces $\gamma \geq 1$.

However, because balancing weights for rich models converge to the inverse propensity weights, and they exceed one, we tend to get negative weights when we are balancing a small—and likely to be misspecified—model. When this is the case, using a richer model can be a better fix than a non-negativity constraint or, at least, a good complement.

The parameter σ^2 in the minimax weights (12) is often described as controlling a bias-variance trade-off: a small σ^2 emphasizes balance and therefore control of bias, and a large σ^2 emphasizes control of the magnitude of the weights and therefore of variance. The minimax interpretation of the weights discussed in Section 3.2 suggests choosing σ^2 to be roughly the conditional variance of the outcomes, $\text{var}[Y_i | X_i, W_i]$. This choice is supported by asymptotic theory (e.g., Hirshberg & Wager, 2021; Hirshberg et al., 2019), which suggests that taking σ^2 to be constant order is a relatively robust choice.

But σ^2 is fundamentally a signal-to-noise ratio, comparing imbalance in model-scale units to outcome noise. The minimax heuristic assumes a particular model scale; changing the units of $\mathcal{M}$ changes what σ^2 should be. Sometimes we have intuition about model scale. For example, in a medical study, a domain expert may say treatment groups are comparable in age if they differ by no more than a few months—that pins down the scale of the age component of $\mathcal{M}$.⁷

For rich models, the effect of σ^2 is limited anyway: weights converge to the inverse propensity weights regardless (e.g., Equation 34). Extreme choices matter in small samples, but there is a range of reasonable values. How to select this parameter automatically is an open question. Some approaches have been proposed (Kallus, 2020a; Wang & Zubizarreta, 2020; Zhao, 2019) but theoretical and empirical evidence supporting them is currently limited.

6 Degree of balance

The discussion above focuses on balancing increasingly large, expressive models. Why would we choose a smaller model over a larger one when the latter is more likely to be correctly specified? The answer lies in a fundamental trade-off: while larger models can capture more complex relationships, we cannot balance them as well, and negligible imbalance in the regression error (in the sense described in Section 4) is crucial for inference.⁸ In this section, we will focus on understanding when this happens.

6.1 Comparing to the true inverse propensity weights

We first focus on the unaugmented estimator, so $\delta m = \mu_1$, and assume our model is valid in the sense that $\mu_1 \in \mathcal{M}$; thus, imbalance in μ_1 is bounded by $\text{imbalance}_{\mathcal{M}}(\hat{\gamma})$. We can bound this by comparing it to $\text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}})$, the maximal imbalance achieved by the true inverse propensity weights. For example, because the minimax weights $\hat{\gamma}$ minimize the objective function in Equation 12,

$$\text{imbalance}_{\mathcal{M}}^2(\hat{\gamma}) + \frac{\sigma^2}{n^2} \sum_{W_i=1} \hat{\gamma}(X_i)^2 \leq \text{imbalance}_{\mathcal{M}}^2(\gamma^{\text{ipw}}) + \frac{\sigma^2}{n^2} \sum_{W_i=1} \gamma^{\text{ipw}}(X_i)^2. \quad (34)$$

⁷It can be more intuitive to minimize the magnitude of the weights subject to a constraint on imbalance, which is equivalent. There exists some data-dependent choice σ^2 that yields a solution of the constrained problem for each bound on the imbalance and vice versa (Boyd & Vandenberghe, 2004, Chapter 5).

⁸However, see Armstrong and Kolesár (2021) for a discussion of bias-aware confidence intervals.

As Hirshberg and Wager (2021) discuss, because the estimated weights $\hat{\gamma}$ converge to the true inverse propensity weights γ^{ipw} , the difference between the variance terms is asymptotically negligible. So up to a negligible term, $\text{imbalance}_{\mathcal{M}}(\hat{\gamma}) \leq \text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}})$, and we can focus on characterizing the latter.

Because the inverse propensity weights have the population balance property (4), $\text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}})$ is the maximum, over all $m \in \mathcal{M}$, of an average of iid mean-zero terms. Consider the model $\mathcal{M} = \{\beta \cdot \phi(x) : \|\beta\|_1 \leq 1\}$ for a basis expansion $\phi(x) \in \mathbb{R}^p$. The maximal imbalance is the imbalance in the least-balanced basis function:

$$\text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}}) = \max_{j=1, \dots, p} \left| \frac{1}{n} \sum_{i=1}^n \phi_j(X_i) - W_i \gamma^{\text{ipw}}(X_i) \phi_j(X_i) \right|. \quad (35)$$

By the central limit theorem, each individual imbalance is approximately normal with standard deviation of order $n^{-1/2}$. So the maximal imbalance behaves like the maximum of p mean-zero normals, which is $O_p(\sqrt{\log(p)/n})$ —depending only weakly on the number of basis functions. Athey et al. (2018) use this approach to analyze balancing weights for high dimensional linear models.

We can extend this to infinite-dimensional models. The maximum of p normals grows like $\sqrt{\log p}$ only when they are sufficiently uncorrelated. If the normals become increasingly correlated as p grows—recycling variation we’ve already seen—the growth stops. Models $\mathcal{M}$ that are “small enough” for this to happen, i.e. models for which $\text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}}) = O_p(n^{-1/2})$, are called Donsker classes in empirical process theory.⁹ One example, discussed in Kallus (2020a) and Wong and Chan (2018), is the unit ball of a Sobolev space with $s > p/2$ bounded derivatives.

Symmetrization and the connection to regression. The same measures of model complexity that govern the difficulty of nonparametric regression also govern balance. The intuition above—maximal imbalance as a maximum of correlated normals—can be made precise. Rather than reasoning asymptotically about what the individual imbalances converge to, we can work with a symmetrized version that *conditionally* has the structure we want. Letting $\xi_1, \dots, \xi_n$ be independent and identically distributed standard normals, independent of the data, we define the *symmetrized imbalance*

$$\text{imbalance}_{\mathcal{M}}^{\xi}(\gamma^{\text{ipw}}) = \max_{m \in \mathcal{M}} \left| \frac{1}{n} \sum_{i=1}^n \xi_i \{m(X_i) - W_i \gamma^{\text{ipw}}(X_i) m(X_i)\} \right|. \quad (36)$$

Conditional on the data, this is normally distributed. And it’s a good approximation of imbalance itself:¹⁰

$$\frac{1}{4\sqrt{\log(n)}} \leq \frac{\mathbb{E} \text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}})}{\mathbb{E} \text{imbalance}_{\mathcal{M}}^{\xi}(\gamma^{\text{ipw}})} \leq \sqrt{2\pi}. \quad (37)$$

The symmetrized imbalance measures how well functions in our model can correlate with noise. This is exactly the quantity that governs the difficulty of nonparametric regression. In least squares, the estimation error of $\hat{\mu}$ depends on how large the noise correlation $\frac{1}{n} \sum_{i=1}^n \varepsilon_i \{\mu(X_i) - m(X_i)\}$ can be relative to the squared error $\|\mu - m\|_{L_2}^2$. To see this, write the loss difference between an arbitrary $m \in \mathcal{M}$ and the true regression function μ :

$$\begin{aligned} \ell(m) &= \frac{1}{n} \sum_{i=1}^n \{Y_i - m(X_i)\}^2 - \frac{1}{n} \sum_{i=1}^n \{Y_i - \mu(X_i)\}^2 \\ &= \|\mu - m\|_{L_2(P_n)}^2 - \frac{2}{n} \sum_{i=1}^n \varepsilon_i \{\mu(X_i) - m(X_i)\} \end{aligned} \quad (38)$$

⁹To be more precise, $\text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}}) = O_p(n^{-1/2})$ when $\mathcal{M}^{\text{ipw}} = \{(1 - W\gamma^{\text{ipw}})m : m \in \mathcal{M}\}$ is Donsker. If the *strict overlap* condition that γ^{ipw} is bounded holds, this is true if $\mathcal{M}$ itself is Donsker, but it can be stronger otherwise. See Hirshberg and Wager (2021, Remark 4).

¹⁰With Gaussian multipliers ξ_i , the expected value of maximal imbalance is at most $\sqrt{2\pi} \approx 2.5$ times larger than its symmetrized counterpart’s and at least $1/(4\sqrt{\log(n)})$ times as large—almost equivalent up to constant factors. The expectation of this symmetrized imbalance is called the Gaussian complexity of the inverse probability-weighted model $\mathcal{M}^{\text{ipw}}$. Using random signs, i.e. independent ξ_i taking on the values ± 1 each with probability 1/2, we can get true equivalence—the ratio of these expectations is between 1/2 and 2. With these multipliers, the expectation is called the Rademacher complexity of $\mathcal{M}^{\text{ipw}}$. See Bartlett and Mendelson (2002) for details.

where $\varepsilon_i = Y_i - \mu(X_i)$ is the regression error. The least squares estimator $\hat{\mu}$ minimizes the loss, so $\ell(\hat{\mu}) \leq 0$. This means the squared error $\|\mu - \hat{\mu}\|^2$ can be at most twice the noise correlation term. Turning this around: if we can show that the noise correlation is small for all m at distance r from μ , we can rule out $\|\hat{\mu} - \mu\| = r$. Concretely,

$$\|\hat{\mu} - \mu\| \neq r \quad \text{if} \quad \frac{r^2}{2} \geq \max_{\substack{m \in \mathcal{M} \\ \|\mu - m\|_{L_2(\mathbb{P}_n)} = r}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \{\mu(X_i) - m(X_i)\}. \quad (39)$$

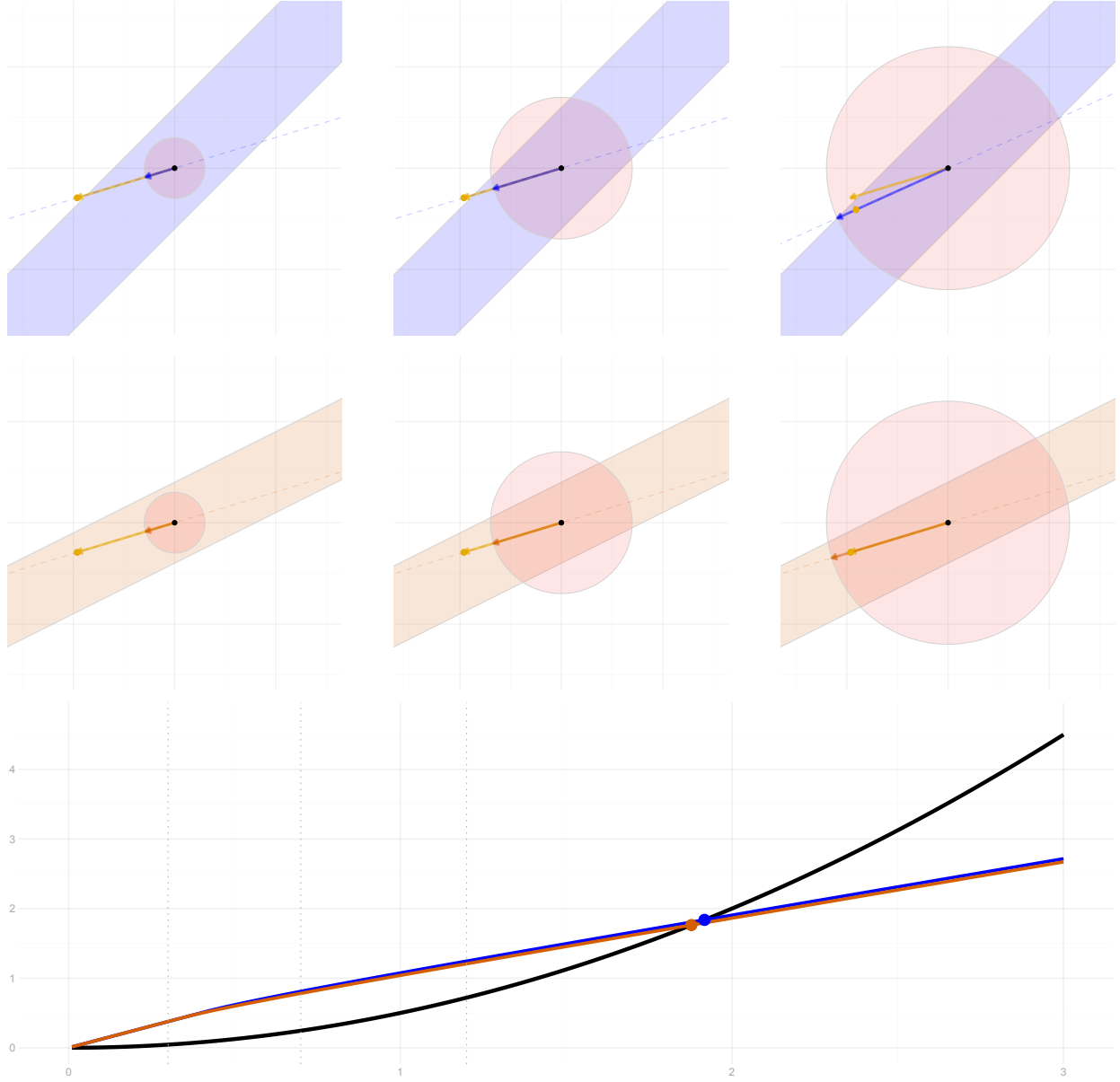


Figure 1: Geometry of the fixed-point argument for two models: the regression model $\mathcal{M}$ (blue, top row) and the IPW-reweighted model $\mathcal{M}^{\text{ipw}}$ (vermillion, middle row). Both rows show the same noise draw; the models differ only in the tube's slope and width, reflecting the coordinate-wise rescaling by propensity score factors. As the ball shrinks (right to left), the maximal noise correlation shrinks too—this is equicontinuity. But in most models—linear ones are an exception—it shrinks at a rate slower than the ball's radius does. What happens is that, as our ball gets smaller, it does more and more of the work in constraining $m - \mu$. In the extreme case shown on the left, where the ball is contained in the model, the model is doing nothing at all. The bottom panel plots the expected maximal correlation (Gaussian width) as a function of the ball's radius r against the curve $r^2/2$; the *fixed point* where they cross bounds the regression error $\|\hat{\mu} - \mu\|$ or the analogous balancing error. The narrower IPW model crosses earlier, reflecting the additional structure imposed by reweighting.

The right-hand side is exactly the same “correlation with noise” structure—just for functions $\mu - m$ in the model $\mathcal{M}$ rather than functions $(1 - W\gamma^{\text{ipw}})m$ in the IPW-reweighted model $\mathcal{M}^{\text{ipw}} = \{(1 - W\gamma^{\text{ipw}})m : m \in \mathcal{M}\}$ that appear in the same role in our symmetrized imbalance (36). Models that are easy to estimate by least squares are easy to balance. And the notions of model size used in the context of least squares—Rademacher complexity, entropy integrals, effective dimension, the Donsker property—are the right notions for talking about balance because they are approximations to or summaries of the maximal correlation (36).

Why $O_p(n^{-1/2})$ is not enough. Unfortunately, even for Donsker models, $O_p(n^{-1/2})$ imbalance is not sufficient for valid inference—we need $o_p(n^{-1/2})$. To see this, consider the extreme case of a model containing a single function: $\mathcal{M} = \{\mu_1\}$. Then

$$\text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}}) = \left| \frac{1}{n} \sum_{i=1}^n \{1 - W_i \gamma^{\text{ipw}}(X_i)\} \mu_1(X_i) \right|. \quad (40)$$

By the central limit theorem, $\sqrt{n} \text{imbalance}_{\mathcal{M}}(\gamma^{\text{ipw}})$ converges not to zero but to the absolute value of a normal with variance $\text{Var}[\{1 - W_i \gamma^{\text{ipw}}(X_i)\} \mu_1(X_i)]$.¹¹ The imbalance is $O_p(n^{-1/2})$, not $o_p(n^{-1/2})$, even when the model contains only the truth.

We see the same phenomenon in least squares. The global complexity bound

$$\|\hat{\mu} - \mu\| \neq r \quad \text{if} \quad \frac{r^2}{2} \geq \max_{m \in \mathcal{M}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \{\mu(X_i) - m(X_i)\} \quad (41)$$

gives $n^{-1/4}$ rates at best—even for a model containing a single function. The problem is that we are bounding the noise correlation over all of $\mathcal{M}$, not just over the small functions near μ that we actually care about. To do better, we need *local* complexity measures like the one in (39), which restrict the maximum to functions at distance r from the truth. The decay in the maximum as r shrinks is called the *equicontinuity* of the empirical process.

And this is exactly what we need for the augmented estimator. Because we care about imbalance in $\delta\mu = \hat{\mu}_1 - \mu_1$, which shrinks as our regression estimate improves, equicontinuity—the fact that we balance small functions better than big ones—is what converts $O_p(n^{-1/2})$ imbalance over the full model into $o_p(n^{-1/2})$ imbalance in the regression error.

6.2 The role of augmentation

We can do better with this approach when analyzing the augmented estimator we discuss in Section 2.4. The key idea is that rather than analyzing balance in a *fixed* model, by augmenting the estimator we can restrict our attention to a *shrinking* sequence of models.

For intuition, we will again consider a model containing a single element. This time, however, we will use the model $\mathcal{M}_n = \{\delta m_n\}$ that contains the regression error $\delta\mu_n = \hat{\mu}_1 - \mu_1$ instead of μ_1 itself. What has changed is that, assuming we use a good estimator $\hat{\mu}_1$, the element $\delta\mu_n$ in this set will shrink as we obtain more observations. It is convenient to think of $\hat{\mu}_1$ as cross-fit, i.e., estimated on a sample other than $\{(W_i, X_i, Y_i) : i \leq n\}$. This allows us to think of $\delta\mu$ as *non-random* in the sense that it does not depend on this sample; formally, we do most of our analysis conditional on $\delta\mu$.¹² The maximal imbalance is

$$\text{imbalance}_{\mathcal{M}_n}(\gamma^{\text{ipw}}) = \left| \frac{1}{n} \sum_{i=1}^n W_i \gamma^{\text{ipw}}(X_i) \delta m_n(X_i) - \delta m_n(X_i) \right| \quad \text{for} \quad \mathcal{M}_n = \{\delta m_n\}.$$

¹¹The IPW estimator with the true weights γ^{ipw} is asymptotically unbiased and normal, because the imbalance term is perfectly centered: we get additional variance, but not bias. However, this delicate centering property is not preserved by estimated weights, even ones close to γ^{ipw} in mean squared error.

¹²This is not necessary for all arguments, e.g., it is not required by those of Hirshberg and Wager (2021). However, it is increasingly common practice, has been shown to be help in some simulation studies, and can be done without loss of efficiency asymptotically by averaging estimators that fit $\hat{\mu}_1$ on different subsamples (see, e.g. Chernozhukov, Chetverikov, et al., 2018; Schick, 1986; Zheng & van der Laan, 2011).

By the same argument we used for $\mathcal{M} = \{\mu_1\}$, if $\text{var}[\{1 - W_i\gamma^{\text{ipw}}(X_i)\}\delta m_n(X)]$ converges to zero, then $\sqrt{n}\text{imbalance}_{\mathcal{M}_n}(\gamma^{\text{ipw}})$ will converge to zero in probability and the estimator will achieve asymptotically negligible imbalance. Given mild assumptions, this quantity converges to zero if $\hat{\mu}_1$ is a consistent estimator.¹³

To view this phenomenon more generally, suppose we have a valid model $\mathcal{M}_n$ with many elements, all of them shrinking like $\delta\mu_n$ in the previous example. In particular, suppose that we have an ‘a priori model’ $\mathcal{M}_0$ specifying sample-size-independent characteristics we expect of $\delta\mu$, e.g. smoothness or approximate sparsity, and that we know the rate of convergence of our estimator $\hat{\mu}_1$, so at any given sample size we are able to rule out the possibility that our regression error $\|\delta\mu_n\|_{L_2(P)}$ exceeds some bound r_n that shrinks to zero.¹⁴ Incorporating this knowledge leaves us with the smaller model $\mathcal{M}_n = \{m \in \mathcal{M}_0 : \|m\|_{L_2(P)} \leq r_n\}$. It is reasonable to ask whether this is enough that the maximal imbalance would still satisfy $\sqrt{n}\text{imbalance}_{\mathcal{M}_n}(\gamma^{\text{ipw}}) \rightarrow 0$, which would imply negligible imbalance in $\delta\mu_n$ if our assumptions were valid in the sense that $\delta\mu_n \in \mathcal{M}_0$ and $\|\delta\mu_n\|_{L_2(P)} \leq r_n$ with probability tending to one as n grows. The answer depends on the size the a-priori model $\mathcal{M}_0$, but essentially a sufficient condition on $\mathcal{M}_0$ is that $\text{imbalance}_{\mathcal{M}_0}(\gamma^{\text{ipw}}) = O_p(n^{-1/2})$.¹⁵

From this, it would appear that we need to know a bound on the regression error $\|\delta\mu_n\|_{L_2(P)}$ so that we can use the model $\mathcal{M}_n$ to solve for our balancing weights. However, Hirshberg and Wager (2021, see Section 2) show that this is unnecessary, as it works well enough to simply use the model $\mathcal{M}_0$ instead. In particular, they show that the minimax weights (12) for an a-priori model $\mathcal{M}_0$ of appropriate size achieve negligible imbalance over any sequence of submodels $\mathcal{M}_n = \{\delta\mu \in \mathcal{M}_0 : \|\delta\mu\|_{L_2(P)} \leq r_n\}$ satisfying regression error bounds r_n tending to zero. Because this will include one for which $r_n \geq \|\delta\mu_n\|_{L_2(P)}$ with probability tending to one if one exists (i.e., if $\hat{\mu}_1$ is consistent), in practice all we need is an a-priori model $\mathcal{M}_0$ that is valid in the sense that $\delta\mu \in \mathcal{M}_0$ with probability tending to one. Concretely, Hirshberg and Wager (2021, Theorem 1) show that when we use minimax balancing weights for a Donsker a-priori model $\mathcal{M}_0$ and augment with a consistent estimator $\hat{\mu}_1$, the AIPW estimator is asymptotically linear with the efficient influence function—requiring only overlap and that $\delta\mu \in \mathcal{M}_0$, with no assumptions on the propensity score.

This is an instance of the equicontinuity phenomenon we discussed above: imbalance over a shrinking sequence of models inherits the $o_p(n^{-1/2})$ rate from shrinking variance, even when the maximal imbalance over the fixed a-priori model is only $O_p(n^{-1/2})$.

One subtlety is that using a model $\mathcal{M}'$ for the outcome μ_1 does not guarantee that $\delta\mu = \mu_1 - \hat{\mu}_1$ is in $\mathcal{M}'$. For example, if $\mathcal{M}'$ consists of smooth functions and $\hat{\mu}_1$ is a regression tree, the difference $\delta\mu$ may not be smooth even when μ_1 is. This mismatch can cause augmentation to fail; see Section 3.2.3 of the arXiv v1 version of Hirshberg and Wager (2021) for a simulation. A simple fix is to take $\mathcal{M}_0$ to be the convex hull of $\mathcal{M}' - \hat{\mu}_1 := \{m - \hat{\mu}_1 : m \in \mathcal{M}'\}$, which contains $\delta\mu$ whenever $\mu_1 \in \mathcal{M}'$ (Hirshberg & Wager, 2021, Section 2.6).

6.3 Doing Better than the True Weights

The approaches above—plug-in comparison and augmentation—ultimately characterize when the AIPW estimator achieves behavior comparable to what we would get with the true inverse propensity weights γ^{ipw} . But estimated weights can do better. Even when one is using the true weights, as in a randomized experiment, neither IPW nor AIPW has any bias, but the AIPW estimator has smaller variance:

$$\text{var}(\hat{\psi}_1^{\text{aipw}}) = \text{var}\left(\frac{1}{n} \sum_i W_i \gamma^{\text{ipw}}(X_i) \varepsilon_i\right) + \text{E} \left[\left(\frac{1}{n} \sum_i W_i \gamma^{\text{ipw}}(X_i) \delta\mu(X_i) - \frac{1}{n} \sum_i \delta\mu(X_i) \right)^2 \right]. \quad (42)$$

¹³For example, sufficient overlap (Assumption 2') and boundedness of $\|\delta\mu_n\|_\infty$ suffice (Hirshberg & Wager, 2021, see Proof of Theorem 1).

¹⁴In discussion of the role of augmentation within the balancing tradition, it has been common to focus on consistency in the strong sense that $\|\delta\mu\|_{\mathcal{M}} \rightarrow 0$ rather than in the mean-square sense $\|\delta\mu\|_{L_2(P)} \rightarrow 0$ that we use here. While simpler, that approach has some disadvantages. See Appendix D for a discussion.

¹⁵To be precise, it is yes when $\mathcal{M}_0$ and $\{f(w, x) = w\gamma^{\text{ipw}}(x)m(x) : m \in \mathcal{M}_0\}$ are Donsker and the maximal variance of a term for an individual element of $\mathcal{M}_n$, $\max_{\delta\mu \in \mathcal{M}_n} \text{var}[(1 - \gamma^{\text{ipw}}(X_i)W_i)\delta\mu(X)]$, shrinks to zero. Generalizing the one-element case discussed above, sufficient overlap (Assumption 2') and boundedness of $\max_{m \in \mathcal{M}_0} \|m\|_\infty$ suffice for the latter. A slight generalization of this case is addressed in (Hirshberg & Wager, 2021, see Proof of Theorem 1). This phenomenon, the inheritance of convergence of $\sqrt{n}\text{imbalance}_{\mathcal{M}_n}(\gamma^{\text{ipw}})$ from the convergence of the largest variance of a term $(1 - \gamma^{\text{ipw}}(X_i)W_i)\delta\mu(X)$ in the averages we are maximizing over, is called the asymptotic equicontinuity of the empirical process, and is the defining feature of Donsker classes (e.g., A. W. van der Vaart & Wellner, 1996, Section 2.1.2).

The first term is irreducible noise. The second is the mean-squared imbalance in $\delta\mu$. IPW is the special case $\hat{\mu}_1 = 0$, so $\delta\mu = \mu_1$. When $\hat{\mu}_1$ is a good estimator, $\delta\mu$ is closer to zero than μ_1 is, so AIPW has lower variance. This is the cost of imbalance of order $n^{-1/2}$: it inflates variance even when it causes no bias.

There is another path to low variance: estimated weights that achieve smaller imbalance than the true weights would. This may seem paradoxical—how can an estimate beat the truth?—but there is a simple explanation. With a k -dimensional model and $k \leq n$, we can achieve exact balance: it is just linear algebra. The true propensity score does not know it needs to balance our particular basis, so it will typically have imbalance on the order of $n^{-1/2}$.

Chan et al. (2016) formalize this for sieves, showing that calibration weights achieving exact balance on a growing basis attain semiparametric efficiency (see also Fan et al., 2016; Wang & Zubizarreta, 2020). For RKHS models, Hirshberg et al. (2019), inspired by Newey and Robins (2018), show that minimax balancing weights achieve $\text{imbalance}_{\mathcal{M}}(\hat{\gamma}) = o_p(n^{-1/2})$ essentially without assumptions on γ^{ipw} . When the RKHS model is correctly specified, we do not need augmentation for asymptotic efficiency. One does not even need to be explicitly balancing to get this benefit. Hirano et al. (2003) showed that the IPW estimator using an estimated propensity score is more efficient than one using the true propensity score. The intuition is that the estimated propensity score is fit to the sample, so it achieves better balance on observed covariates than the population propensity score would.

The results above establish that balancing weights can beat the true weights, but they do not tell us by how much. There is a long tradition of work on minimax estimation of linear functionals (Donoho, 1994; Donoho & Liu, 1991; Ibragimov & Khas'minskii, 1985) that studies precisely this kind of problem: estimating a linear functional of an unknown function known to lie in a convex class. Its central object is the *modulus of continuity* ω , in terms of which the maximal imbalance of the minimax balancing weights satisfies¹⁶

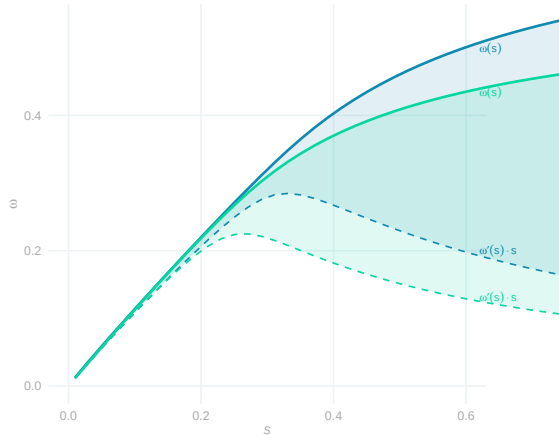


Figure 2: Modulus of continuity $\omega(s)$ (solid) and its linearization $\omega'(s)s$ (dashed) for two models on the same sample ($n = 500$, $d = 1$). The shaded gap is the bias $\omega(s) - \omega'(s)s$. The smoother model (Gaussian RBF, green) has a more linear modulus and smaller bias than the rougher model (Matérn-3/2, blue).

$$\text{imbalance}_{\mathcal{M}}(\hat{\gamma}) = \omega(s) - \omega'(s)s \quad \text{where} \quad \omega(s) = \sup_{m \in \mathcal{M}} \left\{ \frac{1}{n} \sum_{i=1}^n m(X_i) - \frac{1}{2sn} \sum_{i=1}^n m(X_i)^2 \right\}. \quad (43)$$

When ω is approximately linear, the maximal imbalance is approximately zero. Centering ω at γ^{ipw} separates the linear growth from the deviation. Define the *offset complexity*

$$\xi(s) = \sup_{m \in \mathcal{M}} \left\{ \frac{1}{n} \sum_{i=1}^n \{1 - W_i \gamma^{\text{ipw}}(X_i)\} m(X_i) - \frac{1}{2sn} \sum_{i=1}^n m(X_i)^2 \right\}. \quad (44)$$

¹⁶This penalized formulation of ω is equivalent by Lagrangian duality to the constrained formulation used by Donoho (1994) and Armstrong and Kolesár (2018). See Appendix A.

Because $E[\{1 - W_i \gamma^{\text{ipw}}(X_i)\} m(X_i)] = 0$ by the population balance property (4), the integrand is mean-zero: ξ is a centered empirical process penalized by a quadratic. The maximal imbalance depends only on ξ , not on the linear part of ω :

$$\text{imbalance}_{\mathcal{M}}(\hat{\gamma}) = \omega(s) - \omega'(s) s = \xi(s) - \xi'(s) s. \quad (45)$$

When γ^{ipw} is in the model, $\xi = 0$. When it is not, bounding the maximal imbalance reduces to bounding ξ .

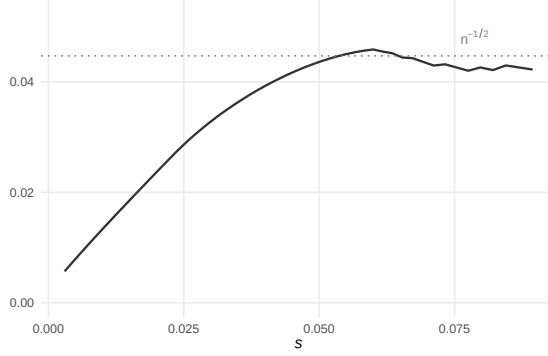


Figure 3: Bias bound $\xi(s) + (\sigma^2/2n) B(1 - sB)$, optimized over the competitor g at each s (Matérn-3/2, $n = 500$, $\sigma = 3$). The dotted line is $n^{-1/2}$. The bound stays below $n^{-1/2}$ for moderate s , locating the operating point in the nearly-linear regime of Figure 2.

Kong (2025) and Hirshberg (2026) bound $\xi(s)$ by two terms. The first is the local complexity of the model at the fixed-point radius r satisfying (39)—the same localization as in Section 6.2, illustrated in Figure 1, and bounded by $C r^2$. The second reflects how well γ^{ipw} can be approximated in the model: for any $g \in \mathcal{M}$ with $\|g\|_{\mathcal{M}} \leq B$ and $\|g - \gamma^{\text{ipw}}\|_{L_2(P)} \leq \varepsilon$, this approximation term is $(\sigma^2/2n) B(1 - sB)$. The bound holds at every s and for any such competitor g ; minimizing over both gives the tightest result. Because the local complexity term grows with s while the approximation term shrinks, the minimum balances the two—and locates the operating point on the modulus curve. For nonparametric models, the approximation term is negligible for any s : it suffices that *some* function in the model approximates γ^{ipw} at finite model norm, which is automatic whenever γ^{ipw} is square-integrable. In this case the binding constraint is just the local complexity $C r^2$, the same quantity that governs regression in Section 6.2. The operating s falls in the range where ω is nearly linear in Figure 2—the shaded gap is small. To match the local complexity rate $C r^2$, the stronger condition $B \leq C r^2 n / \sigma^2$ is needed; this is the approximation condition in Proposition 6.1. When both terms are $O(r^2)$, the total is $C r^2 = o_p(n^{-1/2})$, and the conclusion is stronger than in Section 6.2: there, augmentation with a consistent estimator was needed to convert an $O_p(n^{-1/2})$ bound on maximal imbalance into $o_p(n^{-1/2})$. Here, the minimax balancing weights achieve it without augmentation.

Proposition 6.1 (Maximal imbalance of minimax balancing weights). Let $\hat{\gamma}$ be the minimax balancing weights (12) with tuning parameter σ , and let $r > 0$ satisfy the fixed-point inequality (39). If there exists a function g with $\|g\|_{\mathcal{M}} \leq C r^2 n / \sigma^2$ and $\|g - \gamma^{\text{ipw}}\|_{L_2(P)} \leq C r^2 \sqrt{n} / \sigma$, then $\text{imbalance}_{\mathcal{M}}(\hat{\gamma}) \leq C' r^2$ for universal constants C, C' . If $\mathcal{M}$ is a nonparametric (Definition 1) Donsker class of uniformly bounded functions, such approximations exist for $r \rightarrow 0$, so $\text{imbalance}_{\mathcal{M}}(\hat{\gamma}) = o_p(n^{-1/2})$.¹⁷

This route to bounding imbalance is fundamentally different from the comparison to γ^{ipw} in Section 6.1. That approach—bounding $\text{imbalance}_{\mathcal{M}}(\hat{\gamma})$ via (34)—requires $\hat{\gamma}$ to converge to γ^{ipw} , limiting how aggressively the tuning parameter can prioritize balance over variance. The modulus argument bounds imbalance directly from the geometry of the model, without decomposing it into “true weights plus estimation error.” Because convergence of the weights does not enter the argument, there is no restriction on the degree of undersmoothing. Less smoothing—a smaller σ —means better balance.

But this is an *in-sample* phenomenon. If we fit weights on one sample and evaluate balance on another, we lose the cancellation that makes aggressive tuning work. Figure 4 illustrates this: in-sample imbalance

¹⁷For RKHS models, the approximation condition can be made explicit in terms of eigenvalue decay rates; see Appendix A.

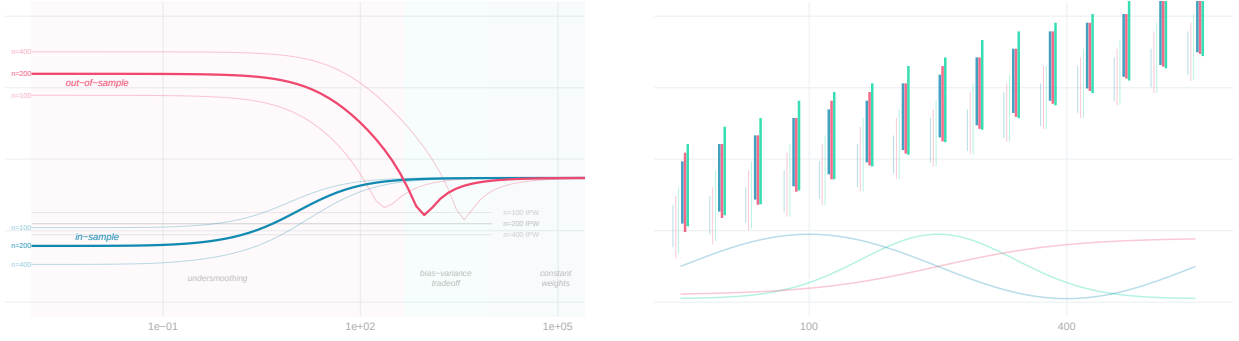


Figure 4: *Left*: In-sample versus out-of-sample maximal imbalance as a function of the tuning parameter σ^2 , for $n \in \{100, 200, 400\}$ (bold: $n = 200$). Balancing weights are fit on sample A and evaluated on both sample A (in-sample, blue) and an independent sample B (out-of-sample, red). Gray lines mark IPW imbalance. Shaded regions indicate three regimes: undersmoothing, bias–variance tradeoff, and constant weights. The gap between the curves is the in-sample advantage exploited by Proposition 6.1. *Right*: Location of the bias–variance tradeoff regime on the σ^2 axis for three propensity score shapes across a range of sample sizes. Each stick spans the σ^2 values in this regime for one propensity at one sample size; the undersmoothing regime lies below each stick and the constant-weights regime lies above. Bold sticks use a Matérn-3/2 model; faint sticks use a Gaussian RBF model on the same data. The smoother model has a wider tradeoff regime. Faint curves in the background show the three propensity score functions.

decreases monotonically as σ shrinks, while out-of-sample imbalance—the imbalance that controls bias when weights are cross-fit—first decreases then increases as the weights overfit the training sample. The gap between the two curves is precisely the in-sample advantage of balancing weights—the advantage that cross-fitting, which evaluates weights on held-out data, cannot access.

The upward shift of the tradeoff regime with n in Figure 4 reflects the distinction between in-sample and out-of-sample balance. In-sample balance improves as σ shrinks, because smaller σ gives the weights more freedom to match the sample. Out-of-sample balance, by contrast, requires the weights to estimate the inverse propensity score function, and the rate-optimal regularization for this estimation grows with n : taking $\sigma \propto \sqrt{n}r$ for the fixed-point radius r (39) gives the minimax rate of convergence for the weight function (Hirshberg & Wager, 2021, Section 3.3). Because $r \rightarrow 0$ slower than $n^{-1/2}$, $\sigma^2 \propto nr^2$ grows with n . This means that tuning for in-sample balance is forgiving—any sufficiently small σ works—while tuning for out-of-sample balance targets a moving window whose location depends on both the sample size and the model.

Remark 3 (Augmentation and convex models). Where does this leave augmentation? For convex models, the minimax linear theory tells us that augmentation is unnecessary for near-optimality, and the results above show this extends to asymptotic efficiency with no restrictions on undersmoothing. The largest benefits of augmentation appear in nonconvex models, such as sparse models, where linear estimators are no longer near-optimal (Cai & Low, 2004). In simulations using sparse models, augmentation can cut root mean squared error by half or more (Hirshberg & Wager, 2021, Section 6, Setup 2).

6.4 The inner product perspective

We have been using an asymmetric framing of the bias, bounding the worst case over the outcome model. The conventional framing of this estimation problem is more symmetric. The connection is the decomposition of the conditional bias in Equation 11 into an inner product of the weight error $\delta\gamma = \hat{\gamma} - \gamma^{\text{IPW}}$ and the outcome error $\delta\mu$, plus a remainder:

$$\frac{1}{n} \sum_{i=1}^n W_i \hat{\gamma}(X_i) \delta\mu(X_i) - \frac{1}{n} \sum_{i=1}^n \delta\mu(X_i) = \frac{1}{n} \sum_{i=1}^n W_i \delta\gamma(X_i) \delta\mu(X_i) + \frac{1}{n} \sum_{i=1}^n \{W_i \gamma^{\text{IPW}}(X_i) - 1\} \delta\mu(X_i). \quad (46)$$

When $\hat{\mu}_1$ is cross-fit, the second term on the right is $o_p(n^{-1/2})$ if $\hat{\mu}_1$ is consistent. The first—the inner product—treats the two nuisance errors symmetrically. And the conditional bias is negligible if and only if this inner product is. An inner product can only be small if the factors are small or nearly orthogonal, so bounding maximal imbalance over the model—as in the preceding sections—amounts to ensuring that $\delta\gamma$ is either small or nearly orthogonal to every function in the model that is not itself small. The preceding sections established this in an asymmetric framing; the inner product formulation makes the underlying symmetry explicit.

A common way to use this decomposition is to bound the inner product by the product of norms $\|\delta\gamma\|_{L_2(P_n)}\|\delta\mu\|_{L_2(P_n)}$, giving *rate double robustness*: the bias is negligible when this product is $o_p(n^{-1/2})$, regardless of which nuisance is estimated more precisely. This captures a real phenomenon—the two nuisances can compensate for each other—but it also suggests a false limitation. When both converge at the same rate, that rate must exceed $n^{-1/4}$, and this is often taken as a fundamental barrier. It is not.

When weights or outcomes are estimated by projection—least squares onto a basis, regularized regression, or the lasso—first-order conditions show that the errors $\delta\gamma$ and $\delta\mu$ are approximately orthogonal to the fitted subspace. Orthogonality makes the inner product small even when neither norm is—something the product-of-norms bound cannot see. This *improved double robustness* (Tan, 2010; Vermeulen & Vansteelandt, 2015) sharpens the rate conditions beyond what Cauchy–Schwarz gives. Taken to its logical conclusion, this leads to *optimal double robustness*. In smooth nonparametric models, Mukherjee et al. (2017) and J. Robins et al. (2009) show that efficient inference is possible whenever *either* γ^{ipw} or μ_1 is smooth enough to be estimable at a rate faster than $n^{-1/4}$, and that this requirement cannot be improved. They achieve it with estimators based on higher-order influence functions; subsequent work showed that much simpler estimators have the same property. Newey and Robins (2018) show this for AIPW with sieve nuisance estimators, and Kennedy (2020) extends it to general linear nuisance estimators. In high-dimensional models, where the story is somewhat different, Bradic et al. (2019) establishes an analogous optimality result.

Remark 4 (Regression for orthogonality). The symmetry of the inner product means the maximal imbalance logic can be reversed. Wang and Shah (2020) observe that rather than choosing weights orthogonal to outcome model errors, we can choose the outcome regression so that its error is orthogonal to weight errors. This yields $\sqrt{n}$ -consistent estimation under assumptions on the propensity score rather than the outcome model. See Appendix F for a sketch in our notation.

7 Other estimands

Thus far, we have considered estimating the average treatment effect $\tau = \psi_1 - \psi_0$, focusing on estimating ψ_1 in particular. These ideas and this framework, however, can be used for many other quantities of interest. In particular, they generalize to estimate many other functions $\psi(m)$ of the conditional mean outcome $\mu_w(x) = E[Y \mid W = w, X = x]$, where the treatment w need not be discrete-valued. This includes many generalizations of the average treatment effect. For example,

$$\psi(m) = E\left[\frac{\partial}{\partial w} m_w(X) \mid_{w=W}\right], \quad \text{average effect of an infinitesimal dose increase;} \quad (47)$$

$$\psi(m) = \int \{\mu_1(x) - \mu_0(x)\} p(x) dx, \quad \text{population-targeted average treatment effect.} \quad (48)$$

All of these are instances of a general estimand that can be estimated with essentially the same approach (see, e.g., Chernozhukov, Newey, & Robins, 2018; Hirshberg & Wager, 2021).

$$\psi(m) = E[h(W_i, X_i, m)] \quad \text{where} \quad h(x, w, m) \text{ is linear in } m. \quad (49)$$

Above, where we have focused on the case that $\psi(m) = \psi_1 = E[\mu_1(X_i)]$, the inverse propensity weights $\gamma_\psi(w, x) = w\gamma^{\text{ipw}}(x)$ have played an essential role. In general, this role is played by the *Riesz representer* of the function ψ , which is the solution $\gamma_\psi(w, x)$ to a balance condition that generalizes the one defining the inverse propensity weights (4):

$$E[h(W_i, X_i, f)] = E[\gamma_\psi(W_i, X_i)f(W_i, X_i)] \quad \text{for all } f \text{ with } E[f(W_i, X_i)^2] < \infty. \quad (50)$$

The solution γ_ψ is guaranteed to exist and be square integrable when ψ is *mean-square-continuous*. For $\psi(m) = \psi_1$, mean-square continuity is equivalent to sufficient overlap.

We can mechanically derive generalized forms of everything we discuss above: the IPW and AIPW estimators, imbalance-focused decompositions of their error, appropriate forms of maximal imbalance, and estimators for balancing weights. To do this, we substitute $\gamma_\psi(W_i, X_i)$ for $W_i\gamma^{\text{IPW}}(X_i)$ and then $h(W_i, X_i, m)$ for $\mu_1(X_i)$ in Equations 6-13 above.¹⁸ All of our discussion above generalizes, whether focused more on maximal imbalance as in Sections 5.1-6.3 (Hirshberg & Wager, 2021) or on estimation of the inverse propensity score as in Section 6.4 (Chernozhukov, Newey, & Robins, 2018).

For the resulting IPW and AIPW estimators $\hat{\psi}^{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \hat{\gamma}_i(W_i, X_i)Y_i$ and $\hat{\psi}^{\text{aug}} = \frac{1}{n} \sum_{i=1}^n h(W_i, X_i, \hat{\mu}) + \hat{\gamma}(W_i, X_i)\{Y_i - \hat{\mu}(W_i, X_i)\}$, the relevant notion of imbalance is

$$\text{imbalance}_{h, \mathcal{M}}(\hat{\gamma}) = \max_{m \in \mathcal{M}} \left| \frac{1}{n} \sum_{i=1}^n \hat{\gamma}(W_i, X_i)m(W_i, X_i) - \frac{1}{n} \sum_{i=1}^n h(W_i, X_i, m) \right|. \quad (51)$$

Hirshberg and Wager, 2021 describes the behavior of $\hat{\psi}^{\text{aug}}$ with weights generalizing the minimax weights (12) described above. The proposition below, which generalizes Proposition 4.1, describes the behavior of the estimators $\hat{\psi}^{\text{IPW}}$ and $\hat{\psi}^{\text{AIPW}}$ with arbitrary design-based weights.

Proposition 7.1. Let $\Gamma := \sqrt{n^{-1} \sum_{i=1}^n \hat{\gamma}(W_i, X_i)^2}$ be the root-mean-squared weight. Suppose that each squared weight is negligible compared to their sum $n\Gamma^2$, and assume that the noise sequence $\varepsilon_1 \dots \varepsilon_n$ is uniformly square-integrable conditional on the design. If the imbalance in the regression error $\delta m = m - \hat{\mu}$ is $o_p(n^{-1/2}\Gamma)$,

$$\hat{\psi}^{\text{aug}} - \tilde{\psi} = \frac{1}{n} \sum_{i=1}^n \hat{\gamma}(W_i, X_i)\{Y_i - m(W_i, X_i)\} + o_p(n^{-1/2}\Gamma) \quad \text{for} \quad \tilde{\psi} = \frac{1}{n} \sum_{i=1}^n h(W_i, X_i, m). \quad (52)$$

Furthermore, if $\hat{\mu}$ is a consistent estimator of μ , then

$$\frac{\hat{\psi}^{\text{aug}} - \tilde{\psi}}{\sqrt{\hat{V}}} \rightarrow N(0, 1) \quad \text{for} \quad \hat{V} = \frac{1}{n^2} \sum_{i=1}^n \hat{\gamma}(W_i, X_i)^2 \{Y_i - \hat{\mu}(W_i, X_i)\}^2. \quad (53)$$

If imbalance in m itself is $o_p(n^{-1/2}\Gamma)$, the same results hold with $\hat{\psi}^{\text{IPW}}$ in place of $\hat{\psi}^{\text{aug}}$.

Some common estimands, such as the average treatment effect on the treated (ATT), are not directly included in the general class defined by (49), but can be expressed as the quotient of something that is in this class and something that is trivial to estimate. For example, the ATT can be written as $E[m(1, X) - m(0, X) \mid W = 1] = E[W\{m(1, X) - m(0, X)\}] / E[W]$. To estimate it, we can start with an estimator $\hat{\psi}^{\text{IPW}}$ or $\hat{\psi}^{\text{aug}}$ for the numerator, then divide by the natural estimator of the denominator, $\bar{W} = \frac{1}{n} \sum_{i=1}^n W_i$, so division by n in the formula for $\hat{\psi}^{\text{IPW}}$ or $\hat{\psi}^{\text{aug}}$ is replaced with division by $N_1 = n\bar{W} = \sum_{i=1}^n W_i$.

Furthermore, a small tweak allows us to estimate some functions $\psi(m)$ of the conditional mean outcome that are not mean-square continuous, like the conditional average treatment effect (CATE) $\tau(x_0) = \mu_1(x_0) - \mu_0(x_0)$ at a point x_0 . The CATE is an instance of the population-targeted average treatment effect (48) in which the target population's covariate distribution $p(x)$ is a spike $\delta_{x_0}(x)$ at $x = x_0$. Here, the lack of overlap of this spiky target distribution δ_{x_0} with the distribution of the observed covariates causes the function not to be continuous. However, we can approximate the spike with a smoother density, taking $p(x)$ to be the density of a normal distribution with mean x_0 and covariance $h^2 I$ for small *bandwidth* h , which will yield good approximation of $\tau(x_0)$ if τ is continuous at x_0 . A refined version of this approach can be used for other estimands like the dose-response curve for a continuous-valued treatment, $\mu(w_0) = E[m_{w_0}(X_i)] = E[\int m_w(X_i)\delta_{w_0}(w)dw]$, which fix some aspects of the treatment or population and average over others (e.g., Chernozhukov, Newey, & Singh, 2018; Kennedy et al., 2017).

Other parameters identified by moment conditions can be estimated by a similar approach as well, including quantile treatment effects or average treatment effects identified by instrumental variables. In

¹⁸Order is important when making these substitutions only because we have followed convention in writing $W_i\mu_1(X_i)$ where the equivalent expression $W_i\mu_{W_i}(X_i)$ is perhaps more natural.

essence, if a parameter θ_0 satisfies the moment condition $E[h(W_i, X_i, m, \theta)] = 0$ for any function m that we can estimate, and we can propose a valid model $\mathcal{M}$ for this estimator's error $\delta\mu = \hat{\mu} - m$, then for any given θ we can form a generalized AIPW estimator $\hat{\psi}_\theta$ of $\psi_\theta(m) = E[h(W_i, X_i, m, \theta)]$. To estimate θ_0 , all we must then do is solve $\hat{\psi}_\theta \approx 0$. See Chernozhukov et al. (2016) and Singh and Sun (2019).

8 Matching for Balance

The analysis of balance we have developed applies to any weights, integer-valued or otherwise: imbalance, augmentation, asymptotic normality, and the empirical process theory connecting model complexity to balance all carry over unchanged. What changes when we move to matching is how the weights are constructed. The optimization methods we have discussed—stable balancing weights, kernel balancing, entropy balancing—solve continuous problems and produce real-valued solutions. Matching uses integer weights, and this comes with both interpretability and efficiency advantages. Because unmatched controls have weight exactly zero, the cost of a study is driven by the number of controls followed up, not the number available. In longitudinal retrospective cohort studies, for instance, matching selects which control subjects to follow up (Stuart & Rubin, 2008), and the savings can be the primary determinant of study cost. The question is whether we can reap the cost savings of integer weights without sacrificing the inferential guarantees of continuous ones.

The answer is essentially yes. We show that any good set of continuous weights can be rounded to integers with negligible change in imbalance (Proposition 8.1), yielding a matching estimator that is asymptotically unbiased and normal. This is both a usable method and a guarantee that extant matching-for-balance approaches like cardinality matching (Zubizarreta et al., 2014) and generalized optimal matching (Kallus, 2020a) can achieve the same inferential guarantees. All results are proven in Appendix E.

To state results precisely, we work with the matched difference in means estimator of the average treatment effect on the treated (e.g., Rosenbaum and Rubin 1985, Abadie and Imbens 2006, Zubizarreta et al. 2014), which can be interpreted as a special case of the inverse probability weighting estimator $\hat{\psi}^{IPW} = \frac{1}{N_1} \sum_{i=1}^n W_i Y_i - \bar{W}_i \theta_i Y_i$. Here we have written our estimator in terms of the rescaled weights $\theta_i = \gamma_i / \bar{W}$ appropriate when dividing by N_1 and written $\bar{W}_i$ for $1 - W_i$. In this context, we write the design-conditional bias and the associated imbalance, which bounds the magnitude of the bias when $m \in \mathcal{M}$, as follows.

$$\begin{aligned} \text{bias} &= \frac{1}{N_1} \sum_{i=1}^n W_i m(0, X_i) - \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i \theta_i m(0, X_i) \\ \text{imbalance}_{h, \mathcal{M}}(\theta) &= \max_{m \in \mathcal{M}} \left| \frac{1}{N_1} \sum_{i=1}^n W_i m(0, X_i) - \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i \theta_i m(0, X_i) \right|. \end{aligned} \tag{54}$$

In the matched difference in means estimator, each control unit has an integer weight $\theta_i \in \{0, 1, \dots\}$, describing the number of treated units to which it is matched.

Remark 5 (Nearest-neighbor matching and the Lipschitz model). Nearest-neighbor matching is, in fact, a matching-for-balance method: it is the $\sigma^2 = 0$ case of the minimax weights (12) for the Lipschitz model $\mathcal{M} = \{f : \|f\|_{\text{Lip}(\rho)} \leq \kappa\}$ (Kallus, 2020a, Theorem 3). The problem is that the Lipschitz model is large. The conditional bias of the nearest-neighbor matching estimator is $O_p(n^{-1/d})$ (Abadie & Imbens, 2006, Theorem 1), which is $o_p(n^{-1/2})$ only when $d \leq 2$. In the language of Section 6, the Gaussian width of the Lipschitz ball grows too quickly with dimension for the fixed point to yield $\sqrt{n}$ rates. Matching with a smaller model—one whose complexity is controlled by stronger smoothness or sparsity assumptions—avoids this barrier; see the discussion in Section 6.

A simple way to turn a weighting estimator into a matching one is through *Bernoulli rounding*. We shave off the fractional parts $\hat{r}_i = \hat{\theta}_i - \lfloor \hat{\theta}_i \rfloor$ of our weights and replace them with independent Bernoullis $B_i \sim \text{Bernoulli}(\hat{r}_i)$ with means matching what we have shaved. This gives us a *randomized estimator* with integer weights $\lfloor \hat{\theta}_i \rfloor + B_i$. The procedure is unbiased, as the expectation of the estimator is unchanged, but

it increases the variance solely to obtain integral weights. By the law of total variance,

$$\text{var} \left(\frac{1}{N_1} \sum_i \bar{W}_i (\lfloor \hat{\theta}_i \rfloor + B_i) Y_i \right) = \text{E} \left(\frac{1}{N_1^2} \sum_i \bar{W}_i \hat{r}_i (1 - \hat{r}_i) Y_i^2 \right) + \text{var} \left(\frac{1}{N_1} \sum_i \bar{W}_i \hat{\theta}_i Y_i \right). \quad (55)$$

This is not the irreducible cost of matching. We can do better. This is what matching costs: the variance of any negligibly-biased estimator using deterministic integer weights $\tilde{\theta}_i$ is

$$\text{var} \left(\frac{1}{N_1} \sum_i \bar{W}_i \tilde{\theta}_i Y_i \right) = \text{E} \left(\frac{1}{N_1^2} \sum_i \bar{W}_i \hat{r}_i (1 - \hat{r}_i) \text{var}\{Y_i \mid W_i, X_i\} \right) + \text{var} \left(\frac{1}{N_1} \sum_i \bar{W}_i \hat{\theta}_i Y_i \right), \quad (56)$$

where $\tilde{\theta}_i$ denotes any deterministic integer rounding of the real-valued weights $\hat{\theta}_i$, given mild conditions on $\hat{\theta}$ (see Proposition E.4). The difference between these two variances—call it ΔV —is the mean-squared *change in imbalance in the conditional mean* μ from random rounding:

$$\Delta V = \text{E} \left(\frac{1}{N_1^2} \sum_i \bar{W}_i \hat{r}_i (1 - \hat{r}_i) \mu(X_i)^2 \right) = \text{E} \left[\left\{ \frac{1}{N_1} \sum_i \bar{W}_i (B_i - \hat{r}_i) \mu(X_i) \right\}^2 \right]. \quad (57)$$

This is the same phenomenon that makes AIPW preferable to IPW with the true weights γ^{IPW} : mean-zero imbalance costs nothing in bias but inflates variance. Here, it is the *change* in μ -imbalance from rounding, rather than the μ -imbalance itself. To avoid this cost, we need to round with negligible change in imbalance. Given a model $\mathcal{M}$ for μ , we can ask for negligible change in imbalance by solving this integer program, which minimizes the rounding-induced change uniformly over $m \in \mathcal{M}$.

$$\begin{aligned} \lfloor \hat{\theta} \rfloor = & \underset{\substack{\theta \in \mathbb{Z}^n \\ \|\theta - \hat{\theta}\|_\infty \leq 1 \\ \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i \theta_i = 1}}{\text{argmin}} \quad \text{rounding}_{\mathcal{M}, \hat{\theta}}(\theta) \quad \text{for} \quad \text{rounding}_{\mathcal{M}, \hat{\theta}}(\theta) = \max_{m \in \mathcal{M}} \left| \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i (\theta_i - \hat{\theta}_i) m(X_i) \right|. \end{aligned} \quad (58)$$

We show that, for models with bounded uniform entropy integral (BUEI)¹⁹, the change in imbalance we get with this rounding scheme is uniformly negligible. To keep things simple, we shall suppose we have started with a translation-invariant estimator. This means that our weights average to one and therefore can be rounded to integers with no change in sum.

This, of course, has implications for bias as well as variance. By the triangle inequality, the maximal imbalance with the rounded weights is no larger than the sum of our maximal rounding error and our maximal imbalance with our original weights.

$$\text{imbalance}_{h, \mathcal{M}}(\lfloor \hat{\theta} \rfloor) \leq \text{imbalance}_{h, \mathcal{M}}(\hat{\theta}) + \text{rounding}_{\mathcal{M}, \hat{\theta}}(\lfloor \hat{\theta} \rfloor).$$

That means that if we start with weights achieving uniformly negligible imbalance over correctly specified model ($\mu \in \mathcal{M}$), the matching estimator we get by rounding has uniformly negligible imbalance too, and as a result we get asymptotics like we discussed in Proposition 4.1 above. Inference is easy—the t -statistic based on our matched difference in means is asymptotically standard normal.

Proposition 8.1. Suppose $\mathcal{M}$ has bounded uniform entropy integral. For any weights $\hat{\theta}$ with $\frac{1}{N_1} \sum_{i=1}^n \hat{\theta}_i = 1$, the maximal rounding error $\text{rounding}_{\mathcal{M}, \hat{\theta}}(\lfloor \hat{\theta} \rfloor)$ resulting from our balance-preserving rounding scheme (58) is $o_p(n^{-1/2})$.

We prove it by working with an apparently simpler quantity. The rounding error in (58) is a scaled version of what is called *linear discrepancy* in the combinatorics and optimization literature: $\text{rounding}_{\mathcal{M}} =$

¹⁹Having a bounded uniform entropy integral is probably the most frequently-used sufficient condition for $\mathcal{M}$ to be Donsker. See (A. W. van der Vaart & Wellner, 1996, Chapter 2.5)

$N_1^{-1} \text{lindisc}_{\mathcal{M}}$. A classical result in discrepancy theory (Lovász et al., 1986) shows that linear discrepancy is bounded by *hereditary discrepancy*, i.e.,

$$\text{lindisc} \leq \text{hdisc} \quad \text{where} \quad \text{hdisc}(T) = \max_{I \subset \{1 \dots n\}} \min_{\epsilon \in \{-1, 1\}^n} \max_{t \in T} \left| \sum_{i \in I} \epsilon_i t_i \right| \quad (59)$$

is the worst-case imbalance when we optimally two-color any subset of our units.

Rounding weights to integers is, at heart, a coloring problem: we’re deciding which units to “round up” ($\epsilon_i = +1$) and which to “round down” ($\epsilon_i = -1$), trying to keep every function in our model balanced. The “hereditary” part—the max over all subsets I —is crucial. Ordinary discrepancy (for the full set) can be small by lucky cancellation, but hereditary discrepancy is small only when the structure genuinely allows good colorings.

To build intuition, consider a model containing all constant functions, so that perfectly balanced weights must satisfy $\sum_{i \in I} \epsilon_i = 0$. If the subset I has even size $|I| = 2k$, this means assigning k units to each color—a matched pair design. The inner minimization then asks: what is the best balance we can achieve when we optimally match the $2k$ units? And hereditary discrepancy asks: what is this worst-case optimal balance, over all subsets I and all functions $m \in \mathcal{M}$?

Two caveats. First, discrepancy is defined in terms of sums $\sum_{i \in I} \epsilon_i m(X_i)$, not averages, so the connection to matched differences in means involves a factor of $|I|/2$. Second, the constraint $\sum_{i \in I} \epsilon_i = 0$ requires $|I|$ even; for odd-sized subsets, one “group” is larger by one, so discrepancy for any model containing all constants is infinite. For the same reason, models like this are not actually BUEI and do not satisfy the conditions of Proposition 8.1.

Why does rounding reduce to coloring? The idea is to round recursively on binary expansions: approximate each weight $q_i \in [0, 1]$ to M binary digits, then round the last digit optimally (choosing $\pm 2^{-M}$ to minimize imbalance), then the second-to-last, etc. Each step is a coloring problem on a subset—the units whose k th digit is 1—and costs at most 2^{-k} times the hereditary discrepancy. Summing the geometric series gives a bound of $\sum_{k=1}^M 2^{-k} \text{hdisc} \rightarrow \text{hdisc}$ as $M \rightarrow \infty$.

We show in Appendix E, following the approach of Mendelson (2011), that $\text{hdisc}(\mathcal{M}) = o_p(n^{1/2})$ for BUEI classes. Because rounding $\leq N_1^{-1} \text{lindisc} \leq N_1^{-1} \text{hdisc}$, this means that rounding $_{\mathcal{M}} = o_p(n^{-1/2})$.

The cost of matching can be characterized precisely. For any regular matching estimator,

$$\text{var}(\hat{\psi}^{\text{match}}) - V^{\text{eff}} \geq \frac{1}{N_1^2} \sum_{i=1}^n \bar{W}_i r(X_i) \{1 - r(X_i)\} v_0(X_i) + o_p(N_1^{-1}), \quad (60)$$

where V^{eff} is the variance of the efficient estimator and $r(x) = \theta(x) - \lfloor \theta(x) \rfloor$ is the fractional part of the propensity odds $\theta(x) = e(x)/\{1 - e(x)\}$.

Proposition 8.2 (Lower bound). Suppose $v_0(x) = \text{var}[Y \mid W = 0, X = x]$ is bounded. For any *regular* matching estimator—one whose integer weights achieve $o_p(1)$ imbalance in every $f \in L^2(P)$ —the variance satisfies the lower bound (60).

Proposition 8.3 (Achievability). Under the conditions of Proposition 8.2, balance-preserving rounding (58) achieves equality in (60):

$$\text{var}(\hat{\psi}^{\text{match}}) - V^{\text{eff}} = \frac{1}{N_1^2} \sum_{i=1}^n \bar{W}_i r(X_i) \{1 - r(X_i)\} v_0(X_i) + o_p(N_1^{-1}).$$

Corollary 8.4. Let $\Gamma := \sqrt{N_1^{-1} \sum_{i=1}^n \bar{W}_i \hat{\theta}(X_i)^2}$ be the root-mean-squared weight. Suppose that each squared weight is negligible compared to their sum $n\Gamma^2$, and assume that the noise sequence $\varepsilon_1 \dots \varepsilon_n$ is uniformly square-integrable conditional on the design. If m is in a BUEI model $\mathcal{M}$ and $\text{imbalance}_{h, \mathcal{M}}(\hat{\theta}) = o_p(n^{-1/2})$, then $\hat{\psi}^{\text{match}} = \frac{1}{N_1} \sum_{i=1}^n W_i Y_i - \bar{W}_i \lfloor \hat{\theta} \rfloor_i Y_i$, where $\lfloor \hat{\theta} \rfloor$ are the uniform-balance-preserving rounded weights (58), satisfies

$$\hat{\psi}^{\text{match}} - \tilde{\psi}(m) = \frac{1}{N_1} \sum_{i=1}^n (W_i + \bar{W}_i \lfloor \hat{\theta} \rfloor_i) \{Y_i - m(W_i, X_i)\} + o_p(n^{-1/2}). \quad (61)$$

Furthermore, if $\hat{\mu}$ is a consistent estimator of μ ,

$$\frac{\hat{\psi}^{match} - \tilde{\psi}(m)}{\sqrt{\hat{V}}} \rightarrow N(0, 1) \quad \text{for} \quad \hat{V} = \frac{1}{N_1^2} \sum_{i=1}^n \{W_i + \bar{W}_i (\lfloor \hat{\theta} \rfloor_i)^2\} \{Y_i - \hat{\mu}(W_i, X_i)\}^2. \quad (62)$$

Remark 6. The most interesting case is when the fractional part is the weight itself (i.e. the weight is in $[0, 1]$), as rounding the whole weight to zero is what provides us the opportunity to avoid follow-up. This will be the case when the treatment probability $e(X_i)$ is well under $1/2$, as the weights $\hat{\theta}(X_i)$ estimate the inverse probability weights $e(X_i)/\{1 - e(X_i)\}$. When this is true for all or almost all units, our rounded weighting estimator will have all or almost all weights in $\{0, 1\}$ and the same asymptotic distribution as a simple difference in means in a randomized experiment in which each of $2N_1$ units received treatment with probability $1/2$. Relative to weighting the full set of N_0 control units, this comes at the cost described in (55). But in cases where follow-up is costly, it is natural to compare its efficiency to other estimators requiring the same amount of follow-up, e.g. an IPW estimator based on our first N_1 control observations. Relative to that estimator, Proposition E.6 in Appendix E.2 shows that our matching approach reduces asymptotic variance by a factor of $(\rho + 1)/\{\rho + \mu_2/\mu_1^2\}$, where $\rho = v_1/v_0$ is the ratio of treated to control noise variance and $\mu_k = E[\theta(X)^k \mid W = 0]$. For equal noise ($\rho = 1$) and small propensity ($e(X) \ll 1/2$, so $\theta \approx e$), this simplifies to $1/\{1 + \text{var}(e(X) \mid W = 0)/(2E[e(X) \mid W = 0]^2)\}$.

This rounding approach, while reasonable, is not necessarily the best way to get an asymptotically unbiased and normal matching estimator. Rather than finding non-integer weights that give us negligible imbalance and then rounding them in a way that controls the rounding-induced change in imbalance, we could simply find integer-valued weights that give us negligible imbalance. We could do this, for example, by solving the following integer program to minimize the estimator's conditional variance subject to an imbalance budget $B_n \ll n^{-1/2}$,

$$\lfloor \hat{\theta} \rfloor = \underset{\substack{\theta \in \mathbb{Z}^n \text{ with} \\ \text{imbalance}_{h, \mathcal{M}}(\theta) \leq B_n}}{\text{argmin}} \quad \frac{1}{n} \sum_{i=1}^n \bar{W}_i \theta_i^2 \quad (63)$$

This integer-constrained balancing weights approach, which was called *generalized optimal matching* by Kallus (2020b), and the related *cardinality matching* approach of Zubizarreta et al. (2014), can be understood using Proposition 8.1 as well. For models in which a non-integer-constrained version of (63) achieves uniformly negligible imbalance—as it does for nonparametric BUEI classes of bounded functions, because BUEI classes are Donsker and Proposition 6.1 applies—Corollary 8.4 implies that there is a feasible solution and the result is, in fact, an optimal matching estimator.

Corollary 8.5. If $\mathcal{M}$ is a nonparametric (Definition 1) class of uniformly bounded functions with bounded uniform entropy integral, then for an appropriately chosen sequence of budgets $B_n \ll n^{-1/2}$, the integer program (63) has a solution with probability tending to one and the corresponding difference in means estimator $\hat{\psi}^{match} = \frac{1}{N_1} \sum_{i=1}^n W_i Y_i - W_i \lfloor \hat{\theta} \rfloor_i Y_i$ satisfies the asymptotic expansions (61) and (62), with variance achieving equality in the matching cost bound (60).

The results in this section show that matching for balance makes statistical inference straightforward and can be more cost-effective than asymptotically efficient estimators by using the responses of a smaller number of units in the analysis. However, if our sole concern is efficiency while strictly limiting the number of responses we need to record, this approach may not be optimal. We could, for example, replace the integrality constraint on our balancing weights (63) with a constraint on the number of nonzero weights. Sparse weighting ideas like these are an interesting area for future work.

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A Modulus of continuity: penalized and constrained forms

The penalized modulus (43) used in Section 6.3 is related by Lagrangian duality to the constrained formulation of Donoho (1994) and Armstrong and Kolesár (2018). The constrained modulus is

$$\bar{\omega}(\delta) = \sup \left\{ \left| \frac{1}{n} \sum_{i=1}^n m_1(X_i) - \frac{1}{n} \sum_{i=1}^n m_2(X_i) \right| : m_1, m_2 \in \mathcal{M}, \|m_1 - m_2\|_{L_2(P_n)} \leq \delta \right\}, \quad (64)$$

where $\|f\|_{L_2(P_n)}^2 = n^{-1} \sum_{i=1}^n f(X_i)^2$. The two formulations are related by the Fenchel–Legendre transform: $\omega(s) = \sup_{\delta \geq 0} \{\bar{\omega}(\delta) - \delta^2/(2s)\}$, or equivalently $\bar{\omega}(\delta) = \inf_{s > 0} \{\omega(s) + \delta^2/(2s)\}$, with the optimal s and δ linked by $\delta = s \bar{\omega}'(\delta)$.

B Asymptotics

We state our asymptotic results in the general AMLE framework of Hirshberg and Wager (2021), then specialize to the treatment-specific mean as a corollary. For a functional $\psi(m) = n^{-1} \sum_{i=1}^n h(W_i, X_i, m)$, the AIPW estimator is

$$\hat{\psi}^{\text{aug}} = \frac{1}{n} \sum_{i=1}^n h(W_i, X_i, \hat{\mu}) + \hat{\gamma}(W_i, X_i) \{Y_i - \hat{\mu}(W_i, X_i)\}.$$

Proposition 7.1’. *Let $\Gamma = (n^{-1} \sum_{i=1}^n \hat{\gamma}(W_i, X_i)^2)^{1/2}$ be the root-mean-squared weight. Suppose that the noise $\varepsilon_i = Y_i - m(W_i, X_i)$ is well behaved in the sense that its conditional variance $\nu(W_i, X_i) := E[\varepsilon_i^2 | W_i, X_i]$ is bounded away from zero and that it satisfies the uniform integrability condition $\max_{i \leq n} E[\varepsilon_i^2 1(\varepsilon_i^2 \geq M) | W_1, X_1 \dots W_n, X_n] \rightarrow_p 0$ as $M \rightarrow \infty$. Suppose that imbalance in $\delta\mu = m - \hat{\mu}$ is negligible, i.e. $o_p(n^{-1/2}\Gamma)$, that the weights are infinitesimal in the sense that $\max_{i \leq n} \hat{\gamma}(W_i, X_i)^2 = o_p(n\Gamma^2)$, and that $\hat{\mu}$ is consistent in the weighted L_2 sense*

$$\sum_{i=1}^n \hat{\gamma}(W_i, X_i)^2 \{\hat{\mu}(W_i, X_i) - m(W_i, X_i)\}^2 = o_p(n\Gamma^2). \quad (65)$$

Then

$$\frac{\hat{\psi}^{\text{aug}} - \tilde{\psi}}{(\hat{V})^{1/2}} \rightarrow N(0, 1) \quad \text{where} \quad \tilde{\psi} = \frac{1}{n} \sum_{i=1}^n h(W_i, X_i, m), \quad \hat{V} = \frac{1}{n^2} \sum_{i=1}^n \hat{\gamma}(W_i, X_i)^2 \{Y_i - \hat{\mu}(W_i, X_i)\}^2.$$

The weighted L_2 consistency condition (65) is implied by sup-norm consistency, i.e. $\max_{i \leq n} |\hat{\mu}(W_i, X_i) - m(W_i, X_i)| = o_p(1)$. It is also satisfied naturally by IPW regression estimators, where $\hat{\mu}$ is fit by weighted least squares using the weights $\hat{\gamma}$ —in this case, the weighted residuals are orthogonal to the fitted values by construction.

Proposition 4.1’. *Let $\Gamma = (n^{-1} \sum_{i=1}^n W_i \hat{\gamma}(X_i)^2)^{1/2}$ be the root-mean-squared weight. Suppose that the noise $\varepsilon_i = Y_i - \mu_1(X_i)$ is well behaved in the sense that its conditional variance $\nu(X_i) = E[\varepsilon_i^2 | X_i, W_i = 1]$ is bounded away from zero and satisfies the uniform integrability condition $\max_{i \leq n} E[\varepsilon_i^2 1(\varepsilon_i^2 \geq M) | W_1, X_1 \dots W_n, X_n] \rightarrow_p 0$ as $M \rightarrow \infty$. Suppose that imbalance in $\delta\mu_1 = \hat{\mu}_1 - \mu_1$ is negligible, i.e. $o_p(n^{-1/2}\Gamma)$, that the weights are infinitesimal in the sense that $\max_{i \leq n} W_i \hat{\gamma}(X_i)^2 = o_p(n\Gamma^2)$, and that $\hat{\mu}_1$ is consistent in the weighted L_2 sense*

$$\sum_{i=1}^n W_i \hat{\gamma}(X_i)^2 \{\hat{\mu}_1(X_i) - \mu_1(X_i)\}^2 = o_p(n\Gamma^2).$$

Then

$$\frac{\hat{\psi}_1^{\text{aug}} - \tilde{\psi}_1}{(\hat{V})^{1/2}} \rightarrow N(0, 1) \quad \text{where} \quad \tilde{\psi}_1 = \frac{1}{n} \sum_{i=1}^n \mu_1(X_i), \quad \hat{V} = \frac{1}{n^2} \sum_{i=1}^n W_i \hat{\gamma}(X_i)^2 \{Y_i - \hat{\mu}_1(X_i)\}^2.$$

Proof of Propositions 7.1' and 4.1'. Proposition 4.1' is an instance of Proposition 7.1' with $h(W, X, m) = m(1, X)$ and $\hat{\gamma}(W, X) = W\hat{\gamma}(X)$, so it suffices to prove Proposition 7.1'. This proof has two steps. We'll show that $Z_n = (\hat{\psi}^{\text{aug}} - \tilde{\psi})/V_n^{1/2}$ is asymptotically standard normal for $V_n = n^{-2} \sum_{i=1}^n \hat{\gamma}(W_i, X_i)^2 \nu(W_i, X_i)$, then we'll show that $\hat{V}/V_n$ converges to one in probability, so Slutsky's theorem implies our claim. For the first step, we work with the decomposition

$$\hat{\psi}^{\text{aug}} - \tilde{\psi} = \left[\frac{1}{n} \sum_{i=1}^n \hat{\gamma}(W_i, X_i) \delta\mu(W_i, X_i) - \frac{1}{n} \sum_{i=1}^n \{h(W_i, X_i, \hat{\mu}) - h(W_i, X_i, m)\} \right] + \left[\frac{1}{n} \sum_{i=1}^n \hat{\gamma}(W_i, X_i) \varepsilon_i \right].$$

The first term is the imbalance in $\delta\mu$, which is $o_p(n^{-1/2}\Gamma)$ by assumption and therefore, because ν is bounded away from zero, negligible relative to the denominator $V_n^{1/2}$, which exceeds $n^{-1}\Gamma\underline{\nu}^{1/2}$ for $\underline{\nu} = \min_{w,x} \nu(w, x)$. It follows that $(\hat{\psi}^{\text{aug}} - \tilde{\psi})/V_n^{1/2}$ converges in probability to $V_n^{-1/2}$ times the second term. Conditional on the design $\mathcal{D}_n = \{W_i, X_i : i \leq n\}$, this is a standardized sum of independent mean zero random variables $\xi_{i,n} = \hat{\gamma}(W_i, X_i) \varepsilon_i / (n^2 V_n)^{1/2}$, so the central limit theorem implies it converges to a standard normal random variable if the sum is not dominated by too few terms. To be more precise, letting $\hat{\gamma}$ refer to the weights estimated at sample size n when defining $\xi_{i,n}$, this is a martingale difference array adapted to the σ -algebras $\mathcal{F}_{i,n}$ generated by $\mathcal{D}_n \cup \{\varepsilon_1 \dots \varepsilon_i\}$, it is standardized in the sense that its sum of conditional variances $\sum_{i=1}^n E[\xi_{i,n}^2 \mid \mathcal{F}_{i-1,n}]$ is one, and this conditional Lindeberg condition suffices to ensure its asymptotic normality (Hall & Heyde, 2014, Corollary 3.1):

$$\sum_{i=1}^n E[\xi_{i,n}^2 1(\xi_{i,n}^2 \geq \epsilon) \mid \mathcal{F}_{i-1,n}] \rightarrow_p 0 \quad \text{for all } \epsilon > 0.$$

To show that our assumptions imply this Lindeberg condition, we decompose it and use Hölder's inequality to bound it as follows. Writing $\hat{\gamma}_i = \hat{\gamma}(W_i, X_i)$ and $\nu_i = \nu(W_i, X_i)$ for brevity,

$$\begin{aligned} \sum_{i=1}^n E[\xi_{i,n}^2 1(\xi_{i,n}^2 \geq \epsilon) \mid \mathcal{D}_n] &= \frac{1}{n^2 V_n} \sum_{i=1}^n \hat{\gamma}_i^2 E[\varepsilon_i^2 1(\varepsilon_i^2 \geq M) \mid \mathcal{D}_n] \\ &\quad + \frac{1}{n^2 V_n} \sum_{i=1}^n \hat{\gamma}_i^2 E[\varepsilon_i^2 1(M \geq \varepsilon_i^2 \geq n^2 V_n) \mid \mathcal{D}_n] \\ &\leq \frac{1}{n^2 V_n} \sum_{i=1}^n \hat{\gamma}_i^2 \cdot \max_{i \leq n} E[\varepsilon_i^2 1(\varepsilon_i^2 \geq M) \mid \mathcal{D}_n] \\ &\quad + \frac{1}{n^2 V_n} \sum_{i=1}^n \hat{\gamma}_i^2 \cdot M \max_{i \leq n} 1(\hat{\gamma}_i^2 M \geq n^2 V_n). \end{aligned}$$

Both terms in this bound have the same leading factor, the ratio of $n^{-2} \sum_{i=1}^n \hat{\gamma}_i^2$ and $n^{-2} \sum_{i=1}^n \hat{\gamma}_i^2 \nu_i$, which is $O_p(1)$ because ν is bounded away from zero. Thus, it suffices to show that the sum of the second factors is $o_p(1)$. Our uniform integrability assumption implies that for large enough M , the probability that the first term exceeds any threshold $\delta > 0$ is arbitrarily small. Moreover, for this M , the second term tends to zero with probability tending to one, as given our assumption that ν is bounded away from zero, the claim that $\max_{i \leq n} \hat{\gamma}_i^2 / \sum_{i \leq n} \hat{\gamma}_i^2 \nu_i \rightarrow_p 0$ reduces to our assumption that the weights are infinitesimal. This concludes the first step of our proof.

For the second step, it is equivalent to show that $(\hat{V} - V_n)/V_n \rightarrow 0$ in probability. Writing $\delta\mu_i = \hat{\mu}(W_i, X_i) - m(W_i, X_i)$, we work with the decomposition

$$\begin{aligned} \frac{\hat{V} - V_n}{V_n} &= \frac{1}{n^2} \sum_{i=1}^n \alpha_i [\{\varepsilon_i - \delta\mu_i\}^2 - \nu_i] \quad \text{for } \alpha_i = \frac{\hat{\gamma}_i^2}{V_n} \\ &= \frac{1}{n^2} \sum_{i=1}^n \alpha_i \delta\mu_i^2 - \frac{2}{n^2} \sum_{i=1}^n \alpha_i \varepsilon_i \delta\mu_i + \frac{1}{n^2} \sum_{i=1}^n \alpha_i \{\varepsilon_i^2 - \nu_i\}. \end{aligned}$$

We will show that each term converges to zero. Note that $n^{-2} \sum_{i=1}^n \alpha_i = \Gamma^2/V_n$ is, as discussed above, bounded because ν is bounded away from zero.

1. The first term equals $(n^2 V_n)^{-1} \sum_{i=1}^n \hat{\gamma}_i^2 \delta \mu_i^2$. The numerator is $o_p(n\Gamma^2)$ by our weighted L_2 consistency assumption (65), and the denominator is $\Theta_p(n\Gamma^2)$ because ν is bounded away from zero. So this term is $o_p(1)$.
2. For the cross-term, we use the Cauchy-Schwarz inequality:

$$\left| \frac{1}{n^2} \sum_{i=1}^n \alpha_i \varepsilon_i \delta \mu_i \right|^2 \leq \frac{1}{n^2} \sum_{i=1}^n \alpha_i \varepsilon_i^2 \cdot \frac{1}{n^2} \sum_{i=1}^n \alpha_i \delta \mu_i^2.$$

The first factor on the right-hand side is $(n^2 V_n)^{-1} \sum_{i=1}^n \hat{\gamma}_i^2 \varepsilon_i^2$, which converges in probability to one by step 3 below. The second factor is the first term in our decomposition, which we have just shown is $o_p(1)$. So the cross-term is $o_p(1)$.

3. The third term is difference between the sum of squares $\sum_{i=1}^n \xi_{i,n}^2$ and the sum of conditional variances $\sum_{i=1}^n E[\xi_{i,n}^2 \mid \mathcal{F}_{i-1,n}]$ of the standardized martingale difference array $\xi_{i,n}$, and the aforementioned conditional Lindeberg condition is sufficient to imply that the two converge in probability (Hall & Heyde, 2014, Theorem 2.23).

□

C More on Models

C.1 Models defined in terms of ℓ_q and L_q norms

In Section 5, we have discussed models defined in terms of ℓ_2 norms on coefficients (22-27). It is common to use models defined analogously in terms of the ℓ_1 norm, or another ℓ_q norm, on the coefficients. For example, the largest imbalance in any coordinate X_{ij} is the maximal imbalance for a variant of the model (22) using the ℓ_1 norm and the sum of absolute imbalances in the coordinates arises from a model based on the ℓ_∞ norm. In models with a low-dimensional basis, such as models without interactions, these are roughly equivalent, in the sense that the unit balls of the ℓ_1 , ℓ_2 , and ℓ_∞ norms on $\mathbb{R}^p$ satisfy $p^{-1}B_\infty \subseteq p^{-1/2}B_2 \subseteq B_1 \subseteq B_2 \subseteq B_\infty$. For models requiring a high dimensional basis (p large), the same relationship does not spell equivalence. Geometrically, B_1 is a ‘diamond’ inscribed in B_2 , a ‘round ball’, inscribed in B_∞ , the ‘cube’ with side-length 2. In $\mathbb{R}^2$ these are of increasing but similar size, whereas in high dimensional spaces almost all the volume of B_∞ is in the corners, outside B_2 , and similarly almost all the volume of B_2 is outside of B_1 . In fact, the size of B_1 is, in a meaningful sense, almost independent of dimension p . This fact lies at the heart of the tractability of some problems in high dimensional statistics and their connection to ℓ_1 regularization, including the problem of balancing high dimensional linear models (e.g., Athey et al., 2018; Tan, 2020).

The RKHS model discussed above (27) is based on a *weighted* ℓ_2 norm on infinitely many scaled coefficients $\beta_j/\lambda_j^{1/2}$. When the scale factors $\lambda_j^{1/2}$ decrease sufficiently fast, the corresponding coefficients β_j must decay rapidly, and as a result this model behaves like one defined in terms of a low-dimensional basis. In particular, models defined analogously in terms of other ℓ_q norms on $\beta_j/\lambda_j^{1/2}$ are of similar size.

Similarly, generalizing the Sobolev balls we have discussed above, based on L_2 norms on derivatives (28), are Sobolev balls defined analogously in terms of other L_q norms. These are of comparable size for different q if the functions in them are sufficiently smooth, e.g., L_2 and L_∞ Sobolev balls of functions on $\mathbb{R}^p$ are of similar size if we include derivatives of order greater than $p/2$ in the norm (e.g., Nickl & Pötscher, 2007, Corollary 4).

C.2 Models defined in terms of variation norms

To check balance in covariate distributions, researchers will sometimes compare histograms of the covariates for the re-weighted treated units to those of the population, often summarizing this comparison by the marginal Kolmogorov-Smirnov (KS) statistics. This measure of imbalance corresponds to an *additive model*

for the outcome, $\mu_1(x) = \sum_{j=1}^d m_{1j}(x_j)$, with a component function $m_{1j}(x_j)$ for the j th covariate. In particular, the maximal KS statistic is the maximal imbalance for an additive model with a bounded summed componentwise *total variation*, $\mathcal{M} = \{\sum_{j=1}^d f_j(x_j) : \sum_{j=1}^d \|f_j\|_{TV} \leq 1\}$ where, for differentiable functions, $\|f_j\|_{TV} = \sum_j \int |f'_j(x)| dx$.²⁰ This is a more general approach to controlling bias than comparing covariate means, but it fails to control potential bias arising from imbalance in covariate interactions. This additive model is, however, essentially contained in a very natural model that does include interactions: the unit ball of the space of functions with bounded *sectional variation* (Bibaut & van der Laan, 2019; Fang et al., 2021). The norm for this space is similar to the mixed-smoothness Sobolev norm from Equation 30, depending on the same sum of derivatives, but evaluates it via mix of L_1 -type averaging and evaluation at zero, so it does not coincide with any Sobolev norm exactly (see Equations 23 and 25 in Fang et al., 2021).

C.3 Sobolev and Reproducing Kernel Hilbert Spaces

The particular norms we state in Section 5.2.2’s Equations 28 and 30 are the norms of the first-order isotropic and dominating mixed smoothness Sobolev spaces of bivariate functions on the cube $[0, 1]^2$ (e.g., Kühn et al., 2015, Section 2.1). Sometimes this is described as a norm on functions on $[-1, 1]^2$ that periodic with period 2 in all coordinates, and in Equation 28 the average over the uniform distribution on $[0, 1]^2$ is replaced with an average over the uniform distribution over $[-1, 1]^2$; the arbitrary functions on $[0, 1]^2$ can be extended to such functions on $[-1, 1]^2$ via ‘reflection’, letting $f(\pm x_1, \pm x_2) = f(x_1, x_2)$. For these ‘reflections’, the norm in Equation 28 and the variant that averages over $[-1, 1]^2$ match. If instead of the cube $[0, 1]^2$, we define our space on all of $\mathbb{R}^2$ by averaging our derivatives over the normal distribution, we get a RKHS with a norm characterized in terms of a Hermite polynomial rather than a cosine basis.

Choosing as the model the unit ball $\mathcal{B}_{\mathcal{H}}$ of an RKHS $\mathcal{H}$ is computationally convenient due to the so-called *kernel trick*, which can be used to reduce optimization problems involving infinite series (as in 27) to finite dimensional problems. What lies behind it is the so-called *reproducing property*: for each point x , there is an element $K(x, \cdot) \in \mathcal{H}$ with the property that we can evaluate functions $f \in \mathcal{H}$ at the point x by taking their inner product with $K(x, \cdot)$, i.e., with the property that $f(x) = \langle K(x, \cdot), f \rangle$ for all $f \in \mathcal{H}$. As a result, if we take regularized optimization problems over $f \in \mathcal{H}$ and rewrite each instance of $f(X_i)$ as an inner product with $K(X_i, \cdot)$, it often becomes clear that the solution must lie in the span of $K(X_1, \cdot) \dots K(X_n, \cdot)$. Thus, knowing the solution has the form $f(\cdot) = \sum_{i=1}^n \alpha_i K(X_i, \cdot)$ for $\alpha \in \mathbb{R}^n$, we can solve our optimization problem by optimizing over a finite-dimensional vector $\mathbb{R}^n$ instead of an infinite-dimensional function $f \in \mathcal{H}$. This is summarized in so-called *representer theorems*. Taking this approach, we can simplify the maximal imbalance over the ball $\mathcal{B}_{\mathcal{H}}$ in terms of its kernel K as follows.

$$\text{imbalance}_{\mathcal{B}_{\mathcal{H}}}(\gamma)^2 = \frac{1}{n^2} \sum_{j=1}^n \sum_{i=1}^n K(X_i, X_j) - 2W_i K(X_i, X_j) \gamma_i + W_i W_j K(X_i, X_j) \gamma_i \gamma_j. \quad (66)$$

This is helpful when we know the kernel of our space $\mathcal{H}$. See, e.g., Novak et al. (2018) for the kernels associated with certain Sobolev spaces and techniques for calculating kernels more generally, and Hirshberg et al. (2019, Section 2.3) and references therein for additional detail on RKHSes more generally.

Balancing weights for Sobolev space models and RKHS models more generally are discussed in Hazlett (2020), Hirshberg et al. (2019), Kallus (2020), Singh et al. (2020), and Wong and Chan (2018).

D Augmentation and strong notions of convergence

In Section 6.2, we discussed the impact of augmentation on imbalance by focusing on mean-square consistency of $\hat{\mu}$ and the equicontinuity of the imbalance process $f \rightarrow n^{-1} \sum_{i=1}^n f(X_i) - W_i \gamma^{\text{ipw}}(X_i) f(X_i)$.

An argument that appeared earlier (e.g., in Athey et al., 2018; Kallus, 2020; Wong & Chan, 2018) relied on the assumption that $\hat{\mu}$ was consistent in a stronger sense: instead of requiring the mean squared error $\|\delta\mu\|_{L_2(P)}$ to vanish, it required the gauge of the error $\|\delta\mu\|_{\mathcal{M}}$ to do so. In other words, it required that

²⁰See Fang et al., 2021 for a thorough discussion of total variation. One useful interpretation is that $\|f\|_{TV}$ is the vertical distance a pen travels when drawing the function f : for differentiable functions, it is $\int |f'(x)| dx$ and the seminorm associated with the model is $\sum_j |f'_j(x)| dx$.

$\delta\mu$ be in a vanishingly small scaled-down version of $\mathcal{M}$, $\mathcal{M}_n = r_n \mathcal{M}$ for $r_n \rightarrow 0$; in the case of the model $\mathcal{M} = \{\beta \cdot x : \|\beta\|_1 \leq 1\}$ (35), this means that we must use an estimate of the form $\hat{\mu}_1(x) = \hat{\beta} \cdot \phi(x)$, and the coefficients must be consistent in the ℓ_1 norm, as $\|\hat{\mu}_1 - \mu_1\|_{\mathcal{M}} = \|\hat{\beta} - \beta\|_1$. When we have consistency in this sense, $\mathcal{M}_n$ is a valid model for $\delta\mu$, and the maximal imbalance $\text{imbalance}_{\mathcal{M}_n}(\hat{\gamma}) = r_n \text{imbalance}_{\mathcal{M}}(\hat{\gamma})$ is negligible if $\text{imbalance}_{\mathcal{M}}(\hat{\gamma})$ is $O_p(n^{-1/2})$; if instead $\text{imbalance}_{\mathcal{M}}(\hat{\gamma})$ is $O_p(\sqrt{\log(p)/n})$ as discussed above for this model, it suffices to have gauge (e.g., ℓ_1) consistency with an $o_p(\sqrt{1/\log(p)})$ rate. While this argument is perhaps easier to internalize, it is often preferable to use the one above, as this stronger form of consistency can distract us with thinking about extraneous aspects of the data. For example, multicollinearity of the basis functions ϕ_j has no effect on our ability to estimate the conditional mean function $\mu_1(x) = \beta \cdot \phi(x)$, but it can make it impossible to accurately estimate its *coefficients* β (see, e.g., Chatterjee, 2013).

E Rounding

E.1 Rounding and Imbalance

In this section, we will prove the rounding error bound from Section 8. We will let $P_{\mathcal{X}_I} \mathcal{F}$ denote the set of *coordinate projections* of the functions in $\mathcal{F}$ on the indices $i_1 \dots i_{|I|}$ in I , i.e., $P_{\mathcal{X}_I} \mathcal{F} = \{[f(X_{i_1}) \dots f(X_{i_{|I|}})] : f \in \mathcal{F}\}$, and $P_{\mathcal{X}}$ the same for the whole set of indices $I = \{1 \dots n\}$. And we define the *linear discrepancy* and *hereditary discrepancy* of a set of vectors $T \subset \mathbb{R}^n$ as follows.

$$\begin{aligned} \text{lindisc}(T) &= \max_{q \in [0,1]^n} \min_{r \in \{0,1\}^n} \max_{t \in T} \left| \sum_{i=1}^n (q_i - r_i) t_i \right| \\ \text{hdisc}(T) &= \max_{I \subset \{1 \dots n\}} \min_{\epsilon \in \{-1,1\}^n} \max_{t \in T} \left| \sum_{i \in I} \epsilon_i t_i \right| \end{aligned} \tag{67}$$

Linear discrepancy bounds the degree to which balance in the vectors in T is impacted when we optimally round a vector of weights $q \in [0,1]^n$ into binary weights $r \in \{0,1\}^n$. Hereditary discrepancy bounds the imbalance we get in those vectors when we optimally color subsets of our units ± 1 , i.e. when we split subsets of our units into two groups and compare the averages of the vectors in T . Lovász et al. (1986, Corollary 1) shows that rounding is no harder than this kind of coloring, i.e. that

$$\text{lindisc}(T) \leq \text{hdisc}(T). \tag{68}$$

The formal proof we will leave to Lovász et al. (1986), but for the sake of offering a complete argument, we will sketch one. The idea is to think of a rounding process that works recursively, term-by-term and ‘from back to front’, on the binary expansions $q_i = \sum_{j=0}^{\infty} 2^{-j} q_{ij}$ of the elements $q_i \in [0,1]$ of the vector we are rounding, and use hereditary discrepancy bound the changes induced by our rounding process at each step.

Proof Sketch for Equation 68. Having approximated the elements q_i to arbitrary precision using an M -digit expansion $q_i^M = \sum_{j=0}^M q_{ij} 2^{-j}$, we’ll first show that we can find $M-1$ digit approximations q_i^{M-1} for which $\max_{t \in T} \sum_{i=1}^n (q_i^{M-1} - q_i^M) t_i$ is small, then similarly find $M-2$ digit approximations to q_i^{M-1} , etc, until we have elements q_i^0 in $\{0,1\}$. To approximate the k -digit expansions by $k-1$ -digit ones, we’ll find the set of coordinates I_k where its last digit q_{ik}^k is one and add $\pm 2^{-k}$. Doing this optimally, the change in imbalance $\max_{f \in \mathcal{F}} \sum_{i=1}^n (q_i^{M-1} - q_i^M) t_i$ is at most $2^{-k} \text{hdisc}(T)$. And we can bound the magnitude of the total change in imbalance over our k rounding steps by summing our bounds at each level, giving the bound $\sum_{k=1}^M 2^{-k} \text{hdisc}(T) \leq \sum_{k=1}^{\infty} 2^{-k} \text{hdisc}(T) = \text{hdisc}(T)$. $\square$

What this tells us is that we can bound our rounding-induced change in imbalance in Equation 58 by the (appropriately scaled) hereditary discrepancy for the vector of coordinate projections of the functions in $\mathcal{F}$, i.e., by $n^{-1} \text{hdisc}(P_{\mathcal{X}} \mathcal{F})$. To show that this $o_p(n^{-1/2})$, it suffices to show that $\text{hdisc}(P_{\mathcal{X}} \mathcal{F}) = o_p(n^{1/2})$. We will do this for classes $\mathcal{F}$ of bounded functions with bounded uniform entropy integral, first proving a finite sample bound then showing that bound is $o_p(n^{1/2})$.

Lemma E.1. Let $\mathcal{F}$ be a class of functions that is star-shaped around zero with $\max_{f \in \mathcal{F}} \|f\|_\infty \leq b$ and let $r > 0$ satisfy the Rademacher complexity fixed point inequality $2b^3 R_n(\{f \in \mathcal{F} : Pf^2 \leq r^2\}) \leq r^2$. With probability $1 - \exp(-nr^2/8b^2)$, for every $I \subseteq \{1 \dots n\}$ and $s \geq 0$,

$$\begin{aligned} \inf_{\epsilon \in \{-1,1\}^n} \frac{1}{\sqrt{n}} \max_{f \in \mathcal{F}} \left| \sum_{i \in I} \epsilon_i f(X_i) \right| &\leq c_1 b \left\{ s \sqrt{\frac{|I|}{n}} \exp\left(-c_2 2^{2^{s+1}}\right) + n^{-1/2} 2^{2^{s+1}} 2^{(s+1)/2} \right\} \\ &\quad + c_3 \int_0^{2\sqrt{\frac{\log(|I|)}{\log(4/3)}} \max(\delta, 2r)} \sup_Q \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(Q)}, u\|F\|_{L_2(Q)})} du \\ &\quad \text{for } \delta \text{ satisfying } \log_2 \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P)}, \delta) \leq 2^{s+1}. \end{aligned}$$

Here F is any function satisfying $F(x) \geq |f(x)|$ for all $f \in \mathcal{F}$, $\mathcal{N}(\mathcal{F}, \|\cdot\|, \varepsilon)$ is $\mathcal{F}$'s $\|\cdot\|$ ε -covering number, $R_n(\mathcal{G})$ is the Rademacher complexity of the class $\mathcal{G}$, $|I|$ denotes the cardinality of the set I , $c_1 \dots c_3$ are universal constants, and $\exp(x) = \exp(x)/\{1 - \exp(x)\}$.

Because this bound is increasing in $|I|$, replacing $|I|$ with n throughout gives a single bound that holds for all I , i.e. a bound on hereditary discrepancy. We summarize its asymptotic implications as follows.

Corollary E.2. If $\mathcal{F}$ is a class of bounded functions with bounded uniform entropy integral that is star-shaped around zero, $\text{hdisc}(\mathcal{P}_{\mathcal{X}} \mathcal{F}) = o_p(n^{1/2})$.

These results are similar to discrepancy bounds in Mendelson (2011, Theorems 5.1 and A) for *subgaussian classes* $\mathcal{F}$, i.e. classes of functions on which the $L_2(P)$ and subgaussian norms are equivalent. Our proofs are essentially simplified versions of those results, as some of the complexity in Mendelson (2011, Theorem 5.1) is in its comparison of empirical and population two-norms, and that is substantially simplified by the uniformity of the bounds we get when working with classes with bounded uniform entropy integral.

Lemma E.3. Let $\mathcal{F}$ be a class of functions that is star-shaped around zero and with $\max_{f \in \mathcal{F}} \|f\|_\infty \leq b$ and let $r > 0$ satisfy the Rademacher complexity fixed point inequality $2b^3 R_n(\{f \in \mathcal{F} : P_n f^2 \leq r^2\}) \leq r^2$. With probability $1 - \exp(-nr^2/8b^2)$, all functions $f \in \mathcal{F}$ satisfy $\|f\|_{L_2(P_n)} \leq 2 \max(\|f\|_{L_2(P)}, 2r)$.

Proof of Lemma E.1. To show that the bound from Lemma E.1 is satisfied, we will show that there exists a sequence $\epsilon \in \{-1, 1\}^n$ that satisfies the bound. We will do this iteratively, constructing ϵ as a sum $\sum_{j' \leq j_0} \eta^{j'}$ of ‘partial assignments’ $\eta^j \in \{-1, 0, 1\}^n$. At each step j , we update only the ‘unassigned’ indices, i.e. we find a vector η^j with the property that $\eta_i^j = 0$ unless $\eta_i^{j'} = 0$ for all $j' < j$. As a result, the sum $\sum_{j' \leq j} \eta^{j'}$ is itself a partial assignment for all j . And we will repeat this process until we have no zeros left, i.e., until $\sum_{j' \leq j} \eta^{j'} \in \{-1, 1\}^n$; that j will be j_0 . Our approach, following Mendelson (2011) uses the entropy method from discrepancy theory — indirectly through Mendelson (2011, Corollary 3.4) — in combination with chaining.

Notation. Throughout, we’ll write $c_1, c_2, \dots$ to denote universal constants.²¹ We’ll write I for a subset of the indices $\{1 \dots n\}$ and $|I|$ for its cardinality. In our iterative process, I will be the set of indices for which we have not yet assigned η_i to ± 1 . $\pi_0, \pi_1, \dots$ will be an admissible sequence of maps from $\mathcal{F}$ to itself, i.e. a sequence of functions for which the image $\pi_s(T)$ is a discrete set taking on at most 2^{2^s} values that are nested in the sense that $\pi_s(f) = \pi_s(g)$ implies $\pi_{s-1}(f) = \pi_{s-1}(g)$. We will choose these maps later, but we can think of $\pi_1(f), \pi_2(f), \dots$ as a sequence of approximations to f at increasing ‘resolution’ s , where we say we are at resolution s if we allow our approximations to take on 2^{2^s} values.

We will keep track of the unassigned indices with a set $I^j = \{i \in I : \sum_{j' < j} \epsilon^{j'} = 0\}$. We start with $I^0 = I$ and proceed inductively. Suppose we are $j \geq 0$ steps into this process and our set of unassigned indices is I^j . Mendelson (2011, Corollary 3.4) says that there exists a vector $\eta^j \in \{-1, 0, 1\}^n$ with (i) $\eta_i^j = 0$

²¹Constants will be renamed for use the lemma’s statement. c_1 here does not correspond to c_1 there, etc.

for $i \notin I^j$, (ii) η_{I^j} having between $|I^j|/4$ and $3|I^j|/4$ zeros, and (iii) for all $t \in T$,

$$\left| \sum_{i \in I} \eta_i^j f(X_i) \right| \leq \sum_{s \geq 1} Q_s^j \|P_{\mathcal{X}_{I^j}} \{\pi_s(f) - \pi_{s-1}(f)\}\|_2 \quad \text{for} \quad Q_s^j = c_1 \begin{cases} \exp(-c_2 \sqrt{|I^j|}), & \text{if } s < s_j, \\ 1, & \text{if } s = s_j, \\ 2^{s/2}, & \text{if } s > s_j \end{cases} \quad (69)$$

and $s_j = \max \left\{ s : 2^{2^{s+1}} \leq c_3 |I^j| \right\}.$

Because the number of zeros $|I^j|$ at each step is at least $1/4$ and at most $3/4$ the number $|I^{j-1}|$ we had in the previous step, $|I^j| \in [(1/4)^j |I|, (3/4)^j |I|]$ and we will have a complete assignment $\epsilon = \sum_{j=0}^{j_0} \eta^j \in \{-1, 1\}^k$ for j_0 satisfying $(3/4)^{j_0} |I| \leq 1$, i.e. for $j_0 \leq \log(1/|I|)/\log(3/4) = \log(|I|)/\log(4/3)$. It follows from Equation 69 and the triangle inequality that this complete assignment ϵ satisfies this bound:

$$\begin{aligned} \max_{f \in \mathcal{F}} \left| \sum_{i \in I} \epsilon_i f(X_i) \right| &= \max_{f \in \mathcal{F}} \left| \sum_{i \in I} \left(\sum_{j=0}^{j_0} \eta_i^j \right) f(X_i) \right| \\ &\leq \max_{f \in \mathcal{F}} \sum_{j=0}^{j_0} \left| \sum_{i \in I} \eta_i^j f(X_i) \right| \\ &\leq \max_{f \in \mathcal{F}} \sum_{j=0}^{j_0} \sum_{s \geq 1} Q_s^j \|P_{\mathcal{X}_{I^j}} \{\pi_s(f) - \pi_{s-1}(f)\}\|_2 \end{aligned} \quad (70)$$

We will bound this by choosing some ‘critical resolution’ s_* and splitting our sum over s into two parts. We will break it into a high-resolution ($s > s_*$) part where we will use the bound $Q_s^j \leq 2^{s/2}$ and use the same chaining argument we would if $\epsilon_1 \dots \epsilon_k$ were chosen by flipping a coin, and a low-resolution ($s \leq s_*$) part where we will take advantage of Q_s^j often being substantially smaller. The parts will be separately bounded uniformly over $\mathcal{F}$.

$$\begin{aligned} \max_{f \in \mathcal{F}} \left| \sum_{i \in I} \epsilon_i f(X_i) \right| &\leq T_{\leq} + T_{>} \quad \text{for} \\ T_{\leq} &= \max_{f \in \mathcal{F}} \sum_{s \leq s_*} \sum_{j=0}^{j_0} Q_s^j \|P_{\mathcal{X}_{I^j}} \{\pi_s(f) - \pi_{s-1}(f)\}\|_2 \\ T_{>} &= \max_{f \in \mathcal{F}} \sum_{s > s_*} 2^{s/2} \sum_{j=0}^{j_0} \|P_{\mathcal{X}_{I^j}} \{\pi_s(f) - \pi_{s-1}(f)\}\|_2. \end{aligned}$$

The Low-Resolution Part.

On the low-resolution part, we will use the bound

$$\begin{aligned} \|P_{\mathcal{X}_{I^j}} \{\pi_s(f) - \pi_{s-1}(f)\}\|_2 &\leq \sqrt{|I_j|} \|\pi_s(f) - \pi_{s-1}(f)\|_{\infty} \\ &\leq \sqrt{|I_j|} \text{diameter}_{\infty}(\mathcal{F}) \quad \text{for} \quad \text{diameter}_{\infty}(\mathcal{F}) = \max_{g, h \in \mathcal{F}} \|g - h\|_{\infty}. \end{aligned}$$

Making this substitution, we can bound the low-resolution part of our discrepancy as follows.

$$T_{\leq} \leq \text{diameter}_{\infty}(\mathcal{F}) \sum_{s \leq s_*} \sum_{j=0}^{j_0} Q_s^j \sqrt{|I_j|}.$$

What remains is to bound this sum. To simplify our handling of the ‘resolution weights’ Q_s^j , we will group terms where s_j takes on the same value and therefore the three cases of Q_s^j occur at the same values of s , defining $U_{\ell} = \{j : s_j = \ell\}$. In these terms, we can rewrite our bound as follows, then bound the values of

Q_s^j uniformly over U_ℓ .

$$\begin{aligned}
\sum_{s \leq s_*} \sum_{j=0}^{j_0} Q_s^j \sqrt{|I_j|} &= \sum_{s=1}^{s_*} \sum_{\ell=1}^{s_n} \sum_{j \in U_\ell} Q_s^j \sqrt{|I_j|} \\
&= \sum_{s=1}^{s_*} \sum_{\ell=1}^{s_n} \bar{Q}_s^\ell \sum_{j \in U_\ell} \sqrt{|I_j|} \\
\text{where } \bar{Q}_s^\ell &= c_1 \begin{cases} \exp(-c_2 c_3^{-1/2} 2^{2^\ell}), & \text{if } s < \ell, \\ 1, & \text{if } s = \ell, \\ 2^{s/2}, & \text{if } s > \ell \end{cases}
\end{aligned} \tag{71}$$

Here our bound for on Q_j^s for $s < \ell$ follows from the definition of s_j as the largest value of s (and therefore a value of s) for which $c_3 |I_j| \geq 2^{2^{s+1}}$:

$$\sqrt{|I_j|} \geq c_3^{-1/2} \sqrt{2^{2^{s_j+1}}} = c_3^{-1/2} 2^{2^{s_j}} \quad \text{and consequently} \quad \exp(-c_2 \sqrt{|I_j|}) = Q_s^j \leq \bar{Q}_s^\ell = \exp(-c_2 c_3^{-1/2} 2^{2^\ell}).$$

Because $|I_j|$ decays geometrically ($|I_{j+1}| \leq 3/4 |I_j|$ and therefore $|I_{j+k}| \leq (3/4)^k |I_j|$), we can bound the sum of $\sqrt{|I_j|}$ over U_ℓ by a multiple of its largest term. Letting j_ℓ be smallest elements of U_ℓ , we have

$$\begin{aligned}
\sum_{j \in U_\ell} |I_j|^{1/2} &= \sum_{k=0}^{|U_\ell|-1} |I_{j_\ell+k}|^{1/2} \\
&\leq \sum_{k=0}^{|U_\ell|-1} \sqrt{3/4}^k \sqrt{|I_{j_\ell}|} \\
&\leq \sqrt{|I_{j_\ell}|} \sum_{k=0}^{\infty} \sqrt{3/4}^k = \sqrt{|I_{j_\ell}|} \frac{1}{1 - \sqrt{3/4}}
\end{aligned}$$

Furthermore, as $s_j = \ell$ if and only if $c_3^{-1} 2^{2^{\ell+1}} \leq |I_j| < c_3^{-1} 2^{2^{\ell+2}}$, we have the bound $\sqrt{|I_{j_\ell}|} \leq \min(c_3^{-1/2} 2^{2^{\ell+1}}, \sqrt{n|I|})$. It follows that

$$\begin{aligned}
T_\leq &\leq \text{diameter}_\infty(\mathcal{F}) \sum_{s=1}^{s_*} \sum_{\ell=1}^{s_k} 2^{2^{\ell+1}} \bar{Q}_s^\ell \\
&\leq \frac{c_1 \text{diameter}_\infty(\mathcal{F})}{1 - \sqrt{3/4}} \sum_{s=1}^{s_*} \sum_{\ell=1}^{s|I|} \begin{cases} \sqrt{|I|} \exp(-c_2 c_3^{-1/2} 2^{2^\ell}) & \text{for } \ell > s_* \\ c_3^{-1/2} 2^{2^{\ell+1}} 2^{s/2} & \text{for } \ell \leq s_* \end{cases}
\end{aligned}$$

We split this into a sum over $\ell > s_*$ and one over $\ell \leq s_*$ and bound the sums as follows. In the first, because the summand does not depend on s , we get s_* copies of the sum over ℓ .

$$\begin{aligned}
\sum_{s=1}^{s_*} \sum_{\ell=s_*+1}^{s|I|} \sqrt{|I|} \exp(-c_2 c_3^{-1/2} 2^{2^\ell}) &= \sqrt{|I|} s_* \sum_{\ell=s_*+1}^{s|I|} \exp(-c_2 c_3^{-1/2} 2^{2^\ell}) \\
&\leq \sqrt{|I|} s_* \frac{\exp(-c_2 c_3^{-1/2} 2^{2^{s_*+1}})}{1 - \exp(-c_2 c_3^{-1/2} 2^{2^{s_*+1}})}
\end{aligned}$$

Here the last bound follows by bounding the sum over ℓ by the sum of a geometric series, observing that the

rate of decrease is at least the sum's first term:

$$\begin{aligned}\frac{\exp(-c_2 c_3^{-1/2} 2^{2^{x+1}})}{\exp(-c_2 c_3^{-1/2} 2^{2^x})} &= \exp\left\{-c_2 c_3^{-1/2} (2^{2^{x+1}} - 2^{2^x})\right\} \\ &= \exp\left\{-c_2 c_3^{-1/2} 2^{2^x} (2^{2^x} - 1)\right\} \\ &\leq \exp(-c_2 c_3^{-1/2} 2^{2^x}).\end{aligned}$$

In the second, we can use the same geometric series approach to bound our sums over s and then over ℓ .

$$\begin{aligned}\sum_{s=1}^{s_*} \sum_{\ell=0}^{s_*} 2^{s/2} 2^{2^{\ell+1}} &= \sum_{\ell=0}^{s_*} 2^{2^{\ell+1}} 2^{s_*/2} \sum_{k=0}^{s_*-1} 2^{-k/2} \\ &\leq \sum_{\ell=0}^{s_*} 2^{2^{\ell+1}} 2^{s_*/2} \sum_{k=0}^{\infty} \sqrt{2}^{-k} \\ &= \sum_{\ell=0}^{s_*} 2^{2^{\ell+1}} 2^{s_*/2} / (1 - \sqrt{1/2}) \\ &\leq 2^{2^{s_*+1}} 2^{s_*/2} \cdot 2 / (1 - \sqrt{1/2}).\end{aligned}$$

Here we've used the following bound on the rate of decrease of our series 2^{2^x} as we count down from $\ell = s_*$ to $\ell = 0$ to justify replacing the sum with two times its largest term.

$$\frac{2^{2^x}}{2^{2^{x+1}}} = 2^{2^x(1-2)} = 2^{-2^x} \leq 1/2 \quad \text{for } x \geq 0.$$

Summarizing, we have the following bound, in which $\text{expo}(x) = e^x/(1 - e^x)$

$$\begin{aligned}T_{\leq} &\leq \frac{c_1 \text{diameter}_{\infty}(\mathcal{F})}{1 - \sqrt{3/4}} \left\{ \sqrt{|I|} s_* \text{expo}\left(-c_2 c_3^{-1/2} 2^{2^{s_*+1}}\right) + c_5 2^{2^{s_*+1}} 2^{(s_*+1)/2} \right\} \\ \text{for } c_5 &= c_3^{-1/2} \cdot 2 / (1 - \sqrt{1/2}).\end{aligned}$$

The High-Resolution Part. Using the Cauchy-Schwarz inequality, we can bound the sum of two-norms that appears in (70) by a weighted two-norm, which we will interpret as a multiple of a weighted empirical norm $\|\cdot\|_{L_2^w(I)}$ with weights proportional to the number of rounds each unit went without being assigned.

$$\begin{aligned}\sum_{j=0}^{j_0} \|P_{\mathcal{X}_{I^j}} g\|_2 &\leq \sqrt{j_0 + 1} \sqrt{\sum_{j=0}^{j_0} \|P_{\mathcal{X}_{I^j}} g\|_2^2} \\ &= \sqrt{\sum_{i \in I} \tilde{w}_i g(X_i)^2} \quad \text{for } \tilde{w}_i = \sum_{j=0}^{j_0} \mathbf{1}(i \in I^j) \\ &= \sqrt{nW} \sqrt{\frac{1}{|I|} \sum_{i=1}^n w_i g(X_i)^2} \quad \text{for } W = \frac{1}{|I|} \sum_{i=1}^n \tilde{w}_i \quad \text{for } w_i = \frac{\tilde{w}_i}{W}.\end{aligned}$$

Here W is no larger than 4 (and at least 1). This follows from the geometric decrease in the number of unassigned units.

$$W = \frac{1}{|I|} \sum_{j=0}^{j_0} |I_j| \leq \sum_{j=0}^{j_0} (3/4)^j \leq \sum_{j=0}^{\infty} (3/4)^j = 1/(1 - 3/4) = 4.$$

It follows that each weight is between $1/4$ and $j_0/4 \leq 4 \log(n)/\log(4/3)$ and therefore that this norm is equivalent to the unweighted empirical norm $\|g\|_{L_2(I)}$ up to log factors. In particular $1/2 \leq \|g\|_w / \|g\|_{L_2(I)} \leq$

$\sqrt{\log(|I|)/\log(4/3)}$. Thus,

$$T_{>} \leq 2\sqrt{|I|} \max_{f \in \mathcal{F}} \sum_{s=s_*+1}^{\infty} 2^{s/2} \|\pi_s(f) - \pi_{s-1}(f)\|_{L_2(w)} \quad \text{where} \quad \frac{\|g\|_{L_2^w(I)}}{\|g\|_{L_2(I)}} \in \left[\frac{1}{2}, \sqrt{\frac{\log(|I|)}{\log(4/3)}} \right]$$

We will bound this in terms of the *entropy numbers*

$$e_s(\mathcal{F}, \|\cdot\|_{L_2^w(I)}) := \inf_{\mathcal{F}_s \subseteq \mathcal{F}, |\mathcal{F}_s| \leq 2^{2s}} \sup_{f \in \mathcal{F}} \inf_{f' \in \mathcal{F}_s} \|f - f'\|_{L_2^w(I)}.$$

By choosing $\pi_s(f)$ to be the closest point to f in the associated optimal cover $\mathcal{F}_s$, the bound $\|\pi_s(f) - \pi_{s-1}(f)\|_{L_2^w(I)} \leq e_s(\mathcal{F}, \|\cdot\|_{L_2^w(I)})$ holds uniformly over $f \in \mathcal{F}$. In those terms,

$$T_{>} \leq 2\sqrt{n} \sum_{s=s_*+1}^{\infty} 2^{s/2} e_s(\mathcal{F}, \|\cdot\|_{L_2^w(I)}).$$

This is a discrete version of Dudley's entropy bound, from which it is a well-known exercise to derive the more common integral version via integral approximation (e.g. Talagrand, 2021, Remarks after Proposition 2.5.1). And we can bound this in terms of a uniform entropy integral, first making the change of variables $u = t/\|F\|_{L_2^w(I)}$ where F is an envelope function for our class $\mathcal{F}$, then maximizing over any set of probability measures Q containing $L_2^w(I)$ (see e.g. the proof of van der Vaart & Wellner, 1996, Theorem 2.5.2).

$$\begin{aligned} T_{>} &\leq c_6 \sqrt{|I|} \int_0^{e_{s_*+1}(\mathcal{F}, \|\cdot\|_{L_2^w(I)})} \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2^w(I)}, t)} dt \\ &= c_6 \sqrt{|I|} \|F\|_{L_2^w(I)} \int_0^{e_{s_*+1}(\mathcal{F}, \|\cdot\|_{L_2^w(I)})/\|F\|_{L_2^w(I)}} \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2^w(I)}, u\|F\|_{L_2^w(I)})} du \\ &\leq c_6 \sqrt{|I|} \int_0^{e_{s_*+1}(\mathcal{F}, \|\cdot\|_{L_2^w(I)})/\|F\|_{L_2^w(I)}} \sup_Q \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(Q)}, u\|F\|_{L_2(Q)})} du. \end{aligned} \quad (72)$$

We conclude the section by replacing our integral's upper limit with a deterministic bound. Because bounds on entropy numbers follow from bounds on norms, i.e. $e_s(\mathcal{F}, \|\cdot\|_a) \leq \min\{ce_s(\mathcal{F}, \|\cdot\|_b), d\}$ if $\|f\|_a \leq \min\{c\|f\|_b, d\}$ for all $f \in \mathcal{F}$, it suffices to find a deterministic upper bound on $\|\cdot\|_{L_2^w(I)}$. To do this, we combine our aforementioned bound $\|\cdot\|_{L_2^w(I)} \leq \sqrt{\log(|I|)/\log(4/3)} \|\cdot\|_{L_2(I)}$ with (i) the bound $\|\cdot\|_{L_2(I)} = \frac{1}{\sqrt{|I|}} \|\mathcal{P}_{\mathcal{X}} \cdot\|_2 \leq \frac{1}{\sqrt{|I|}} \|\mathcal{P}_{\mathcal{X}} \cdot\|_2 = \sqrt{n/|I|} \|\cdot\|_{L_2(\{1 \dots n\})}$ and (ii) Lemma E.3's bound on the ratio $\|\cdot\|_{L_2(\{1 \dots n\})}/\|\cdot\|_{L_2(P)}$. Together, these imply that on an event of probability $1 - \exp(-nr^2/8b^2)$,

$$\begin{aligned} \|f\|_{L_2^w(I)} &\leq 2\sqrt{\log(|I|)/\log(4/3)} \max\{\|f\|_{L_2(P)}, 2r\} \text{ for all } f \in \mathcal{F} \text{ and therefore} \\ e_s(\mathcal{F}, \|\cdot\|_{L_2^w(I)}) &\leq 2\sqrt{\log(|I|)/\log(4/3)} \max\{e_s(\mathcal{F}, \|\cdot\|_{L_2(P)}), 2r\} \text{ for all } s. \end{aligned}$$

Making this substitution in our bound above for $s = s_* + 1$ implies that

$$\begin{aligned} T_{>} &\leq c_6 \sqrt{n} \int_0^{2\sqrt{\frac{\log(|I|)}{\log(4/3)}} \max(\delta, 2r)} \sup_Q \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(Q)}, u\|F\|_{L_2(Q)})} du \\ &\text{for } \delta \geq e_{s_*+1}(\mathcal{F}, \|\cdot\|_{L_2(P)}) \text{ i.e. } \delta \text{ satisfying } \log_2 \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P)}, \delta) \leq 2^{s_*+1}. \end{aligned}$$

Summing our bounds on $T_{\leq}$ and $T_{>}$ and dividing by $\sqrt{n}$ gives us the claimed bound. As our only probabilistic bound is on a quantity that does not involve I , our bound holds simultaneously for all $I \subseteq \{1 \dots n\}$ on the aforementioned event.

To simplify our statement of the bound, we replace the infinity norm diameter with a multiple (2) of our infinity norm bound b , rename s_* to s , and simplify and rename our universal constants. $\square$

Proof of Corollary E.2. Substituting the upper bound n for $|I|$ in the discrepancy bounds Lemma E.1 gives

us this finite sample bound on hereditary discrepancy. Letting F be any envelope function for $\mathcal{F}$, in terms of the uniform entropy integral

$$J(x, \mathcal{F}, F) := \int_0^x \sup_Q \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(Q)}, u \|F\|_{L_2(Q)})} du,$$

we have the bound

$$\begin{aligned} \frac{1}{\sqrt{n}} \text{hdisc}(\mathcal{P}_{\mathcal{X}} \mathcal{F}) &\leq c_1 b \left\{ s \exp(-c_2 2^{2^{s+1}}) + n^{-1/2} 2^{2^{s+1}} 2^{(s+1)/2} \right\} \\ &\quad + c_3 J \left(2 \sqrt{\frac{\log(n)}{\log(4/3)}} \max(\delta, 2r), \mathcal{F}, F \right) \\ &\quad \text{for } \delta \text{ satisfying } \log_2 \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P)}, \delta) \leq 2^{s+1} \text{ and } b \geq \max_{f \in \mathcal{F}} \|f\|_{\infty}. \end{aligned}$$

This bound holds for all s with probability $1 - \exp(-nr^2/8b^2)$ for any r satisfying the inequality $r^2 \geq 2b^3 R_n(\{f \in \mathcal{F} : \|f\|_{L_2(P)} \leq r\})$. To establish that $\text{hdisc}(\mathcal{P}_{\mathcal{X}} \mathcal{F})$ is $o_p(n^{-1/2})$ we will show that we can choose s and r as functions of n so that bound above is $o(1)$ and holds with probability tending to one.

We can take r to be on the order of $n^{-1/4}$. A standard chaining argument van der Vaart and Wellner (see e.g. 1996, Theorem 2.1.41)), gives us a bound on this Rademacher complexity in terms of our uniform entropy integral: for some universal constant c ,

$$R_n(\{f \in \mathcal{F} : Pf^2 \leq r^2\}) \leq cn^{-1/2} \|F\|_{L_2(P)} J(1, \mathcal{F}, F).$$

Thus, our inequality is satisfied for $r^2 = cn^{-1/2} \|F\|_{L_2(P)} J(1, \mathcal{F}, F)/(2b^3)$, i.e. for $r = c_{\mathcal{F}} n^{-1/4}$ for some constant depending on our class $\mathcal{F}$.

To choose s , observe that because our uniform entropy integral is bounded, it must be the case that it vanishes when we integrate over arbitrarily small neighborhoods of zero, i.e. that $J(\delta', \mathcal{F}, F) \rightarrow 0$ as $\delta' \rightarrow 0$. Thus, the integral term in our bound tends to zero if its upper limit $\delta' = \sqrt{\frac{\log(n)}{\log(4/3)}} \max(\delta, 2r) \rightarrow 0$ which, given our characterization of r , happens if $\sqrt{\log(n)} \delta \rightarrow 0$. Considering the other terms as well, our bound will tend to zero if the following conditions on s are satisfied for some sequence s_n .

1. $s_n \exp(-c_2 2^{2^{s_n+1}})$ is $o(1)$.
2. $2^{2^{s_n+1}} 2^{(s_n+1)/2} = o(\sqrt{n})$.
3. There exists $\delta_n = \delta(s_n)$ satisfying $\sqrt{\log(n)} \delta_n \rightarrow 0$ and $\log_2 \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P)}, \delta) \leq 2^{s_n+1}$.

Let ρ be any number in $(0, 1/2)$. We will choose s_n to be the integer s for which $2^{2^{s+1}} \leq n^{\rho} < 2^{2^{s+2}}$. With this choice, condition (1) holds because it does for any sequence s_n tending to ∞ . And condition (2) holds by almost by construction, as $2^{(s_n+1)/2}/n^{1/2-\rho} \rightarrow 0$ as $s \rightarrow \infty$. This leaves condition (3).

Because $\mathcal{F}$ has bounded uniform entropy integral, we know that the integrand, $\sup_Q \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(Q)}, u \|F\|_{L_2(Q)})}$, is $o(1/u)$ as $u \rightarrow 0$; if it were not, our integral would diverge. This means that for any constant c , we can choose δ small enough that $\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P)}, \delta) < c/\delta^2$ and therefore $\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L_2(P)}, \delta) < 2^{s_n+1}$ if $c < \delta^2 2^{s_n+1}$. Having chosen s_n so that $n^{\rho/2} \leq 2^{s_n+1}$, it follows that $\delta_n = c^{1/2} n^{-\rho'/4}$ for $\rho' < \rho$ works and, in addition, satisfies $\sqrt{\log(n)} \delta_n \rightarrow 0$, so condition (3) holds as well. $\square$

Proof of Lemma E.3. We will use Bartlett et al. (2005, Theorem 3.3) which, applied to the set of functions $\mathcal{F}^2 = \{f^2 : f \in \mathcal{F}\}$ with $K \downarrow 1$ and $B = b^2$, gives the probability $1 - e^{-x}$ bound

$$\mathbb{P}_n f^2 \leq 2 \mathbb{P} f^2 + 6b^{-2} s^2 + 16b^2 x/n \quad \text{if} \quad s^2 \geq b^2 R_n(\{f^2 : f \in \mathcal{F} : \mathbb{P} f^4 \leq s^2\}).$$

We will use the boundedness of $\mathcal{F}$ to simplify our characterization of r . It implies that $f^4 \leq b^2 f^2$ and consequently that $\{f \in \mathcal{F} : \mathbb{P} f^4 \leq s^2\} \subseteq \{f \in \mathcal{F} : \mathbb{P} f^2 \leq b^2 s^2\}$ and therefore, because $\phi(x) = x^2$ has

Lipschitz constant $2b$ on $x \in [0, b]$, via the contraction principle for Rademacher processes (e.g. Bartlett et al., 2005, Theorem A.6), that

$$\begin{aligned} b^2 R_n(\{f^2 : f \in \mathcal{F}, P_n f^4 \leq s^2\}) &\leq b^2 R_n(\{f^2 : f \in \mathcal{F}, P_n f^2 \leq b^2 s^2\}) \\ &\leq b^2 \cdot 2b R_n(\{f \in \mathcal{F}, P_n f^2 \leq b^2 s^2\}). \end{aligned}$$

Thus, letting $r = bs$, we have

$$P_n f^2 \leq 2P f^2 + 6r^2 + 16b^2 x/n \quad \text{if} \quad 2b^3 R_n(\{f : P_n f^2 \leq r^2\}) \leq r^2.$$

Taking $x = nr^2/8b^2$ gives us the probability $1 - \exp(-nr^2/8b^2)$ bound

$$P_n f^2 \leq 2P f^2 + 8r^2 \leq \max(4P f^2, 16b^{-2}r^2)$$

and taking square roots gives the bound we claimed. $\square$

E.2 Rounding and Efficiency

Here we analyze the variance cost of rounding in more detail. Let $v_0(x) = \text{var}(Y_i \mid W_i = 0, X_i = x)$ denote the conditional variance among controls, and let $\hat{r}(x) = \hat{\theta}(x) - \lfloor \hat{\theta}(x) \rfloor$ denote the fractional part of the weight. Define the rounding error functional

$$R(f) = \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i \{\hat{\theta}(X_i) - \lfloor \hat{\theta} \rfloor(X_i)\} f(X_i).$$

The excess variance from rounding can be decomposed as follows.

$$\begin{aligned} N_1 \times (V^{\lfloor \hat{\theta} \rfloor} - V^{\hat{\theta}}) &= \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i \{\lfloor \hat{\theta} \rfloor_i - \hat{\theta}(X_i)\}^2 v_0(X_i) + 2R(\hat{\theta}v_0) \\ &= \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i \hat{r}(X_i) \{1 - \hat{r}(X_i)\} v_0(X_i) + R\{(1 - 2\hat{r})v_0\} + 2R(\hat{\theta}v_0). \end{aligned} \tag{73}$$

The first term is the variance from independent Bernoulli rounding. The remaining terms involve the rounding error functional $R(\cdot)$ applied to functions that may not be in our model $\mathcal{M}$.

Proposition E.4. Suppose $\mathcal{M}$ has bounded uniform entropy integral and rounding $_{\mathcal{M}, \hat{\theta}}(\lfloor \hat{\theta} \rfloor) = o_p(n^{-1/2})$. If $\|\hat{\theta}v_0\|_{\mathcal{M}} = O_p(n^{1/2})$ and $\|(1 - 2\hat{r})v_0\|_{\mathcal{M}} = O_p(n^{1/2})$, then $R(\hat{\theta}v_0) = o_p(1)$ and $R\{(1 - 2\hat{r})v_0\} = o_p(1)$.

Proof. By Proposition 8.1, the rounding error is $o_p(n^{-1/2})$ uniformly over the unit ball of $\mathcal{M}$. For f with $\|f\|_{\mathcal{M}} = O_p(n^{1/2})$, we can write $f = \|f\|_{\mathcal{M}} \cdot (f/\|f\|_{\mathcal{M}})$ where $f/\|f\|_{\mathcal{M}}$ is in the unit ball. Because the uniform bound holds for all elements of the unit ball simultaneously, it applies to the random element $f/\|f\|_{\mathcal{M}}$. Thus

$$R(f) = \|f\|_{\mathcal{M}} \cdot R(f/\|f\|_{\mathcal{M}}) = O_p(n^{1/2}) \cdot o_p(n^{-1/2}) = o_p(1).$$

$\square$

Like our analysis of the efficiency of weighting in Section 4, the conditions of Proposition E.4 do not require the functions $\hat{\theta}v_0$ and $(1 - 2\hat{r})v_0$ to be in $\mathcal{M}$ itself—only that they can be approximated by sequences $m_n \in \mathcal{M}$ with $\|m_n\|_{\mathcal{M}} = O_p(n^{1/2})$. For nonparametric models (Definition 1), this holds whenever these functions are square integrable.

For RKHS models, we can verify these conditions using the results of Hirshberg et al. (2019).

Corollary E.5. Let $\mathcal{M}$ be the unit ball of an RKHS $\mathcal{H}$ that is dense in $L^2(P)$, and let $\hat{\theta}$ be derived from the minimax linear weights of Hirshberg et al. (2019). If $\hat{\theta}v_0$ and $(1 - 2\hat{r})v_0$ are in $L^2(P)$, then $R(\hat{\theta}v_0) = o_p(1)$ and $R\{(1 - 2\hat{r})v_0\} = o_p(1)$.

Proof. The proof has two parts. First, we use results from Hirshberg et al. (2019) to establish that the rounding error $R(\cdot)$ is $o_p(n^{-1/2})$ uniformly over the unit ball of $\mathcal{H}$. Second, we show that this implies $R(f) = o_p(1)$ for any $f \in L^2(P)$.

Part 1: Rounding error over the model. By Hirshberg et al. (2019, Theorem 2), the minimax linear weights $\hat{\gamma}$ achieve $o_p(n^{-1/2})$ imbalance uniformly over $\mathcal{M}$. Combined with Proposition 8.1, which shows the rounding scheme (58) preserves this rate, we have

$$\sup_{m \in \mathcal{M}} |R(m)| = o_p(n^{-1/2}).$$

Part 2: Extension to $L^2(P)$. We now show $R(f) = o_p(1)$ for any $f \in L^2(P)$. The idea is to approximate f by elements of $\mathcal{H}$ whose norms grow slowly enough that we can apply Part 1.

Because $\mathcal{H}$ is dense in $L^2(P)$, there exists a sequence $g_1, g_2, \dots$ in $\mathcal{H}$ with $\|g_k - f\|_{L^2(P)} \rightarrow 0$. We construct a sequence $f_n \in \mathcal{H}$ with $\|f_n - f\|_{L^2(P)} \rightarrow 0$ and $\|f_n\|_{\mathcal{H}} = O(n^{1/2})$ by setting $k(n) = \max\{k : \|g_k\|_{\mathcal{H}} \leq n^{1/2}\}$ and $f_n = g_{k(n)}$. Because each g_k has finite RKHS norm, $k(n) \rightarrow \infty$ as $n \rightarrow \infty$, so $\|f_n - f\|_{L^2(P)} = \|g_{k(n)} - f\|_{L^2(P)} \rightarrow 0$.

Now we bound $R(f)$ by the triangle inequality:

$$|R(f)| \leq |R(f_n)| + |R(f - f_n)|.$$

For the first term, because $f_n/\|f_n\|_{\mathcal{H}}$ is in the unit ball of $\mathcal{H}$, Part 1 gives

$$|R(f_n)| = \|f_n\|_{\mathcal{H}} \cdot |R(f_n/\|f_n\|_{\mathcal{H}})| = O(n^{1/2}) \cdot o_p(n^{-1/2}) = o_p(1).$$

For the second term, we use the fact that $|\hat{\theta} - \lfloor \hat{\theta} \rfloor| \leq 1$:

$$|R(f - f_n)| \leq \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i |f(X_i) - f_n(X_i)|.$$

This is a weighted empirical average of $|f - f_n|$. By the law of large numbers, it converges in probability to $E[|f(X) - f_n(X)|]$, which is bounded by $\|f - f_n\|_{L^2(P)} \rightarrow 0$.

Combining these bounds, $R(f) = o_p(1)$. $\square$

The main term in (73) is an average of $\hat{r}(1 - \hat{r})v_0$. Combined with the $\hat{\theta}^2 v_0$ term from the variance of the unrounded estimator, the total variance is proportional to $\{\hat{\theta}^2 + \hat{r}(1 - \hat{r})\}v_0$. When weights are close to binary ($\hat{\theta} \approx \hat{r}$), this simplifies to $\hat{r}v_0$ —larger than $\hat{r}^2 v_0$ by a factor of $1/\hat{r}$, but we only need outcomes from $N_1 \approx nE[\hat{r}]$ individuals rather than all n controls.

Proof of Corollary 8.4. By Proposition 8.1, the rounding scheme (58) satisfies $\text{rounding}_{\mathcal{M}, \hat{\theta}}(\lfloor \hat{\theta} \rfloor) = o_p(n^{-1/2})$, so the triangle inequality gives $\text{imbalance}_{h, \mathcal{M}}(\lfloor \hat{\theta} \rfloor) \leq \text{imbalance}_{h, \mathcal{M}}(\hat{\theta}) + o_p(n^{-1/2}) = o_p(n^{-1/2})$. The matched estimator $\hat{\psi}^{\text{match}}$ is an IPW estimator with weights $\lfloor \hat{\theta} \rfloor$ in place of $\hat{\theta}$. We verify the conditions of Proposition 7.1' for these rounded weights.

The normalization condition $N_1^{-1} \sum_i \bar{W}_i \lfloor \hat{\theta} \rfloor_i = 1$ is preserved by the balance-preserving rounding scheme (58). For infinitesimality, observe that rounding changes each weight by at most one, so $\lfloor \hat{\theta} \rfloor_i^2 \leq 2\hat{\theta}_i^2 + 2$. In particular, $\max_i \bar{W}_i \lfloor \hat{\theta} \rfloor_i^2 \leq 2 \max_i \bar{W}_i \hat{\theta}_i^2 + 2$. For the denominator, because $\lfloor \hat{\theta} \rfloor_i$ are non-negative integers, $\lfloor \hat{\theta} \rfloor_i^2 \geq \lfloor \hat{\theta} \rfloor_i$ for all i , and so $n\Gamma_{\text{round}}^2 = nN_1^{-1} \sum_i \bar{W}_i \lfloor \hat{\theta} \rfloor_i^2 \geq nN_1^{-1} \sum_i \bar{W}_i \lfloor \hat{\theta} \rfloor_i = n$. Combining: $\max_i \bar{W}_i \lfloor \hat{\theta} \rfloor_i^2 / n\Gamma_{\text{round}}^2 \leq (2 \max_i \bar{W}_i \hat{\theta}_i^2 + 2)/n \rightarrow_p 0$, using infinitesimality of $\hat{\theta}$ and $n\Gamma_{\text{round}}^2 \geq n$.

For weighted L_2 consistency, use $\lfloor \hat{\theta} \rfloor_i^2 \leq 2\hat{\theta}_i^2 + 2$ to write

$$\sum_i \bar{W}_i \lfloor \hat{\theta} \rfloor_i^2 (\hat{\mu}_i - m_i)^2 \leq 2 \sum_i \bar{W}_i \hat{\theta}_i^2 (\hat{\mu}_i - m_i)^2 + 2 \sum_i \bar{W}_i (\hat{\mu}_i - m_i)^2.$$

The first term is $o_p(N_1\Gamma^2)$ by the weighted L_2 consistency of $\hat{\mu}$ with $\hat{\theta}$; the second is $o_p(N_1\Gamma^2)$ by L_2 consistency. The asymptotic linearity (61) and central limit theorem (62) then follow from Proposition 7.1'. $\square$

Proof of Corollary 8.5. Because $\mathcal{M}$ has bounded uniform entropy integral, it is Donsker, so Proposition 6.1 applies: the non-integer version of (63) achieves $o_p(n^{-1/2})$ imbalance. By Proposition 8.1, rounding these weights to integers via (58) incurs an additional imbalance of $o_p(n^{-1/2})$, so the rounded weights are feasible for (63) with budget B_n equal to the sum of these two rates (see below for explicit rate calculations). Feasibility with probability tending to one follows.

Let $[\hat{\theta}]^*$ denote the solution to (63). Because the integer program minimizes $N_1^{-1} \sum_i \bar{W}_i \theta_i^2$ over feasible integer solutions, and the rounded continuous solution is feasible, the optimal value satisfies $N_1^{-1} \sum_i \bar{W}_i ([\hat{\theta}]_i^*)^2 \leq N_1^{-1} \sum_i \bar{W}_i [\hat{\theta}]_i^2 = O_p(1)$. Moreover, because the functions in $\mathcal{M}$ are uniformly bounded, the continuous weights are bounded, so $\max_i \bar{W}_i [\hat{\theta}]_i = O_p(1)$; any integer solution with a component exceeding this bound would have larger objective than the rounded continuous solution and hence be suboptimal, so $\max_i \bar{W}_i [\hat{\theta}]_i^* = O_p(1)$. Since the weights $[\hat{\theta}]_i^*$ are non-negative integers summing to N_1 , we have $n\Gamma^{*2} = nN_1^{-1} \sum_i \bar{W}_i ([\hat{\theta}]_i^*)^2 \geq n$, and therefore $\max_i \bar{W}_i ([\hat{\theta}]_i^*)^2 / n\Gamma^{*2} = O_p(1/n) \rightarrow 0$: the infinitesimality condition of Corollary 8.4 holds.

The imbalance condition $\text{imbalance}_{h,\mathcal{M}}([\hat{\theta}]^*) \leq B_n = o_p(n^{-1/2})$ holds by the feasibility constraint. All remaining conditions of Corollary 8.4 are inherited from the hypotheses, so the asymptotic expansions (61) and (62) follow. $\square$

Choosing the budget B_n : explicit rates for Sobolev models. We illustrate with Sobolev models with smoothness $s > d/2$ in dimension d . The budget B_n is the sum of two rates: the imbalance rate from Hirshberg et al. (2019, Theorem 2) and the rounding error rate from Corollary E.2.

For the imbalance rate, the key quantity is the effective dimension $\sum_{j=1}^{\infty} \min(\mu_j, n^{-1})$, where μ_j are the eigenvalues of the kernel. For Sobolev spaces, these decay as $\mu_j \asymp j^{-2s/d}$ (see, e.g., Kühn et al., 2014). The effective dimension is therefore

$$\sum_{j=1}^{\infty} \min(\mu_j, n^{-1}) \asymp \sum_{j \leq n^{d/(2s)}} n^{-1} + \sum_{j > n^{d/(2s)}} j^{-2s/d} \asymp n^{d/(2s)-1}.$$

Substituting into Theorem 2 of Hirshberg et al. (2019) with $\kappa_m = 1$ and $\sigma^2 \asymp 1$ gives an imbalance rate of $n^{-1/2} \cdot n^{(d/(2s)-1)/2} = n^{d/(4s)-1}$.

For the rounding error rate, we use the covering number $\log \mathcal{N}(\mathcal{M}, \|\cdot\|_{L_2}, \epsilon) \asymp \epsilon^{-d/s}$ for the Sobolev unit ball (see, e.g., Kühn et al., 2015, Section 2.1). The entropy integral in Lemma E.1 then gives a hereditary discrepancy bound of order $n^{d/(4s)}$, and the rounding error is $n^{-1} \cdot n^{d/(4s)} = n^{d/(4s)-1}$.

Both rates match, giving $B_n \asymp n^{d/(4s)-1}$. This is $o(n^{-1/2})$ precisely when $s > d/2$, which is the standard condition for bounded uniform entropy integral.

Binary search in practice. In practice, one can find a valid budget by binary search without computing this rate explicitly. Start at $B_n = n^{-1/2}$ and halve repeatedly until the integer program (63) becomes infeasible, then back off one step. Because feasibility is monotone in B_n , this terminates in $O(\log n)$ steps and yields a budget within a factor of two of the tightest feasible value. When each optimization is costly—as it can be in large samples, integer programming being hard—the explicit rate $n^{d/(4s)-1}$ provides a better starting point than $n^{-1/2}$.

If you find yourself needing to *double* from $n^{-1/2}$ to achieve feasibility, rather than halve, that is a sign of trouble: the imbalance you can achieve is worse than $n^{-1/2}$, likely indicating poor overlap.

The probabilistic validity of this procedure follows from a union bound over the $O(\log n)$ candidate budgets. The imbalance bound from Hirshberg et al. (2019) has exponential concentration, so each candidate budget fails with probability at most $\exp(-cn)$ for some $c > 0$. The hereditary discrepancy bound requires only a norm equivalence that holds uniformly over all budgets, contributing a single event to the union. The total failure probability remains exponentially small.

Matching vs. IPW on a subsample. Here we prove the variance comparison claimed in Remark 6. Let $\mu_k = E[\theta(X)^k \mid W = 0]$ for $k = 1, 2$.

Proposition E.6. Suppose $e(X) < 1/2$ almost surely, $\nu(w, x) = v_w$ is constant in x for each $w \in \{0, 1\}$, and $N_1/N_0 \rightarrow 0$. Let $\hat{\psi}^{\text{match}}$ be the integer-SBW matching estimator from Corollary 8.5, and let $\hat{\psi}^{\text{IPW}}$ be an IPW estimator using N_1 randomly selected controls with weights $\theta(X_i)/\bar{\theta}_n$, where $\bar{\theta}_n = N_1^{-1} \sum_{i \in S_{\text{IPW}}} \theta(X_i)$. Then

$$N_1 V^{\text{match}} \rightarrow_p v_1 + v_0, \quad N_1 V^{\text{IPW}} \rightarrow_p v_1 + v_0 \frac{\mu_2}{\mu_1^2},$$

and

$$\frac{V^{\text{match}}}{V^{\text{IPW}}} \rightarrow_p \frac{\rho + 1}{\rho + \mu_2/\mu_1^2}, \quad \rho := \frac{v_1}{v_0}.$$

In particular, $V^{\text{match}} \leq V^{\text{IPW}}$, with equality if and only if $\theta(X)$ is $P(\cdot \mid W = 0)$ -a.s. constant.

Proof. Matching estimator. By Corollary 8.5, $\hat{\psi}^{\text{match}}$ is asymptotically linear with influence function $N_1^{-1}(W_i + (1 - W_i)[\hat{\theta}]_i^*)\{Y_i - m(W_i, X_i)\}$. Since $e(X) < 1/2$, the rounded weights lie in $\{0, 1\}$ with $|S_0| = N_1$ matched controls (by normalization), and the asymptotic variance is

$$N_1^2 V^{\text{match}} = \sum_{W_i=1} v_1 + \sum_{i \in S_0} v_0 = N_1(v_1 + v_0).$$

IPW subsample estimator. The IPW estimator on a random subsample of N_1 controls with normalized propensity weights is asymptotically linear with influence function $N_1^{-1}(W_i - (1 - W_i)(\theta(X_i)/\bar{\theta}_n))\mathbf{1}\{i \in S_{\text{IPW}}\}\{Y_i - m(W_i, X_i)\}$. The asymptotic variance is

$$N_1^2 V^{\text{IPW}} = \sum_{W_i=1} v_1 + \sum_{i \in S_{\text{IPW}}} \frac{\theta(X_i)^2}{\bar{\theta}_n^2} v_0.$$

Since the selected controls are i.i.d. from the control population, $N_1^{-1} \sum_{i \in S_{\text{IPW}}} \theta(X_i)^2 \rightarrow_p \mu_2$ and $\bar{\theta}_n \rightarrow_p \mu_1$ by the law of large numbers. Hence $N_1 V^{\text{IPW}} \rightarrow_p v_1 + v_0 \mu_2/\mu_1^2$.

Comparison. Dividing and applying Slutsky's theorem gives the stated ratio. Since $x \mapsto x^2$ is convex, Jensen's inequality gives $\mu_2 = E[\theta^2 \mid W = 0] \geq E[\theta \mid W = 0]^2 = \mu_1^2$, so $\mu_2/\mu_1^2 \geq 1 \geq 1$ and $\rho + \mu_2/\mu_1^2 \geq \rho + 1$, hence $V^{\text{match}} \leq V^{\text{IPW}}$. Equality holds iff $\mu_2 = \mu_1^2$, i.e., $\text{Var}(\theta \mid W = 0) = 0$. $\square$

For $e(X) \ll 1/2$, $\theta(X) \approx e(X)$ and $\mu_k \approx E[e(X)^k \mid W = 0]$, so $\mu_2/\mu_1^2 \approx 1 + \text{Var}(e \mid W = 0)/E[e \mid W = 0]^2$. With $\rho = 1$ (equal noise), the ratio simplifies to $1/\{1 + \text{Var}(e)/(2E[e]^2)\}$, the expression in Remark 6. The general formula shows the gain diminishes as $\rho \rightarrow \infty$ (treatment noise dominates) and strengthens as $\rho \rightarrow 0$.

Efficiency of integer-weight estimators. The variance formulas (56) and (73) characterize the cost of rounding in terms of the fractional parts $\hat{r}_i = \hat{\theta}_i - [\hat{\theta}_i]$. We now ask: is this cost intrinsic to integer weights, or an artifact of rounding? The answer is that for *regular* estimators—those balancing all L^2 functions—the variance formula is exact, not a bound. Any regular integer-weight estimator achieves the same asymptotic variance, regardless of how the weights are constructed.

Write $r(x) = \theta(x) - [\theta(x)]$ for the fractional part of the population propensity odds $\theta(x) = e(x)/\{1 - e(x)\}$.

Definition 2. A matching estimator with integer weights $\hat{\theta}_i \in \mathbb{Z}$ is *regular* if it achieves $o_p(1)$ imbalance in every $f \in L^2(P)$:

$$\frac{1}{N_1} \sum_{i=1}^n \bar{W}_i \hat{\theta}_i f(X_i) - \frac{1}{N_1} \sum_{i=1}^n W_i f(X_i) = o_p(1) \quad \text{for all } f \in L^2(P).$$

Proposition 8.2'. Suppose $v_0(x) = \text{var}[Y \mid W = 0, X = x]$ is bounded and $\theta v_0, (1 - 2r)v_0 \in L^2(P)$. Any

regular matching estimator satisfies

$$\text{var}(\hat{\psi}^{\text{match}}) \geq V^{\text{eff}} + \frac{1}{N_1^2} \sum_{i=1}^n \bar{W}_i r(X_i) \{1 - r(X_i)\} v_0(X_i) + o_p(N_1^{-1}). \quad (74)$$

The bound is sharp when each $\hat{\theta}_i \in \{\lfloor \theta(X_i) \rfloor, \lceil \theta(X_i) \rceil\}$.

Proof of Proposition 8.2'. The proof uses the $a^2 - b^2$ decomposition twice: first to separate the variance excess into a quadratic term and a cross term, then to decompose the quadratic term into the Bernoulli cost and another cross term. Both cross terms vanish by regularity.

Step 1: Variance excess decomposition. The treated-side contributions to $V^{\hat{\theta}}$ and V^{eff} are identical, so

$$N_1^2(V^{\hat{\theta}} - V^{\text{eff}}) = \sum_{i=1}^n \bar{W}_i (\hat{\theta}_i^2 - \theta(X_i)^2) v_0(X_i).$$

Apply $a^2 - b^2 = (a-b)^2 + 2(a-b)b$ with $a = \hat{\theta}_i$ and $b = \theta(X_i)$:

$$= \sum_{i=1}^n \bar{W}_i (\hat{\theta}_i - \theta_i)^2 v_0(X_i) + 2 \sum_{i=1}^n \bar{W}_i (\hat{\theta}_i - \theta_i) \theta_i v_0(X_i). \quad (75)$$

Step 2: First cross term vanishes. We show the second sum in (75) is $o_p(N_1)$. Set $f = \theta v_0$, which is in $L^2(P)$ by assumption. The sum equals $2N_1$ times

$$\frac{1}{N_1} \sum_{i=1}^n \bar{W}_i (\hat{\theta}_i - \theta_i) f(X_i) = \underbrace{\frac{1}{N_1} \sum_i \bar{W}_i \hat{\theta}_i f_i}_{o_p(1) \text{ by regularity}} - \underbrace{\frac{1}{N_1} \sum_i W_i f_i}_{\rightarrow_p 0 \text{ by LLN}} - \frac{1}{N_1} \sum_i \bar{W}_i \theta_i f_i - \frac{1}{N_1} \sum_i W_i f_i. \quad (76)$$

For the second term, we compute both averages. The control average is

$$\frac{1}{N_1} \sum_i \bar{W}_i \theta_i f_i = \frac{N_0}{N_1} \cdot \frac{1}{N_0} \sum_{W_i=0} \theta(X_i)^2 v_0(X_i) \rightarrow_p \frac{1-p}{p} \mathbb{E}[\theta^2 v_0 \mid W=0].$$

Using $\mathbb{E}[g \mid W=0] = \mathbb{E}[g(1-e)]/(1-p)$ and $\theta^2(1-e) = e\theta$:

$$\frac{1-p}{p} \cdot \frac{\mathbb{E}[\theta^2 v_0(1-e)]}{1-p} = \frac{\mathbb{E}[e\theta v_0]}{p} = \mathbb{E}[\theta v_0 \mid W=1].$$

The treated average $N_1^{-1} \sum_i W_i f_i = N_1^{-1} \sum_{W_i=1} \theta(X_i) v_0(X_i) \rightarrow_p \mathbb{E}[\theta v_0 \mid W=1]$ by the law of large numbers. Both limits match, so the second term in (76) vanishes.

Hence from (75):

$$N_1(V^{\hat{\theta}} - V^{\text{eff}}) = \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i (\hat{\theta}_i - \theta_i)^2 v_0(X_i) + o_p(1). \quad (77)$$

Step 3: Algebraic identity for integer weights. For integer $\hat{\theta}_i$ and $\theta_i = \lfloor \theta_i \rfloor + r_i$, the squared error decomposes as

$$(\hat{\theta}_i - \theta_i)^2 = r_i(1-r_i) + (\hat{\theta}_i - \theta_i)(1-2r_i) + \delta_i(\delta_i - 1), \quad (78)$$

where $\delta_i = \hat{\theta}_i - \lfloor \theta_i \rfloor$ is the integer offset. The remainder $\delta_i(\delta_i - 1) \geq 0$, with equality iff $\delta_i \in \{0, 1\}$, i.e., $\hat{\theta}_i \in \{\lfloor \theta_i \rfloor, \lceil \theta_i \rceil\}$.

To verify: $\hat{\theta}_i - \theta_i = \delta_i - r_i$, so expanding the right-hand side:

$$r_i(1-r_i) + (\delta_i - r_i)(1-2r_i) + \delta_i(\delta_i - 1) = r_i - r_i^2 + \delta_i - 2\delta_i r_i - r_i + 2r_i^2 + \delta_i^2 - \delta_i = \delta_i^2 - 2\delta_i r_i + r_i^2 = (\delta_i - r_i)^2.$$

Step 4: Second cross term vanishes. Summing (78) weighted by v_0 :

$$\sum_i \bar{W}_i (\hat{\theta}_i - \theta_i)^2 v_0 = \sum_i \bar{W}_i r_i (1-r_i) v_0 + \sum_i \bar{W}_i (\hat{\theta}_i - \theta_i) (1-2r_i) v_0 + \sum_i \bar{W}_i \delta_i (\delta_i - 1) v_0.$$

The second sum is N_1 times the imbalance of $\hat{\theta}$ in $(1-2r)v_0 \in L^2(P)$, which is $o_p(1)$ by regularity. The third sum is non-negative.

Combining with (77):

$$N_1(V^{\hat{\theta}} - V^{\text{eff}}) \geq \frac{1}{N_1} \sum_{i=1}^n \bar{W}_i r_i (1-r_i) v_0(X_i) + o_p(1),$$

with equality when all $\delta_i \in \{0, 1\}$. □

The following proposition shows that balance-preserving rounding achieves equality in (74), and provides a consistent variance estimator.

Proposition 8.3'. *Under the conditions of Proposition 8.2', let $\mathcal{M}$ be a model with bounded uniform entropy integral and suppose:*

- (i) *Real-valued weights $\hat{\theta}$ achieve $o_p(n^{-1/2})$ imbalance uniformly over $\mathcal{M}$.*
- (ii) *Balance-preserving rounding (58) preserves this rate: $\text{rounding}_{\mathcal{M}, \hat{\theta}}(\lfloor \hat{\theta} \rfloor) = o_p(n^{-1/2})$.*
- (iii) *$\|\hat{\theta} v_0\|_{\mathcal{M}} = O_p(n^{1/2})$ and $\|(1-2\hat{r})v_0\|_{\mathcal{M}} = O_p(n^{1/2})$.*
- (iv) *$\hat{\theta} \rightarrow_p \theta$ in $L^2(P)$.*

Then the matching estimator $\hat{\psi}^{\text{match}} = N_1^{-1} \sum_i W_i Y_i - \bar{W}_i \lfloor \hat{\theta} \rfloor_i Y_i$ is regular and asymptotically linear:

$$\hat{\psi}^{\text{match}} - \tilde{\psi} = \frac{1}{N_1} \sum_{i=1}^n (W_i + \bar{W}_i \lfloor \hat{\theta} \rfloor_i) \epsilon_i + o_p(N_1^{-1/2}),$$

with variance achieving equality in (74):

$$\text{var}(\hat{\psi}^{\text{match}}) - V^{\text{eff}} = \frac{1}{N_1^2} \sum_{i=1}^n \bar{W}_i r(X_i) \{1 - r(X_i)\} v_0(X_i) + o_p(N_1^{-1}).$$

A consistent variance estimator is

$$\hat{V} = \frac{1}{N_1^2} \sum_{i=1}^n (W_i + \bar{W}_i \lfloor \hat{\theta} \rfloor_i^2) \{Y_i - \hat{m}(W_i, X_i)\}^2, \quad (79)$$

with $\hat{V} / \text{var}(\hat{\psi}^{\text{match}}) \rightarrow_p 1$ and $(\hat{\psi}^{\text{match}} - \tilde{\psi}) / \sqrt{\hat{V}} \rightsquigarrow N(0, 1)$.

Proof of Proposition 8.3'. We verify each conclusion in turn.

Regularity. Condition (ii) gives $o_p(n^{-1/2})$ imbalance uniformly over $\mathcal{M}$. Because $\mathcal{M}$ has bounded uniform entropy integral, $\mathcal{M}$ is Donsker, so $o_p(n^{-1/2})$ imbalance uniformly over $\mathcal{M}$ implies $o_p(1)$ imbalance in every $f \in L^2(P)$ —the definition of regularity (Definition 2).

Asymptotic linearity. By condition (ii) and the triangle inequality (54), $\text{imbalance}_{h, \mathcal{M}}(\lfloor \hat{\theta} \rfloor) \leq \text{imbalance}_{h, \mathcal{M}}(\hat{\theta}) + \text{rounding}_{\mathcal{M}, \hat{\theta}}(\lfloor \hat{\theta} \rfloor) = o_p(n^{-1/2})$. The asymptotic linearity (61) then follows from Corollary 8.4.

Variance formula. We show $V^{\hat{\theta}} = V^{\text{eff}} + o_p(N_1^{-1})$ for the unrounded weights, then add the rounding cost from (73).

For the unrounded weights, the variance decomposition (75) gives

$$N_1^2(V^{\hat{\theta}} - V^{\text{eff}}) = \sum_i \bar{W}_i(\hat{\theta}_i - \theta_i)^2 v_0(X_i) + 2 \sum_i \bar{W}_i(\hat{\theta}_i - \theta_i) \theta_i v_0(X_i).$$

The cross term vanishes by the same argument as (76): regularity of $\hat{\theta}$ (which holds by condition (i) and Donskerity of $\mathcal{M}$) gives $o_p(1)$ imbalance in $\theta v_0 \in L^2(P)$. The quadratic term is

$$\frac{1}{N_1} \sum_i \bar{W}_i(\hat{\theta}_i - \theta_i)^2 v_0(X_i) \leq \|\hat{\theta} - \theta\|_{L^2(P_0)}^2 \cdot \|v_0\|_{\infty} = o_p(1)$$

by condition (iv).

Second, from (73):

$$N_1(V^{\lfloor \hat{\theta} \rfloor} - V^{\hat{\theta}}) = \frac{1}{N_1} \sum_i \bar{W}_i \hat{r}_i(1 - \hat{r}_i) v_0(X_i) + R\{(1 - 2\hat{r})v_0\} + 2R(\hat{\theta}v_0).$$

By Proposition E.4, condition (iii) ensures $R(\hat{\theta}v_0) = o_p(1)$ and $R\{(1 - 2\hat{r})v_0\} = o_p(1)$.

Combining and using $\hat{r}_i \rightarrow_p r_i$ from condition (iv):

$$N_1(V^{\text{match}} - V^{\text{eff}}) = \frac{1}{N_1} \sum_i \bar{W}_i r_i(1 - r_i) v_0(X_i) + o_p(1).$$

Variance estimation. The estimator (79) uses squared residuals $\{Y_i - \hat{m}(W_i, X_i)\}^2$ in place of $v_{W_i}(X_i)$. By uniform consistency of $\hat{m}$, $(Y_i - \hat{m}_i)^2 \rightarrow_p v_{W_i}(X_i)$, so

$$\frac{\hat{V}}{V^{\text{match}}} = \frac{\sum_i (W_i + \bar{W}_i \lfloor \hat{\theta} \rfloor_i^2) (Y_i - \hat{m}_i)^2}{\sum_i (W_i + \bar{W}_i \lfloor \hat{\theta} \rfloor_i^2) v_{W_i}(X_i)} \rightarrow_p 1$$

by the law of large numbers. The CLT follows from asymptotic linearity (61) and this variance consistency. $\square$

For nonparametric BUEI classes of bounded functions, the conditions of Proposition 8.3' are satisfied by the minimax balancing weights.

Corollary 8.5'. *Let $\mathcal{M}$ be a nonparametric (Definition 1) class of uniformly bounded functions with bounded uniform entropy integral. Assume $E[1/e(X)] < \infty$ and $\hat{\theta}v_0, (1 - 2\hat{r})v_0 \in L^2(P)$.*

For an appropriately chosen sequence of budgets $B_n \ll n^{-1/2}$, the integer program (63) has a solution $\lfloor \hat{\theta} \rfloor^$ with probability tending to one. The matching estimator $\hat{\psi}^{\text{match}} = N_1^{-1} \sum_i W_i Y_i - \bar{W}_i \lfloor \hat{\theta} \rfloor_i^* Y_i$ satisfies*

$$\hat{\psi}^{\text{match}} - \tilde{\psi} = \frac{1}{N_1} \sum_{i=1}^n (W_i + \bar{W}_i \lfloor \hat{\theta} \rfloor_i^*) \epsilon_i + o_p(N_1^{-1/2})$$

with variance achieving equality in (74). The variance estimator (79) satisfies $\hat{V}/\text{var}(\hat{\psi}^{\text{match}}) \rightarrow_p 1$ and $(\hat{\psi}^{\text{match}} - \tilde{\psi})/\sqrt{\hat{V}} \rightsquigarrow N(0, 1)$.

Proof of Corollary 8.5'. We verify the conditions of Proposition 8.3'.

Condition (i). By Hirshberg et al. (2019, Theorem 2), the minimax linear weights achieve $o_p(n^{-1/2})$ imbalance uniformly over the unit ball of $\mathcal{H}$.

Condition (ii). By Proposition 8.1, balance-preserving rounding preserves the $o_p(n^{-1/2})$ rate.

Condition (iii). By Corollary E.5, any $f \in L^2(P)$ can be approximated by $f_n \in \mathcal{H}$ with $\|f_n\|_{\mathcal{H}} = O(n^{1/2})$. Since $\hat{\theta}v_0, (1-2\hat{r})v_0 \in L^2(P)$ by assumption, both have $\mathcal{H}$ -norm $O_p(n^{1/2})$.

Condition (iv). L^2 consistency of $\hat{\theta}$ toward θ follows from Hirshberg et al. (2019, Theorem 1): for bounded continuous RKHS kernels dense in $L^2(P)$, the minimax linear weights converge to the IPW weights in $L^2(P)$ at a rate depending on the kernel’s eigenvalue decay.

Feasibility. The rounded minimax linear weights are feasible for (63) with budget B_n equal to the sum of the imbalance and rounding rates (see the explicit rate calculations earlier in this section). The integer-optimal solution $[\hat{\theta}]^*$ minimizes $\sum_i \bar{W}_i \theta_i^2$ over feasible integer weights, so has at least as good imbalance as the rounded continuous solution. In particular, $[\hat{\theta}]^*$ satisfies conditions (i)–(iv) of Proposition 8.3’.

Applying Proposition 8.3’: asymptotic linearity follows from (61), the variance achieves equality in (74), and the variance estimator (79) is consistent. $\square$

F Regression for Orthogonality

Wang and Shah (2020) observe that the balancing logic can be reversed. Rather than choosing weights $\hat{\gamma}$ so that the error $\delta\gamma = \hat{\gamma} - \gamma^{\text{ipw}}$ is approximately orthogonal to a model $\mathcal{M}$ for the outcome, we can choose the outcome regression $\hat{\mu}$ so that its error $\delta\mu = \hat{\mu} - \mu$ is approximately orthogonal to the weight error. Here we sketch the idea in our notation.

Recall that the bias of the AIPW estimator can be written as

$$n^{-1} \sum_{i=1}^n W_i \delta\gamma(X_i) \delta m(X_i) + n^{-1} \sum_{i=1}^n \{W_i \gamma^{\text{ipw}}(X_i) - 1\} \delta m(X_i). \quad (80)$$

When the weights are the true inverse propensity weights, the second term is an average of mean-zero terms and is $O_p(n^{-1/2})$. The first term—the inner product of the errors—is what we need to control.

The balancing approach bounds this by controlling $\sup_{m \in \mathcal{M}} n^{-1} \sum_i W_i \delta\gamma(X_i) m(X_i)$, ensuring approximate orthogonality to a model $\mathcal{M}$ containing $\delta\mu$. But we could instead control the other factor: choose $\hat{\mu}$ so that

$$\sup_{\gamma \in \Gamma} n^{-1} \sum_{i=1}^n \gamma(X_i) \{\hat{\mu}_1(X_i) - Y_i\} = O_p(n^{-1/2}) \quad (81)$$

for a class Γ containing $\delta\gamma$. This makes the regression error approximately orthogonal to the types of errors we expect in estimating the propensity score.

For the treatment-specific mean, this can be implemented using sample splitting. Let (X'_i, Y'_i) be an independent copy of (X_i, Y_i) among treated units. Choose $\hat{\mu}_1(\cdot)$ to satisfy

$$\sup_{\gamma \in \Gamma} \left| n^{-1} \sum_{i=1}^n \gamma(X_i) \hat{\mu}_1(X_i) - n^{-1} \sum_{i=1}^n \gamma(X'_i) Y'_i \right| = O_p(n^{-1/2}). \quad (82)$$

Because $\hat{\mu}_1(\cdot)$ is independent of $W_1 \dots W_n$, the equicontinuity term in the error decomposition is $O_p(n^{-1/2})$ automatically. This yields $\sqrt{n}$ -consistent estimation when the Riesz representer γ^{ipw} is regular in the sense that $\|\hat{\gamma} - \gamma^{\text{ipw}}\|_{\Gamma} = O_p(1)$ for a Donsker class Γ —with essentially no assumptions on the outcome model.

This is the reverse of the balancing approach: there we assume the outcome model is regular and make no assumptions on the propensity score; here we assume the propensity score model is regular and make no assumptions on the outcome.

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